

# **AF Design Guidelines**

UBS Forms

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# 1 1) Introduction

The following section should provide an overview and help to orientate throughout the documentation. The documentation will use the forms “Change of address” and “Change of domicile” as example forms.



Each “Adaptive Form” will be created based on a «Form template». Each form template is described in detail later on. Here’s a short description of the different adaptive form types:

| Adaptive form type | Description |
|--------------------|-------------|
|--------------------|-------------|

|                   |                                                                                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form              | An adaptive form for a certain purpose. This is the most common type of adaptive form. For example, the “Change of address” and “Change of domicile” are separate Forms.                                                                                                                       |
| FormSet           | A FormSet includes multiple “Form” in an accordion style. For example, the Forms “Change of address” and “Change of domicile” can be referenced in a FormSet and therefore integrated into one adaptive form.                                                                                  |
| Confirmation page | Both the “Form” and the “FormSet” will land on a confirmation page after the submit. Each adaptive form can configure, on which confirmation page they should land. Based on the “Confirmation page template”, multiple different confirmation pages can be created and referenced as desired. |

## 2 2) Form templates

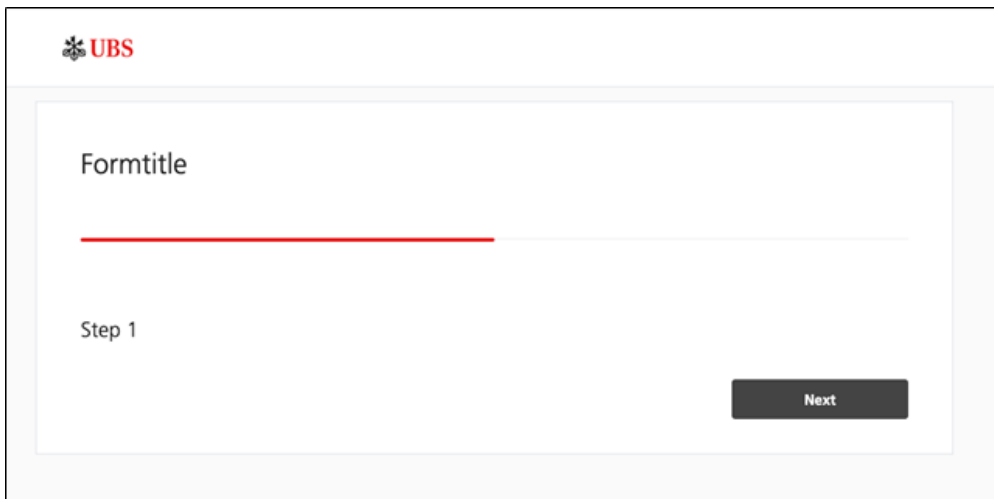
This section gives you an overview over the different form templates. A form template gives the author a basic structure for a specific purpose.

For more information, visit the Adobe documentation:

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/adaptive-forms-advanced-authoring/template-editor.html>

### 2.1 Basic template

The basic template is the source for every “Single Adaptive Form”. It creates a basic structure with two steps. Here’s the structure a basic template has:

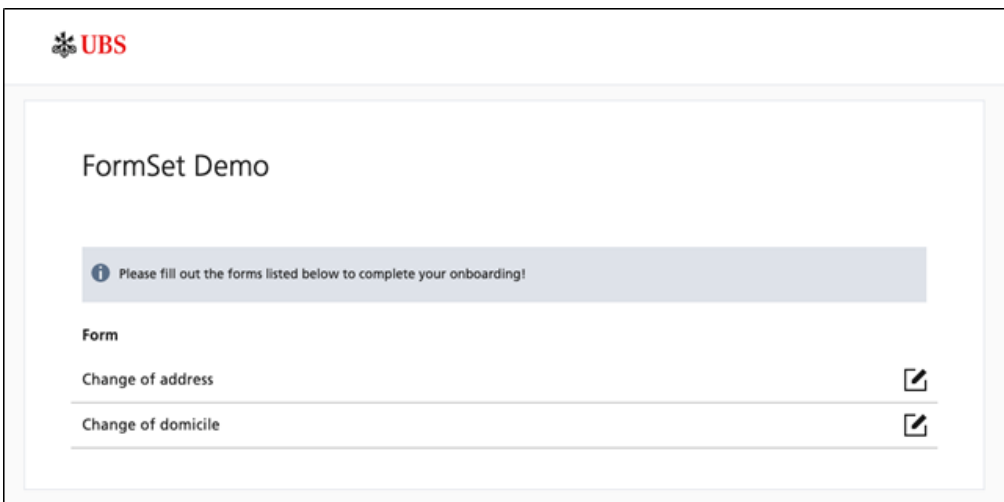
The screenshot shows a form template layout. At the top left is the UBS logo. Below it is a large rectangular area containing the text 'Formtitle'. A red horizontal line is positioned below the title. Underneath the line is the text 'Step 1'. In the bottom right corner of the form area is a dark grey button labeled 'Next'.

| Element             | Description                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Formtitle           | The form title will remain the same throughout every step of the form.                                        |
| Progress bar        | The progress bar indicates the progress while filling the form and will be visible in every step of the form. |
| Step title (Step 1) | This is the title of the current step. Each step of the form will have its own title.                         |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Toolbar</p> | <p>The toolbar consists of four buttons: Next, Back, Preview and Submit.</p> <ul style="list-style-type: none"> <li>• The next button is visible as long as there is a further step. When reaching the last step, it will be replaced with the submit button. Clicking the next button will validate the current step.</li> <li>• The back button is visible when there is a predecessor step and will take the user back to it.</li> <li>• The preview button is visible on the last step of the form and will generate a preview which is displayed in a carousel. For further information, please visit the “Preview” chapter.</li> <li>• The submit button is visible on the last step of the form. Clicking it will validate the form and submit it. The user will be redirected to the configured confirmation page.</li> </ul> |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2.2 FormSet

The form set template is the source for every “Form0” and “Business Formset”. It creates a basic structure with the overview of all subforms, panels for referring the subforms and a preview section. The following example uses the “Change of Address” and “Change of Domicile” form. The entry point in every Form Set is the overview:



In the overview, the user can easily identify all forms he needs to fill out in order to complete the form set. To enter a subform, the user can click the “Edit” button on the right side of a subform entry. When entered a subform, the look and feel is the same as in a single form:

UBS

### Change of Address

Address old

Additional address details \*

Street \*

Postal code/City \*

Country \*

Home phone no. \*      Office phone no. \*

Next

The user can fill out the form and, if configured in the single form, create a preview on the last step. By clicking the “Finish” button on the last page, the subform is completed. When a subform is completed, the overview will look like this:

UBS

## FormSet Demo

Please fill out the forms listed below to complete your onboarding!

| Form               |   |  |
|--------------------|---|--|
| Change of address  | ✓ |  |
| Change of domicile |   |  |

The tick indicates to the user, which forms are already completed. He can always reenter the subform by clicking the “Edit” button. When all subforms are completed, a “Next” button will appear on the Overview page. By clicking “Next”, the user will have the possibility to create a preview for the whole form set.



**FormSet Demo**

By clicking the "Generate Preview" button, you can have a look at your document before finally submitting it. This step is optional and can take some time.

Back Preview Submit

The user can go back to the overview by clicking "Back" and adjust the subforms as he wishes. The "Submit" button works as in the single adaptive forms.

## 2.2.1 Configuration

Follow the steps below to reference an adaptive form in a form set.

|   |                                                                                                                                                                                                                                                                                                                                                                                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Create an adaptive form and select the "UBS FormSet" template.                                                                                                                                                                                                                                                                                                                    |
| 2 | The naming in adaptive forms is important. Select a name for your form. This name will be a part of the following configurations. For example for "Change of Address" it could be: changeOfAddress. It will be referred to as <i>NAME</i> in the following parts.                                                                                                                 |
| 3 | <p>Go to the "Subform1 overview" panel</p> <ol style="list-style-type: none"> <li>1. Rename the panel to <i>NAMEOverview</i> (exp. changeOfAddressOverview) and add a proper title.</li> <li>2. Rename the subform1Title panel to <i>NAMETitle</i> (exp. changeOfAddressTitle)</li> <li>3. Rename the static text "Subform 1" to the subform name the user should see.</li> </ol> |

|   |                                                                                                                                                                                                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Close the “FormList” panel and open the configuration of the panel “Subform Reference 1”</p> <ol style="list-style-type: none"> <li>1. Rename it to <i>NAME</i> (exp. changeOfAddress) and add a proper title.</li> <li>2. Reference your desired subform in the field “Form or Fragment”.</li> </ol>  |
| 5 | <p>Open the rule editor of the edit button in the overview and add the following code when a click happens:</p> <pre> window.forms.ubs.formset.enterSubform(NAME, overviewPanel);NEW : window.forms.ubs.formset.enterSubform(NAME, overviewPanel); </pre> <p>Replace the NAME (exp. changeOfAddress).</p> |
| 6 | <p>Repeat the steps for other subforms. For more subforms, copy one of the example panels.</p>                                                                                                                                                                                                            |

## 2.2.2 Options

| Option                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document of record generation | <p>In a FormSet you can choose between two Document of Record generation types:</p> <ul style="list-style-type: none"> <li>• <b>Default DoR rendering:</b> In this case, every subform is included in a single Document of Record. Each subform is rendered into the template configured in the formset configuration. This means, that a subform with a custom XDP won't be rendered into the custom XDP. The generated PDF which includes all subforms can be downloaded by the user.</li> <li>• <b>Form0:</b> Each subform will create its own Document of record. The generated DoR's will be merged into one PDF. Hint: Each subform is rendered according to its configuration (difference to the default DoR rendering)!</li> <li>• <b>Business Formset:</b> In this case, every subform will create its own Document of Record and store it in CLP. All generated PDF's (one per subform) will be added to a .zip file. The user will be able to download the .zip file. Hint: Each subform is rendered according to its configuration (difference to the default DoR rendering)!</li> </ul> <p><b>You can find this option in the RootPanel configuration of your form set form.</b></p> |

### 2.2.3 Conditional forms

If you want to dynamically include or exclude subforms, use the following javascript functions:

```
window.forms.ubs.formset.includeSubform(panel);

window.forms.ubs.formset.excludeSubform(panel);
```

Refer to the panel which referenecs the adaptive form. In that case, the subform doesn't need to be filled and won't show up in the document of record.

### 2.2.4 Subform completion indicator

If we have to remove the subform completion indicator based on some requirement, we can use below function:

```
$('#.'.concat(subform.name).concat('Overview')).removeClass('subformCompleted');
```

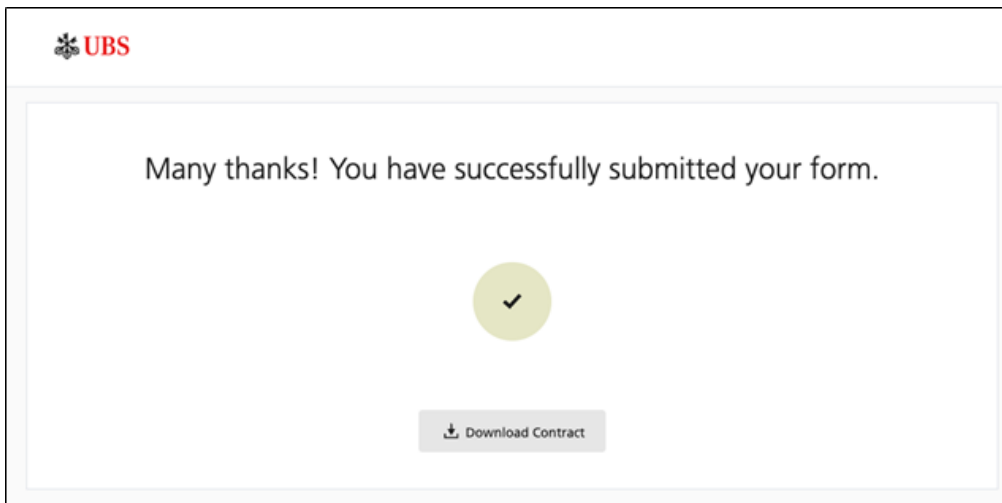
### 2.2.5 External documents on Formset level

External documents (preliminary documents / attachments) on a formset level can only be applied to formsets, which have the “Merged PDF” option on the rootPanel. Other options will not apply the attachments, as it was not needed on the first implementation.

The external documents of the formset will be numbered according to the checkbox on the rootPanel (“Consecutive Numbering”).

## 2.3 Confirmation page template

The confirmation page template can be used for pages the user should land on after submitting the form. Here's the basic structure of a confirmation page template:



| Element         | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Formtitle       | The formtitle is prefilled with the text “Many thanks!...” text. It can be changed to whatever title desired. |
| Success icon    | The success icon is displayed before every component the author might add to the confirmation page.           |
| Download button | The download button is displayed in the toolbox and will allow the user to download the generated document.   |

## 2.4 Using a template

When creating an adaptive form, you can choose upon which template you want to create it. The templates will be presented as follows during the adaptive form creation:



## 3 3) Adaptive form

Every form template described will generate an adaptive form. This is important to know, as all the following configuration possibilities apply to all of them (whether it's from basic template, form set template or configuration page template).

The next chapters will briefly describe important configurations in the UBS context and not every configuration possible. For detailed information about the configuration of adaptive forms, please visit the Adobe documentation:

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/home.html>

### 3.1 Browser Tab Title

If a Formcode and a corresponding mandator is specified in the metadata component, the tab title will include the master CDOK and the formcode:



The mandator to identify the master CDOK is extracted from the URL parameter "mandator":

*HOST:PORT/content/dam/formsanddocuments/afforms\_global\_all/af\_ab/AF\_ABCQ/jcr:content?  
wcmmode=disabled&afAcceptLang=fr-ch&mandator=001*

If no parameter is passed or if its wrong then the current form name will take place as browser tab title.

### 3.2 Multi column form

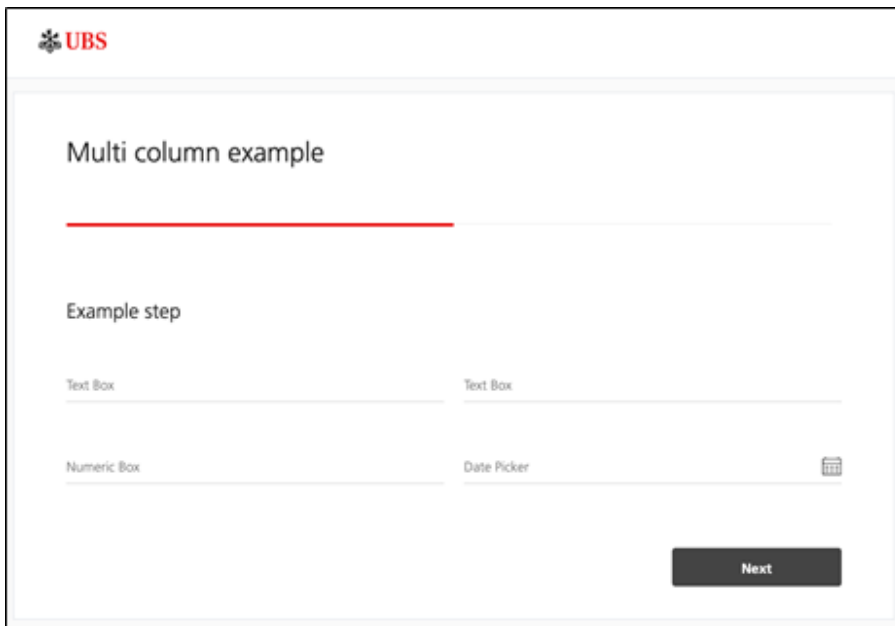
In the beginning of the project, the multi column layout was achieved by setting the "Number of Columns" and "Colspan" property. Adobe changed the approach of this and introduced the "Layout" mode. So the previous fields in the configuration are removed and every newly created adaptive form will not have them. This applies also to adaptive fragments. In the following section, the "Layout" mode is briefly described. For more detailed information, please visit the Adobe documentation:

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/adaptive-forms-basic-authoring/resize-using-layout-mode.html>

**Important:**

1. The column configurations made in the "Layout" mode will only have an effect on the adaptive forms. The document of record is not affected by it. For the configuration of multi columns in the DoR, please visit the chapter in the document of record section.
2. Previously created forms will not be affected and can be configured as before. The form needs no migration.

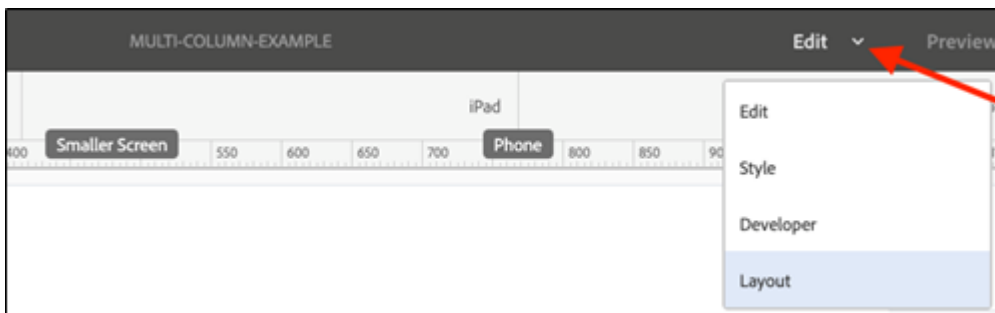
A multi column form can look like this:



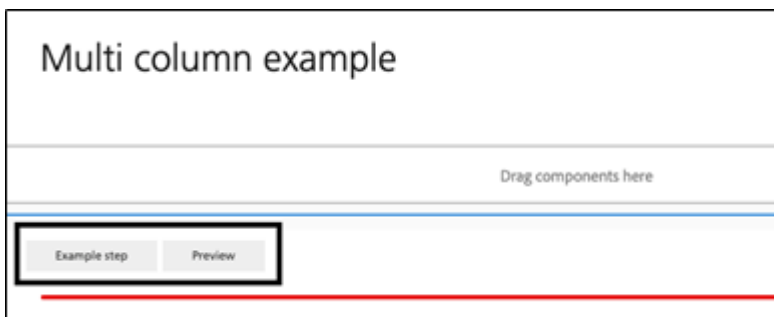
The image shows a form titled "Multi column example" with a red horizontal line below the title. Below the line is the text "Example step". The form contains four input fields arranged in two rows: "Text Box", "Text Box", "Numeric Box", and "Date Picker". A "Next" button is located at the bottom right.

### 3.2.1 Accessing the layout mode

In order to achieve such a multi column layout, access the layout mode by clicking the action bar and selecting "Layout":



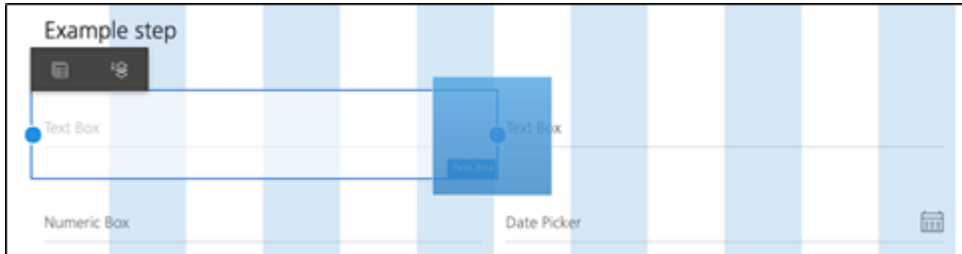
**Hint:** The sidepanel with the steps is not visible in the Layout mode. In order to maneuver through the steps, an additional navigation is added when editing the adaptive form. This navigation shown in the picture below will not be shown to the user:



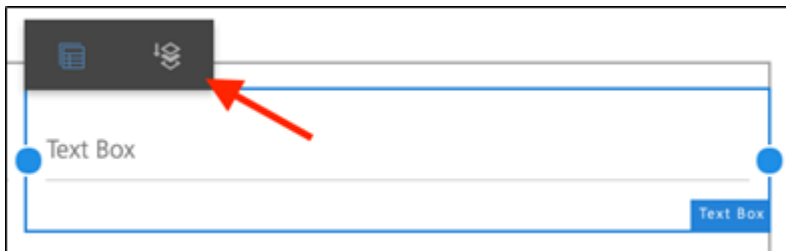
The image shows the form titled "Multi column example" with a red horizontal line below the title. Below the line is the text "Drag components here". At the bottom, there is a navigation bar with two buttons: "Example step" and "Preview".

### 3.2.2 Using the layout mode

When clicking on a component, there are two blue dots shown at each side of it. When clicking on one of the blue dots, the component width can be changed by dragging it on one side. The size can be adjusted inside the 12 columns shown.



When multiple components fit on one row, AEM will align them next to each other. If the components should not be aligned next to each other, the “Float to new line” button can be used. When this button is clicked on a component, it will be moved to the next line:

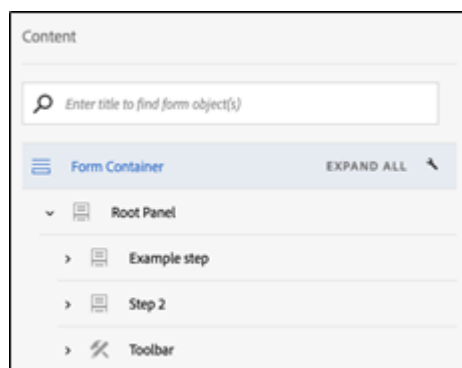


### 3.3 Submission

Every adaptive form can configure actions which will be executed after the submission of a form. The main configurations are the “Redirect URL” and “Submit Action”. Complete the following steps to configure a proper form submission:

1

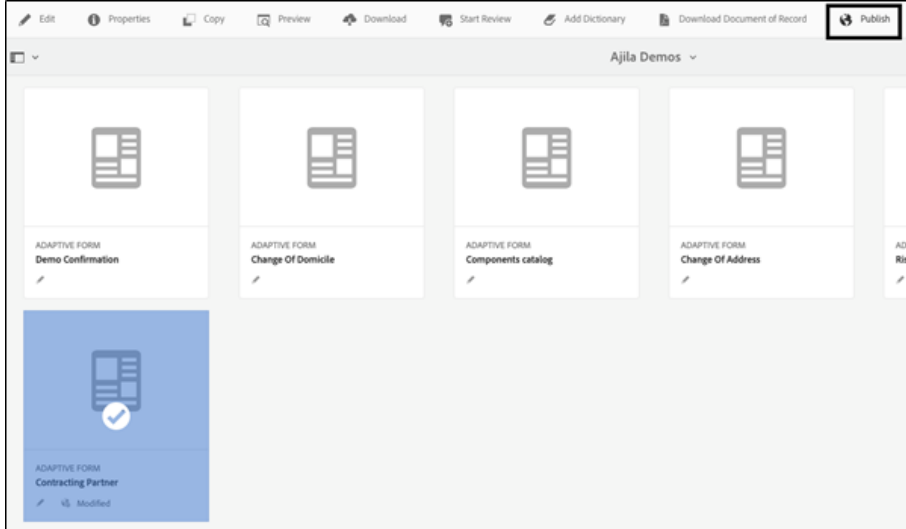
Open the configuration of the adaptive form container.



|   |                                                                                                                                                                                             |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | Open up the “Submission” tab.                                                                                                                                                               |
| 3 | In the “Redirect URL/Path” you can select a confirmation page where the user should land on after the submission.<br>The basic path for the UBS forms is: /content/forms/af/ajila-forms-ubs |
| 4 | The “Submit Action” is set to UBS Submit by default and will trigger the generation of the PDF.                                                                                             |

## 3.4 Publishing

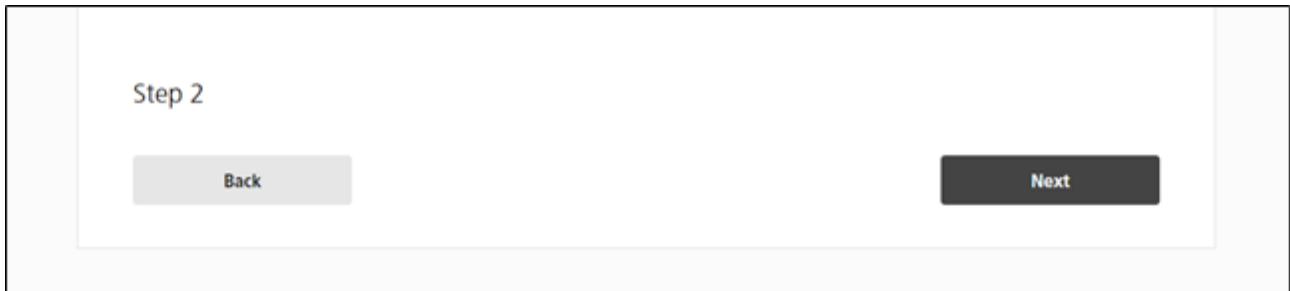
Forms are created in the author instance of AEM. The user will access the form on the publish instance. In order to publish an adaptive form from the author to the publish instance, complete the following steps:

|   |                                                                                                                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Select the adaptive form in the author instance.                                                                                       |
| 2 | Click the button “Publish” in the options bar.<br> |
| 3 | Wait until the Success pop up is shown that the item is queued for publishing.                                                         |



## 3.5 Multistep separation

The UBS standard theme comes with a multistep logic which allows the author to split the form into multiple pages (steps) to offer a better overview to the enduser.



Stick to these guidelines addressing the split up of the form:

- Conform to the existing groupings (sections) of the PDF form.
- If a section only got three components or less, see if you can put together multiple sections to one bigger section. You can therefore use static bold text for the existing section title and make up a title for the new section.
- Keep the steps small, if you have to scroll within a step it is too large (exceptions possible).
- Sections which appear in several forms (such as a address block) should always be build up the same (maybe through fragments?)

## 3.6 3.1) Adaptive Form Types

This section describes the different Adaptive Form Types. Each type has different characteristics. For the naming convention of the PDF per type, please visit this page: [7.1\) PDF File Naming Convention \(see page 101\)](#)

### 3.6.1 Single Adaptive Form

A Single Adaptive Form results from the "UBS" template. It can be used as a standalone form or be included in a Form0 or Business Formset. The Single Adaptive Form has the following characteristics:

- Handles one main CDOK
- Only has one main document produced
- Can have consecutive attachments that are merged into the main document
- Can have individual attachments that result in separate PDF outputs (Annexes)

### 3.6.2 Form0

A Form0 is based on the Formset technology. It comes along with the overview page and includes Single Adaptive Forms. The Form0 has the same characteristics as the Single Adaptive Form.

The Form0 approach can be used when:

- The Single Adaptive Form approach has technical limitations (Multiple titles, pages breaks, ...)
- To achieve a better user experience with the overview page

The Form0 in the end is just another way of presenting a Single Adaptive Form. All "Main" documents of the subforms will be merged together to achieve the single main document output since it handles the same CDOK.

### 3.6.2.1 Example

An example for a Form0 is the adaptive form ABKA. It includes for Single Adaptive Forms as subforms. But all of these subforms have the same CDOK in the metadata component (63126). Therefore, those subforms can be merged together to one document and be processed as one by CLP/DPE.

## 3.6.3 Business Formset

A Business Formset is also based on the Formset technology. It also comes along with the overview page and includes Single Adaptive Forms. The Business Formset has the following characteristics:

- Handles multiple CDOKs
- Produces one "Main" document per subform (CDOK)
- Includes all individual attachments of its subforms

The Business Formset is what is considered a "real" Formset. Meaning it will also appear as such in Kiosk for example (31.05.2021: Formsets are not yet displayed in Kiosk). Whereas the Single Adaptive Form and Form0 will be presented as normal forms.

### 3.6.3.1 Example

An early example of a Business Formset is the adaptive form ACBU. This Business Formset includes the following Single Adaptive Forms:

1. Change of Name (61501)
2. Change of Domicile (61502)

Since the CDOK is different between the subforms, multiple Main documents need to be created and processed.

## 3.6.4 DoR Configuration for Form0 approach

On the Formset Root Panel, we have to set the configuration as below:

Root Panel

▼ **Basic**

Name \*

Panel Layout \*

Toolbar Position

CSS Class

☐ Hide Object

☐ Disable Object

Document of Record generation

☐ Consecutive Numbering ?

> Document of Record

Based on the requirement, we can have continuous paginations for all subforms or separate pagination for the each subform of the Formset by selecting or deselecting the checkbox.

Root Panel

▼ **Basic**

Name \*

Panel Layout \*

Toolbar Position

CSS Class

☐ Hide Object

☐ Disable Object

Document of Record generation

☒ Consecutive Numbering ?

> Document of Record

## 4 4) Components

Every adaptive form consists of components. AEM Forms has a number of standard components. These standard components are the basis for the components inside the “UBS” component group. The components inside the “UBS” component group can be used as they support the UBS look and feel.

For further information about the AEM standard components and the basic configuration of them, please visit the Adobe documentation below. Because the UBS components extend these standard components, the configurations apply to the UBS components as well.

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/getting-started/introduction-forms-authoring.html>

### 4.1 Basic configuration

The basic configuration apply to all UBS form components.

#### 4.1.1 Basic tab

| Property               | Description                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                   | Provide a name for the component. This name is more technical and should be unique inside the adaptive form. It will be used for the XML tag in the form output when no schema is referenced. |
| Title                  | The title will be displayed in the form as component title and therefore be visible to the user.                                                                                              |
| Hide title             | Activate this checkbox when you don't want the title to be shown in the form.                                                                                                                 |
| Placeholder Text       | Define a placeholder text which will be displayed.                                                                                                                                            |
| Required field         | Activate this checkbox if the component needs to be filled by the user. The form will validate the field on every step change or before the submission.                                       |
| Required Field Message | Provide a message displayed to the user when the component is required but not filled.                                                                                                        |

|                           |                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Script Validation Message | Provide a message displayed to the user when a validation script fails. This script can be created by the rule editor for example. |
| Bind Reference            | Bind the component to a schema node. This will be mainly used for custom XDP's.                                                    |
| Default Value             | Provide an initial value for the user.                                                                                             |
| Hide Object               | Activate this checkbox when the component shouldn't be displayed to the user. For example a placeholder in a multi column form.    |
| Disable Object            | Activate this checkbox when the component should not be editable.                                                                  |
| CSS class                 | Add custom CSS classes to the component.                                                                                           |

### 4.1.2 Help tab

| Property          | Description                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Short description | Add a short description to the component which will be placed underneath it.                                                                     |
| Long description  | Add a long description. In that case, a question mark icon will be presented to the user. By clicking on it, the long description will be shown. |

The screenshot shows a form component titled "Text Box". Below the title is a "Short description" field. To the right of the "Short description" field is a button labeled "Long description" with a question mark icon next to it.

### 4.1.3 Patterns tab

On various components you can define patterns for displaying and validating the user input. This section will be a separate tab. The documentation of Adobe for defining patterns can be found here:

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/adaptive-forms-advanced-authoring/adaptive-form-expressions.html>

## 4.2 Static title

The static title component can be used for informative titles with whom the user cannot interact. While editing the content, AEM offers the following typographical emphasis:

- Bold, italic and underlined
- Text alignment (left, center, right)
- Lists
- Links

In addition, the static title can be used as a step title alternative. This is used to achieve a different result in the document of record. In order to have the same styling, use the CSS class “stepTitle” on the static title.

## 4.3 Static text

The static text component can be used for informative text with whom the user cannot interact. While editing the content, AEM offers the following typographical emphasis:

- Bold, italic and underlined
- Text alignment (left, center, right)
- Lists
- Links

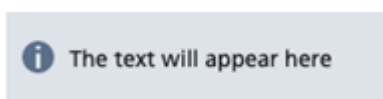
### 4.3.1 Listings

To add special listing stylings (roman, alpha) do the following steps:

- click the bullet listing symbol in the static text editor
- switch to html view
- in the <ol> tag, add this: style="list-style-type: \*STYLE\*;"
- replace \*STYLE\* with one of the following styles:
  - lower-roman
  - upper-roman
  - lower-alpha
  - upper-alpha

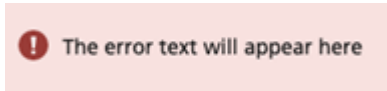
## 4.4 Information text

The information text component offers the same features as the static text but has a specialized appearance which looks as follows:



## 4.5 Error text

The error text component offers the same features as the static text but has a specialized appearance which looks as follows:



## 4.6 Text Box

The Text Box component can be used for user input on one line. If multiple lines are needed, look for the Text Box Multiline.

The text box offers the following additional configurations:

| Property                     | Description                                                                    |
|------------------------------|--------------------------------------------------------------------------------|
| Maximum Number of Characters | Define the maximum number of characters the user can enter into the text box.  |
| Minimum Number of Characters | Define the minimum number of characters the user must enter into the text box. |
| Exact Number of Characters   | Define the exact number of characters the user has to enter.                   |

## 4.7 Text Box Multiline

The text box multiline component can be used for user input with more than one line. The multiline text box offers three lines visible to the user. If the user types more than three lines, a scroll bar is shown. The text box multiline offers the same additional configurations as the text box.

## 4.8 Numeric Box

The numeric box allows the user to only enter numbers. Other inputs such as alphabetic characters will be ignored.

The numeric box offers the following additional configurations:

| Property | Description |
|----------|-------------|
|----------|-------------|

|                       |                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type             | <p>Choose between three datatypes:</p> <ul style="list-style-type: none"> <li>• Empty: Will behave like option “Decimal”.</li> <li>• Decimal: Two decimal places are allowed. Further inputs will be ignored.</li> <li>• Integer: No decimal places are allowed.</li> <li>• Float: No restriction of decimal places.</li> </ul> |
| Lead digits           | <p>Not available with data type integer.</p> <p>Define how many digits before the decimal point are allowed. Any further inputs will be ignored.</p>                                                                                                                                                                            |
| Frac digits           | <p>Not available with data type integer.</p> <p>Define how many digits after the decimal point are allowed. Any further inputs will be ignored.</p>                                                                                                                                                                             |
| Minimum value         | Define the minimum value the user has to enter.                                                                                                                                                                                                                                                                                 |
| Exclude minimum value | <p>Excludes the given minimum value. If the minimum value is 7 and excluded, 7 is not valid.</p> <p>This option is relevant for the validation message the user receives when he doesn’t meet the requirements!</p>                                                                                                             |
| Maximum value         | Define the maximum value the user can enter.                                                                                                                                                                                                                                                                                    |
| Exclude maximum value | <p>Excludes the given maximum value. If the maximum value is 10 and excluded, 10 is not valid.</p> <p>This option is relevant for the validation message the user receives when he doesn’t meet the requirements!</p>                                                                                                           |

## 4.9 Email

The email component offers the user an optimized field for entering an email address. It is preconfigured with validations for a valid email address. It derives from the text box component, which means that the text box configurations apply also to the email component.

The email component offers the following additional configurations:

| Property | Description |
|----------|-------------|
|----------|-------------|



|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| Enable autofill | If activated, the browser will suggest entries from the browser's autofill history. |
|-----------------|-------------------------------------------------------------------------------------|

### 4.9.1 Pattern

In the pattern section, the email validation is preconfigured along with the error text.

## 4.10 Telephone

The telephone component offers the user an optimized field for entering a telephone number. It has preconfigured display and validation patterns to ensure proper input from the user. It derives from the text box component, which means that the text box configurations apply also to the telephone component.

The telephone component offers the following additional configurations:

| Property        | Description                                                                         |
|-----------------|-------------------------------------------------------------------------------------|
| Enable autofill | If activated, the browser will suggest entries from the browser's autofill history. |

### 4.10.1 Pattern

In the pattern section, the telephone display and validation is preconfigured along with the error text.

## 4.11 Datepicker

The datepicker offers the user an intuitive calendar to choose a date from. It is also possible to enter a date via the input field of the datepicker. The datepicker translates its content according to the given locale and supports the following languages:

- English
- German
- Italian
- French

If the user provides a locale the datepicker doesn't support, it will fall back to english as default. In case of german, french and italian the week starts on monday. In case of english it starts on sunday.

The datepicker offers the following additional configurations:

| Property | Description |
|----------|-------------|
|----------|-------------|

|                                   |                                                                                                                                                                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Set current date as default value | Sets the current date as the default value when the form is loaded.                                                                                                                                                                                                    |
| Year range (from)                 | Define how many years back from the current year should be offered in the dropdown year selection.                                                                                                                                                                     |
| Year range (to)                   | Define how many years forth from the current year should be offered in the dropdown year selection.                                                                                                                                                                    |
| Minimum value                     | The minimum date the user has to enter. For example:<br>2000-02-01<br>In that case, 2000-01-31 would throw an error.                                                                                                                                                   |
| Exclude minimum value             | Activate the checkbox, when the minimum value date itself isn't an option. In case of the above example, 2000-02-01 would throw a validation error.<br><br>This option is relevant for the validation message the user receives when he doesn't meet the requirements! |
| Maximum value                     | The maximum date the user can enter. For example:<br>2020-12-31<br>In that case, 2021-01-01 would throw an error.                                                                                                                                                      |
| Exclude maximum value             | Activate the checkbox, when the maximum value date itself isn't an option. In case of the above example, 2020-12-31 would throw a validation error.<br><br>This option is relevant for the validation message the user receives when he doesn't meet the requirements! |

## 4.12 Drop-down List

The drop-down list offers the user a way to select a single choice option from a list. The options can be configured in the authoring section. The options can also be grouped into option groups.

The drop-down list offers the following additional configurations:

| Property | Description |
|----------|-------------|
|----------|-------------|

## Items

Define the options from which the user can choose. The pattern is as follows:

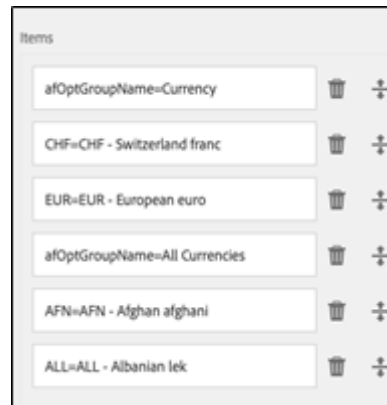
{key}={value}

For example: USD=United States Dollar

If you want to group options together, add an option group following this pattern:

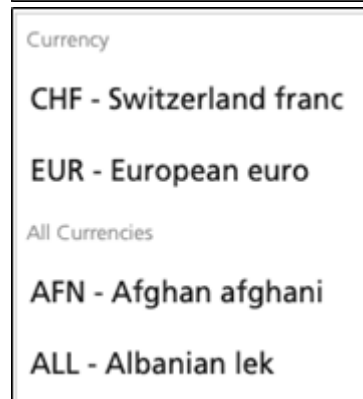
afOptGroupName={groupName}

For example when grouping currencies (left the configuration, right the result):



Items

|                               |   |   |
|-------------------------------|---|---|
| afOptGroupName=Currency       | ✖ | ✚ |
| CHF=CHF - Switzerland franc   | ✖ | ✚ |
| EUR=EUR - European euro       | ✖ | ✚ |
| afOptGroupName=All Currencies | ✖ | ✚ |
| AFN=AFN - Afghan afghani      | ✖ | ✚ |
| ALL=ALL - Albanian lek        | ✖ | ✚ |



Currency

**CHF - Switzerland franc**

**EUR - European euro**

All Currencies

**AFN - Afghan afghani**

**ALL - Albanian lek**

| Sorting                   | <p>Select how the items provided should be sorted and displayed to the user in the adaptive form. There are five possibilities:</p> <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>Default (Items Order)</td><td>The items will be displayed in the order they appear in the author section.</td></tr> <tr> <td>Key ASC</td><td>The items are sorted ascending according to the key (left hand side on the definition)</td></tr> <tr> <td>Key DESC</td><td>The items are sorted descending according to the key (left hand side on the definition)</td></tr> <tr> <td>Value ASC</td><td>The items are sorted ascending according to the value (right hand side on the definition). This works for the translated values as well.</td></tr> <tr> <td>Value DESC</td><td>The items are sorted descending according to the value (right hand side on the definition). This works for the translated values as well.</td></tr> </table> | Value | Description | Default (Items Order) | The items will be displayed in the order they appear in the author section. | Key ASC | The items are sorted ascending according to the key (left hand side on the definition) | Key DESC | The items are sorted descending according to the key (left hand side on the definition) | Value ASC | The items are sorted ascending according to the value (right hand side on the definition). This works for the translated values as well. | Value DESC | The items are sorted descending according to the value (right hand side on the definition). This works for the translated values as well. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----------------------|-----------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Value                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Default (Items Order)     | The items will be displayed in the order they appear in the author section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Key ASC                   | The items are sorted ascending according to the key (left hand side on the definition)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Key DESC                  | The items are sorted descending according to the key (left hand side on the definition)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Value ASC                 | The items are sorted ascending according to the value (right hand side on the definition). This works for the translated values as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Value DESC                | The items are sorted descending according to the value (right hand side on the definition). This works for the translated values as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Allows multiple selection | If checked, the user can select multiple options from the list.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Number of selections      | When multiple selection is activated, the number of selectable elements can be configured with this field. After the user selected this number, an information is shown that no more elements can be selected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Allow filtering           | Check if the user should be able to filter the dropdown or not (applicable for single select and multiselect).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |
| Items Load Path           | Define the path from where the items should be loaded. This could be a file in the CRX for example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |                       |                                                                             |         |                                                                                        |          |                                                                                         |           |                                                                                                                                          |            |                                                                                                                                           |

## 4.13 Radio Button

Radio buttons offer the user the possibility to select a single item from a list of possible options. These options will be visible to the user right away, other than the drop-down list, where the user first needs to open up the list. Selecting another option will uncheck the previously selected option.

The radio button offers the following additional configurations:

| Property       | Description                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Items          | Define the options from which the user can choose. The pattern is as follows:<br>{value}={text}<br>For example: USD=United States Dollar |
| Item Alignment | Define whether the radio button options should be aligned horizontally or vertically.                                                    |

## 4.14 Check Box

Check boxes are used when the user is allowed to select multiple options. Each option can be selected independent from the other options.

The check box offers the following additional configurations:

| Property       | Description                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Items          | Define the options from which the user can choose. The pattern is as follows:<br>{value}={text}<br>For example: USD=United States Dollar |
| Item Alignment | Define whether the radio button options should be aligned horizontally or vertically.                                                    |

#### 4.14.1 Checkbox indentation DoR

To indent the checkbox in the document of record, use a content panel and set the property “Number of Columns in Document of Record” to six. Put another panel in it with colspan one. Then you can place the checkbox after the panel with colspan five. Now the checkbox is indented.

### 4.15 Terms and conditions

The terms and conditions component consists of a title, text box and checkbox to create individual terms and conditions.

The terms and conditions component offers the following additional configurations:

| Property                         | Description                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Consent text                     | This text will be shown next to the checkbox the user has to enable in order to agree to the terms and conditions. |
| Content For Terms and Conditions | This text contains the terms and conditions and are shown before the consent text.                                 |

### 4.16 Image

The image component displays images in an adaptive form.

The image component offers the following additional configurations:

| Property       | Description                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------|
| Image          | Select the image which should be displayed. The image can be chosen from the given assets or directly be uploaded. |
| Alternate Text | Define a text which will be shown if the image cannot be displayed.                                                |

### 4.17 Line Separator

The Line separator component adds a horizontal line to separate two components.

The Line separator component offers the following additional configurations:

| Property  | Description                                                    |
|-----------|----------------------------------------------------------------|
| Thickness | Define the thickness of the separator line. Size is in Pixels. |

## 4.18 Content Panel

The Content Panel can be used to group the Components and to align them in a column Grid.

Also, components within a Content Panel will stick together in the document of records, as long as the Panel got a panel title. You have to keep in mind that panels with a panel title are not allowed to be higher than a document of Records page. If this is the case, the document of records will not be generated properly. To avoid that problem, split up the content of a panel into multiple panels.

## 4.19 Chart

The chart component can be used to visualize data in a well-organized graphic. It supports multiple chart types such as pie, column, line, etc. It can be used to visualize existing data or data which are entered by the user. The “Ajila Demo” contains a “Chart Demo” form to give an example on how the chart component works.

The chart component has the following configurations:

| Property                    | Description                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chart type                  | Select how the data should be visualized (as a pie, column, line, ... diagram).                                                                                                                                                                    |
| Repeating Row or Panel Name | Reference to the panel / row where the data will be entered. Please visit the “Chart Demo” for a working example.                                                                                                                                  |
| X Axis                      | <p><b>Title:</b> Will be shown as title for the x axis.</p> <p><b>Field:</b> The name of the component inside the repeating panel / row which holds the values.</p> <p><b>Function:</b> Select the function used to compute the x axis values.</p> |

|         |                                                                                                                                                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y Axis  | <p><b>Title:</b> Will be shown as title for the y axis.</p> <p><b>Field:</b> The name of the component inside the repeating panel / row which holds the values.</p> <p><b>Function:</b> Select the function used to compute the y axis values.</p> |
| Tooltip | Configure the message the user receives when hovering over a value. You can use \${x} and \${y} to access the values for including into the tooltip information.                                                                                   |

For further information, here the official documentation by Adobe: <https://helpx.adobe.com/in/experience-manager/6-3/forms/using/chart-component.html>

## 4.20 Accordion

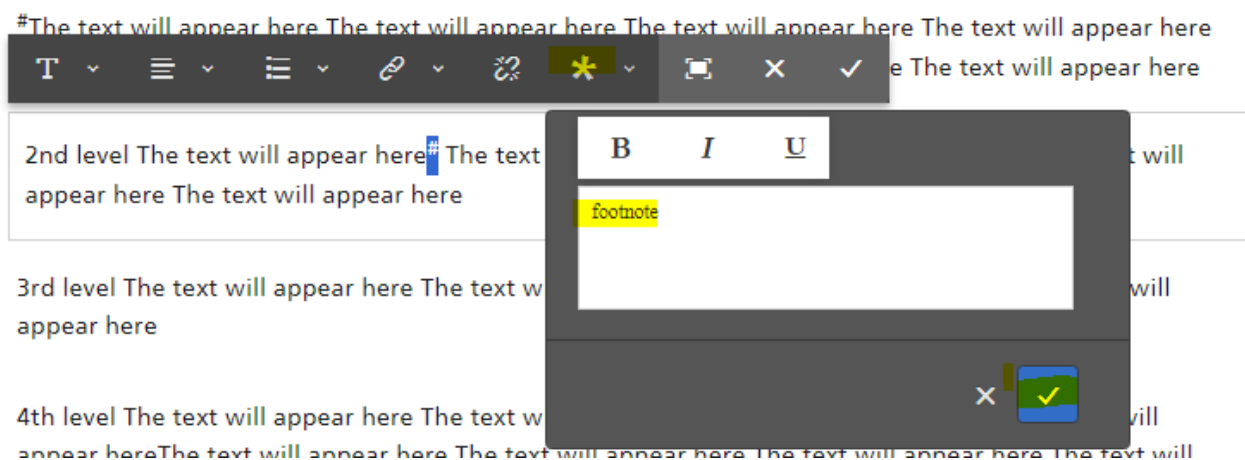
An accordion is panel with the layout “Accordion”. The user can expand an accordion and see the content when he wishes to.

### 4.20.1 Footnote Placeholder

Footnote Placeholder is a component introduced with Adobe's hotfix to handle all footnote related issues and allow designers to create multiple footnotes on a step and link them dynamically to same placeholder.

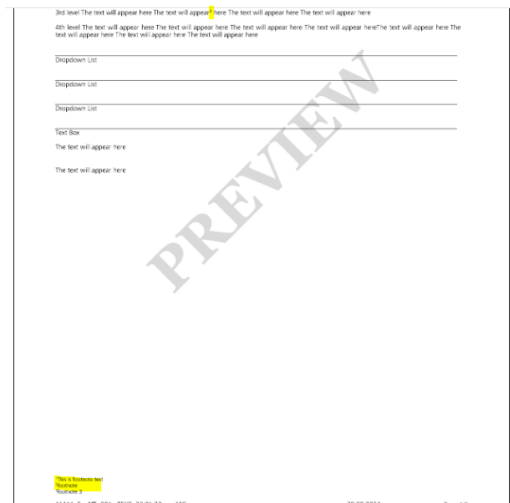
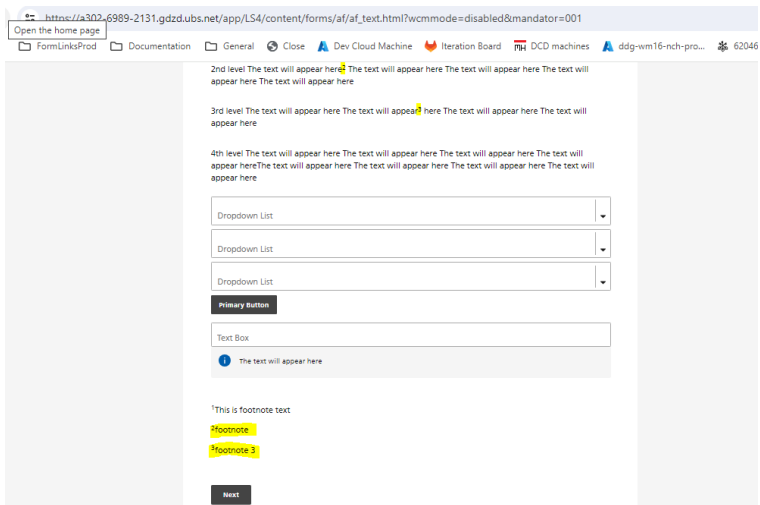
Steps to use:

- 1) Add a footnote to any static text component, add the footnote text using star icon, it will dynamically update numbering based on its placement





2) At the bottom on the step, add a component footnote placeholder. It will appear as hyperlink, taking user to footnote placeholder upon click.



## 4.20.2 Footnote

A normal accordion panel can be styled with the CSS class “ubsAccordionFootnote”. Place it on the panel with the “Accordion” layout and the background is styled similar to the information box.

## 4.20.3 DoR configurations

The descriptions of accordions are not shown in adaptive forms. Because of this the same value can be added for title and description. In the configuration for the document of record can be defined whether the title or the description should be shown. The title is shown if a line above, the description is without a line.

| Configuration | Result in Document of Record |
|---------------|------------------------------|
|---------------|------------------------------|

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| «Exclude title from Document of Record» is unchecked       | <b>5. Use of Third Parties</b><br><small>5.1 The parties agree that UBS may utilize the services of a Delegate to facilitate the submission of Relevant Data under this Agreement (including but not limited to any platform, system, interface or other technology developed by such Delegates for such purpose).<br/> 5.2 The provisions of Clauses 8 (Liability), 10 (Indemnity), and 11 (Confidentiality Waiver) apply in respect of such a Delegate in the same way as they apply to UBS.</small> |
| «Exclude description from Document of Record» is unchecked | <b>3. Unique Trade ID</b><br><small>UBS agrees to specify a Unique Trade ID (as required by item 8 of the Relevant Data) in respect of a Relevant Transaction.</small>                                                                                                                                                                                                                                                                                                                                 |

If description should be shown, in global Document of Record configuration the checkbox “Hide description of panels” must be unchecked.

## 4.21 Footer

The “Footer” component for adaptive forms is included in every template and the footer text doesn’t need to be updated. The footer text is dynamically added when an adaptive form is loaded. The text is part of the following dictionary, which is delivered with the UBS Apps Bundle:

*/apps/ajila-forms-customers/ajila-forms-ubs/i18n/dictionary*

The key is: *ajila-forms-ubs-footer-text*

The text entered here per language is loaded into the adaptive form. When the text is changed in the dictionary, it will have an impact on all adaptive forms. When the text needs to be updated, it has to be properly escaped.

## 4.22 Generic Rule Editor Functions

### 4.22.1 Mandatory

The following function should be used when a component needs to be set to mandatory or not mandatory during runtime (Rule Editor). It takes into account the active or inactive state of the incomplete form mode. In order to change the mandatory property of a component it is necessary to add the following function in the rule editor:

```
window.forms.ubs.components.setMandatory(this, true);
```

The first parameter for the function is the component context, it can be the current component (this) or a different component. The second parameter defines whether the given component is set to mandatory or not mandatory.

## 4.23 Forced line break

If there is a requirement to use the forced line break between the text in Form title, use this "&#x2029;" Html code under the Master page configurations in the DoR configuration for the DoR and shift + enter in AF for the AF.

Example forms: BAMA, BAMB

Master Page Properties

txtBankingRelationship

ShowBankingRelationship

Logo Image (width: 28.955mm, height: 10.475mm)

Use Template Logo Image

FormType

Use Custom

Header Text

Use Template Header Text

Form Title

Use Custom

Equities/Fixed Income Direct Access Clients (DAC)[2020](#)Service Activation Form – DAC Night

## 5 5) Configuration components

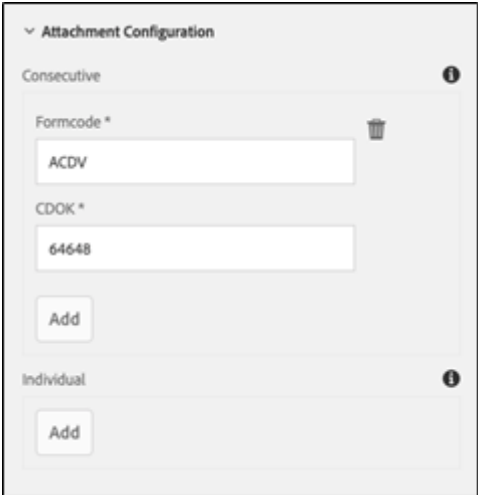
Configuration components are used for enhanced configuration of an adaptive form. They won't be shown in the adaptive form nor in the document of record.

### 5.1 Document of Record Configuration

This component is part of the “UBS” template. It's placed on the second step. Keep this component on the last step. If no additional DoR configuration is needed, the component can be deleted and added later on.

The Document of Record Configuration component offers the following configurations:

| Property                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Custom XDP Configuration           | <p>You can enable the external retrieval of the custom XDP by activating the checkbox “Enable external custom XDP.”</p> <p>In addition, a reference can be configured by which the XDP is retrieved (CDOK or formcode). The XDP will be retrieved the same way the external documents are retrieved.</p>                                                                                                                                     |
| Preliminary Document Configuration | <p>You can add multiple preliminary documents here with the formcode and CDOK reference. The preliminary documents will be retrieved from the UBS Forms API.</p> <p><b>Individual:</b> Those documents will not be included in the DoR PDF. They are added to the ZIP with a own UUID and own storage in CLP.</p> <p><b>Consecutive:</b> Preliminary documents under “Consecutive” will be included in the document of record numbering.</p> |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attachment Configuration | <p>You can add multiple attachments which will be merged into the document of record. This can be used for further terms and conditions, etc. They will be added after the document of record. Add the formcode and the CDOK of the attachment and the attachment will be retrieved from the UBS Forms API.</p> <p><b>Individual:</b> Those documents will not be included in the DoR PDF. They are added to the ZIP with a own UUID and own storage in CLP.</p> <p><b>Consecutive:</b> Attachments under “Consecutive” will be included in the document of record numbering.</p>  <p>The screenshot shows a UI titled 'Attachment Configuration'. It has two sections: 'Consecutive' and 'Individual'. The 'Consecutive' section has a 'Formcode *' field with the value 'ACDV', a 'CDOK *' field with the value '64648', and an 'Add' button. There is a trash icon next to the 'Formcode' field. The 'Individual' section has an 'Add' button. Both sections have an information icon (i) in the top right corner.</p> |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 5.1.1 Attachment exclusion

The attachment references in the configuration are fix. In order to be able to dynamically exclude an attachment or preliminary document, the following two functions were implemented:

#### Exclusions

```

window.forms.ubs.excludeAttachments(["ABCD", "EFGH"]);

window.forms.ubs.excludeAttachments("ABCD");

```

With this code, the given attachment references will not be retrieved and are therefore excluded from the document of record. You can either add an array of references or a single reference.

#### Inclusions

```

window.forms.ubs.includeAttachments(["ABCD", "EFGH"]);

window.forms.ubs.includeAttachments("ABCD");

```

Use the include function to remove references from the list of exclusions. **Important:** Including an attachment reference that is not in the configuration will not work.

If you want to check the excluded attachments, run the following code snippet in the web console:

```
guideBridge.getData({success: function(data) {console.log(data.data)}}})
```

## 5.2 Banking Relationship and form ID

To set a value of the banking relationship and the form ID (footer left and footer center), create text boxes in the adaptive form. The logic for the values of those text boxes can be written in the rule editor of each form (for the form ID). The text boxes should be excluded of the document of records since the values will be displayed in the header and footer. The textboxes need to have the following names:

- txtBankingRelationship
- txtFormId
- txtMiddleFormId

## 5.3 Address Multiline Text

To set the value of the address in the address Template, create a multiline textbox in the adaptive form and set the name of the text box to “txtAddressHeader”.

## 6 6) Component combinations

The components described in the previous chapter can be combined to achieve more sophisticated component structures. The following chapters describe those component combinations and their goal.

### 6.1 Single option with accordion detail

This section covers how to create a panel, where only one checkbox can be selected and where each option has an accordion panel with detail information. The example will be the selection of a risk tolerance option. The result will look like the picture below.

**Risk tolerance asset options**

**Important**  
The information you provide in Section A1 and Section A2 should cover your assets and liabilities with us and with other banks. This information will be applied to all your Banking Relationships to all portfolios under this Investor Profile Questionnaire.

What risk tolerance best describes your investment objective?

☒ Asset preservation - Very low losses ?

**Details** ^

|                                    |             |                                                |                     |
|------------------------------------|-------------|------------------------------------------------|---------------------|
| Expected annual return             | <b>3%</b>   | Longest period of a historical drawdown        | <b>2 years</b>      |
| Maximum historical drawdown return | <b>10%</b>  | Your final risk tolerance investment objective | <b>A - Very Low</b> |
| Expected annual volatility         | <b>3.5%</b> |                                                |                     |

☐ Asset preservation - Low losses ?

**Details** v

☐ Moderate asset appreciation - Moderate losses ?

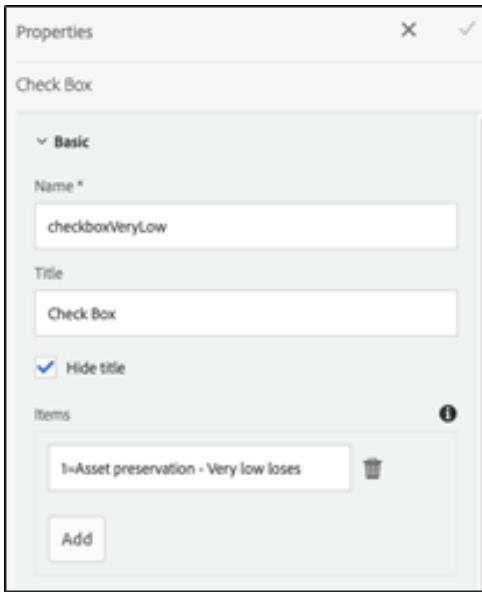
**Details** v

#### 6.1.1 Creating the basic structure

|   |                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|
| 1 | Create a new Adaptive Form and select the UBS template. Your form will initially look like this: |
| 2 | Delete Step 2 as we don't need it in this example.                                               |

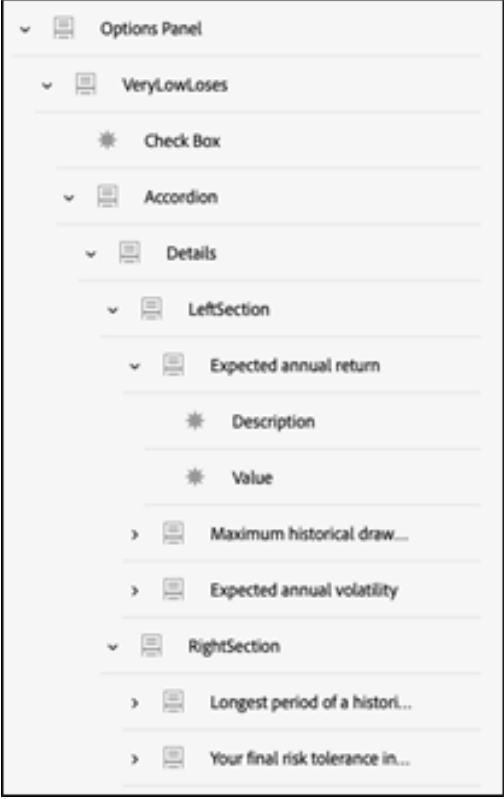
|   |                                                                                                                                                                                                                                                                                                                                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Add a Child Panel to the Root Panel and give it an appropriate name (for example “Options Panel”).                                                                                                                                                                                                                                           |
| 4 | <p>Open the configuration of the “Options Panel” and add the following CSS classes:</p> <ul style="list-style-type: none"> <li>• <code>ubsAccordionPanel</code>: This adds the gray background to the Panel</li> <li>• <code>ubsAccordionCollapsed</code>: To have all the accordions inside the panel collapsed when initialized</li> </ul> |

### 6.1.2 Add a checkbox option with accordion detail information

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Add a Child Panel to the “Options Panel” and name it (for example “VeryLowLoses”).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2 | <p>Add a Check Box to the “VeryLowLoses” Panel and perform the following configurations:</p> <ul style="list-style-type: none"> <li>• Add a name (for example “checkboxVeryLow”). <b>This naming will be important for the rules!</b></li> <li>• Select the “Hide title” checkbox so no title is shown.</li> <li>• Add an option with a proper description.</li> <li>• Add the following CSS class: <code>validationErrorNotVisible</code><br/>This class doesn’t show the individual validation error.</li> </ul>  |
| 3 | Add a Child Panel to the “VeryLowLoses” Panel and name it (for example “Accordion”).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4 | Open the configuration of the “Accordion” Panel and switch the Layout to “Accordion”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | <p>Add a Child Panel to the “Accordion” Panel and name it (for example “Details”). The title of this Panel will be shown as Accordion Title.</p> <p>We need this extra panel to enable multiple columns</p>                                                                                                                                                                                                                                                                                                                           |
| 6  | <p>Open the configuration of the “Details” Panel and perform the following configurations:</p> <ul style="list-style-type: none"> <li>• Colspan: Set value “2”</li> <li>• Number of Columns: Set value “2”</li> </ul>                                                                                                                                                                                                                                                                                                                 |
| 7  | <p>Add two Child Panels to the “Details” Panel.</p> <ol style="list-style-type: none"> <li>1. Add a Child Panel and name it “LeftSection”</li> <li>2. Add a Child Panel and name it “RightSection”.</li> </ol>                                                                                                                                                                                                                                                                                                                        |
| 8  | <p>For both Panels (LeftSection and RightSection), perform the following configuration:</p> <ul style="list-style-type: none"> <li>• Colspan: Set value “1”</li> <li>• Number of columns: Set value “1”</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| 9  | <p>Add a Child Panel for each row you want to add to the “LeftSection” and “RightSection” and configure it as follows:</p> <ul style="list-style-type: none"> <li>• Colspan: Set value “1”</li> <li>• Number of Columns: Set value “3”</li> <li>• CSS Class: ubsAccordionPanelRow (This will add the underline to the row)</li> </ul>                                                                                                                                                                                                 |
| 10 | <p>Add two “Static Text” Components to your row panel:</p> <ol style="list-style-type: none"> <li>1. A “Static Text” with the row description (for example “Expected annual return”). Configure it as follows: <ul style="list-style-type: none"> <li>- Colspan: Set value “2”</li> <li>- CSS Class: ubsPlaceholderFontColor (To add the gray font color)</li> </ul> </li> <li>2. A “Static Text” with the row value. Configure it as follows: <ul style="list-style-type: none"> <li>- Colspan: Set value “1”</li> </ul> </li> </ol> |
| 11 | <p>Repeat the previous two steps for each row you want to add.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|    |                                                                                                                                                          |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | <p>Your form content should now look like this (with one option):</p>  |
| 13 | Repeat the steps above for every option you need.                                                                                                        |

### 6.1.3 Adding the error text

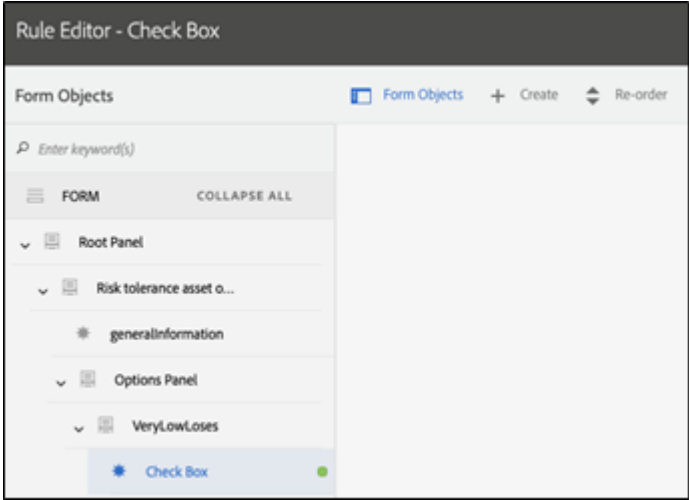
After adding all options, we need an error text component. This error text will be displayed, when no option is selected and the validation fails.

|   |                                                                                                                                                                                                                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Add a “Error Text” component to the “Options Panel” Panel (as the last component).                                                                                                                                                                                                                                                       |
| 2 | <p>Open the configuration and perform the following steps:</p> <ul style="list-style-type: none"> <li>• Add a name (for example “riskToleranceErrorBox”) → <b>This naming will be important for the rules!</b></li> <li>• Select the “Hide Object” option. The Error Text should only be displayed when the validation fails.</li> </ul> |
| 3 | Add the error message to the Error Text component.                                                                                                                                                                                                                                                                                       |

## 6.1.4 Configure the rules

We need to configure rules in the Rule Editor for the following functionalities:

- To ensure that the user can only select one option
- To validate that one option is selected

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>Open the Rule Editor for the “Very Low Loses” Check Box.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2 | <p>Create a new rule by pressing the “+ Create” Button.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3 | <p>Switch to the Code Editor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4 | <p>Select “Value Commit” as your Event and add the following code:</p> <pre data-bbox="470 1442 1152 1532">var otherCheckboxes = [checkbox1, checkbox2]; window.forms.ubs.handleSingleOption(<b>this</b>.value, otherCheckboxes);</pre> <p><b>Line 1</b> declares a variable which needs to have all other relevant checkboxes in it. Please add every other checkbox name to this array declaration. The Rule Editor will support you with an autocomplete function. For example:</p> <pre data-bbox="470 1729 1267 1787">var otherCheckboxes = [checkboxVeryLow, checkboxLow, checkboxModerate, checkboxMedium, checkboxAboveAverage]</pre> <p><b>Line 2</b> calls the provided function “handleSingleOption”. The first parameter is the value the user has selected. The second is the array list. For every element in this array, a checkbox in the DOM is being searched and disabled.</p> |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Create another rule on the checkbox by pressing the “+ Create” Button.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6 | <p>Select “Validate” as your Event and add the following code:</p> <pre>var checkboxList = [checkboxbox1, checkbox2]; window.forms.ubs.validateSingleOption(checkboxboxList, riskToleranceErrorBox);</pre> <p><b>Line 1</b> declares a variable which needs to have all relevant checkbox names in it (including the one where the rule is placed on). Please add every checkbox name to this array declaration. The Rule Editor will support you with an autocomplete function (Hint: The rule editor won't suggest the checkbox you're currently editing). For example:</p> <pre>var checkboxList = [checkboxboxVeryLow, checkboxLow, checkboxboxModerate, checkboxMedium, checkboxAboveAverage, checkboxboxHigh];</pre> <p><b>Line 2</b> calls the provided function “validateSingleOption”. The first parameter is the array with all checkbox names. The logic validates, if one option is selected. The second parameter is the name of the Error Text. Please provide this exact name as second parameter. If the validation fails, this error text component will be shown.</p> |
| 7 | <p>Add these rules to all other relevant checkboxes. Be aware:</p> <ul style="list-style-type: none"> <li>• The handleSingleOption Rule needs to be adjusted on every checkbox. Because all other checkboxes need to be in the array list.</li> <li>• The validateSingleOption Rule is the same on every checkbox. Therefore, just copy and paste it.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 6.2 Dynamic repeating panels

Repeating Panels can be used everywhere the user needs to dynamically add or remove elements. Such elements can contain several components. For example a contracting partner declaration. **Note:** This approach has a limitation. It's not possible for the elements to have a count (Contracting partner 1, Contracting partner 2, ...). There is a solution for adaptive forms, but the DoR cannot display it. A request will be created at Adobe.

**Establishing of the controlling**

---

**Contracting partner declaration**

☐ The person(s) listed below is/are holding 25% or more of the contracting partner's shares (capital shares on voting rights)

☐ If there're no capital shares or voting rights of 25% or more, that the following person(s) listed below is/are controlling the contracting partner in other ways

☐ In case no person(s) exist(s) who exercise(s) control over the contracting partner in a different capacity, the contracting partner hereby declares that listed below is/are the managing director(s)

---

**Contracting Partner 1**

Company

Last name First name

Actual address of domicile / registered office Country

Date of birth

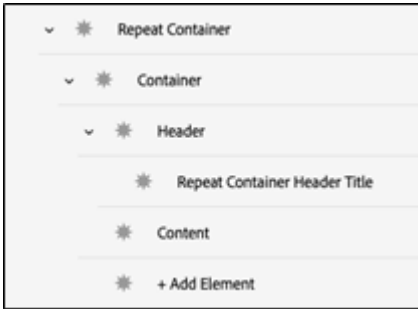
+ Add Element

## 6.2.1 Adding the basic structure

|   |                                                                                                                                                                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Add the <i>Repeat Container Panel</i> Component to your Adaptive Form.                                                                                                                                                                                                                         |
| 2 | Add a <i>Content Panel</i> Component with Title "Container" into your <i>Repeat Container Panel</i> .                                                                                                                                                                                          |
| 3 | <p>Open the properties of the "Container" <i>Content Panel</i> and define the "Minimum" and "Maximum" number of repeats in the "Repeat Settings" section.</p> <p>Please provide both values to make it work. If an infinite number of repeats is needed, set the "Maximum" property to -1.</p> |
| 4 | <p>Add the following components into your "Container" <i>Content Panel</i>:</p> <ol style="list-style-type: none"> <li><i>Repeat Container Header Panel</i> with the Title "Header"</li> <li><i>Content Panel</i> with the Title "Content"</li> </ol>                                          |
| 5 | Add a Tertiary Button to the "Repeat Container" <i>Repeat Container Panel</i> and give it an appropriate Title (in this case "+ Add Element").                                                                                                                                                 |

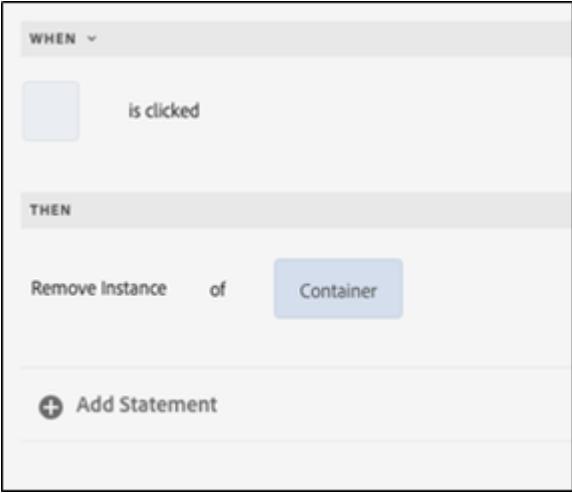
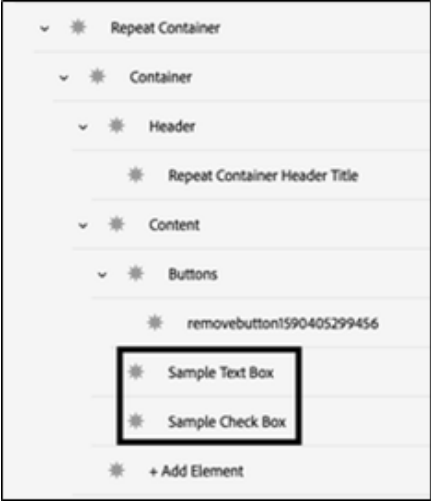
|   |                                                                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | <p>The structure of your <i>Repeat Container Panel</i> should now look like this:</p>  |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 6.2.2 Header section

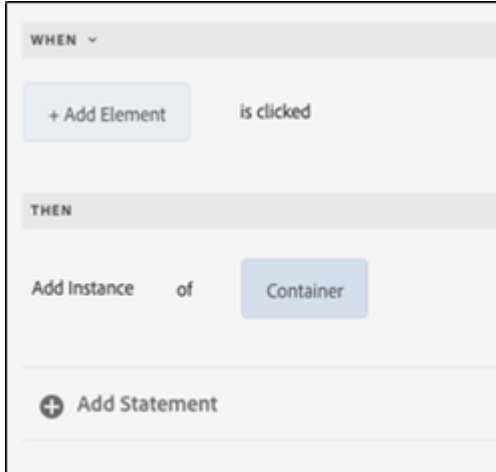
|   |                                                                                                                                                                           |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Add a <i>Repeat Container Header Title</i> to the “Header” Section.                                                                                                       |
| 2 | <p>The structure of your <i>Repeat Container Panel</i> should now look like this:</p>  |

### 6.2.3 Content section

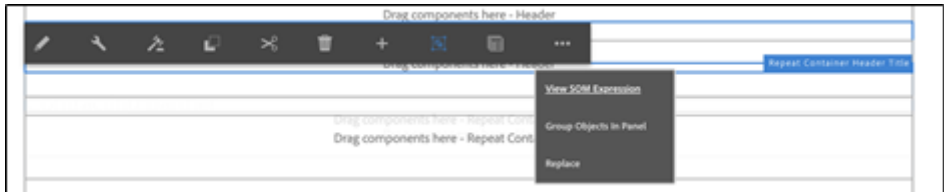
|   |                                                                                               |
|---|-----------------------------------------------------------------------------------------------|
| 1 | Add a <i>Repeat Container Button Panel</i> to the “Content” Section with the Title “Buttons”. |
| 2 | Add a <i>Remove Button</i> to the “Buttons” Section.                                          |
| 3 | Click on the button and open the rule editor by clicking “Rule Editor”.                       |

|   |                                                                                                                                                                                                                                                                                                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>In the “Rule Editor”, add a rule by clicking “+ Create” and configure it as follows:</p>  <p>Click “Done” and close the “Rule editor” by clicking “Close”.</p>                                               |
| 5 | <p>Add all necessary components to your content section (Text Boxes, Check Boxes, etc.)</p>                                                                                                                                                                                                       |
| 6 | <p>The structure of your <i>Repeat Container Panel</i> should now look like this:</p>  <p>The marked components inside the “Content” Section indicate the Form Components which are needed for each panel.</p> |

## 6.2.4 Add Button Configuration

|   |                                                                                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Click on the “+ Add Element” Button and open the rule editor by clicking “Rule Editor”.                                                                                                                                                              |
| 2 | <p>In the “Rule Editor”, add a rule by clicking “+ Create” and configure it as follows:</p>  <p>Click “Done” and close the “Rule editor” by clicking “Close”.</p> |

## 6.2.5 Add counter on repeatable title

|   |                                                                                                                                                                                                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>Focus on your “Repeat Container Header Titel” Component and select “View SOM Expression” by clicking the 3 dots</p>  <p>Copy the SOM Expression. It should look like this:</p>  |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|   |                                                                                         |
|---|-----------------------------------------------------------------------------------------|
| 2 | Click on the “+ Add Element” Button and open the rule editor by clicking “Rule Editor”. |
|---|-----------------------------------------------------------------------------------------|

3

In the “Rule Editor”, add two rules by clicking “Create” and configure it as follows:

**Rule 1**

Event: Click

Code: See Code Snippet Below

**Rule 2**

Event: Initialize

Code: See Code Snippet Below

**Code Snippet**

```
var title = guideBridge.resolveNode('guide[0].guide1[0].guideRootPanel[0].contracting-partner-declaration[0].repeatcontainer1590476712102[0].partnerContainer[0].header[0].repeatcontainerheader1590476944274[0]').nonLocalizedTitle;

for(var i = 0; i < partnerContainer.instanceManager.instances.length; i++) {
    var num = i + 1;

    partnerContainer.instanceManager.instances[i].header.headerTitle.title = title + " " + num;
}
```

First replace the String in Line 1 with the copied SOM expression. Then replace the variables partnerContainer, header & headerTitle with the components binding name. For example the partnerContainer:

The screenshot shows the UBS Forms Rule Editor interface. The top section displays a tree view of components: 'Repeat Container' (expanded), 'Container' (expanded), 'Header' (expanded), and 'Contracting partner'. The bottom section shows the 'Basic' configuration panel with two input fields: 'Name \*' containing 'partnerContainer' and 'Title' containing 'Container'.

|   |                                                                                                                                                                |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Add the same code snippet to the Remove Button. For the Remove Button, we only need one rule:</p> <p>Event: Click</p> <p>Code: Code Snippet from step 3</p> |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Options

With the code snippet above, the titles are going to have a numbering starting with 1. If you don't want a number on the first instance, use this code snippet instead:

```
var title = guideBridge.resolveNode('guide[0].guide1[0].guideRootPanel[0].contracting-partner-declaration[0].repeatcontainer1590476712102[0].partnerContainer[0].header[0].headerTitle[0]').nonLocalizedTitle;

for(var i = 0; i < partnerContainer.instanceManager.instances.length; i++){
  if (i === 0) {
    partnerContainer.instanceManager.instances[i].header.headerTitle.title = title;
  } else {
    var num = i + 1;
    partnerContainer.instanceManager.instances[i].header.headerTitle.title = title + "
" + num;
  }
}
```

### Scroll to Top on new instance of repeatable panel

We have a new method called ScrollToTop which when used along with dynamic repeatable panel will allow the forms to scroll to top. Parameter needed is name of panel.

**PanelName.enabled=true;**

**window<sup>1</sup>.forms.ubs.components.scrollToTop(PanelName);**

```
PN_Receipient_RP.enabled=true;
window.forms.ubs.components.scrollToTop(PN_Receipient_RP);
```

| <input type="checkbox"/> | Title                        | Content                                                                                                |
|--------------------------|------------------------------|--------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Add recipient - Click        | WHEN <a href="#">Add recipient</a> is clicked THEN Add Instance of <a href="#">PN_Receipient_RP</a>    |
| <input type="checkbox"/> | <b>Add recipient - Click</b> | PN_Receipient_RP.enabled=true;<br>window.com.ajila.forms.ubs.components.scrollToTop(PN_Receipient_RP); |

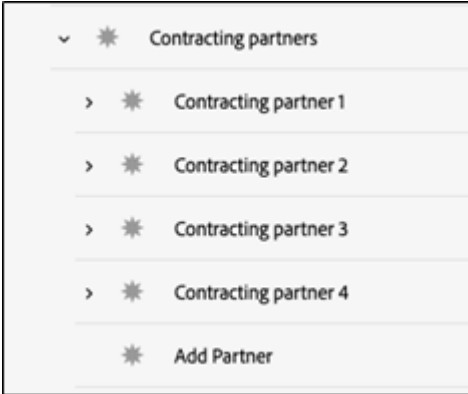
<sup>1</sup> <http://window.com>

## 6.3 Static repeating panels

The static repeating panels were introduced to enable the numeration for repeating panels. The disadvantage here is, that the maximum number of entries needs to be preconfigured and dynamically be shown and hidden. But as discussed with business, most cases only allow four entries. For the user, the approach looks the same as the dynamic repeating panels. The following section describes how to achieve a static repeating panel. The “Contracting partner” demo form is used as a guide.

### 6.3.1 Basic structure

|   |                                                                                                                                                                                                                                                                                                                                                                               |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>Add a “Content Panel” component and name it. In the example:</p> <p><b>Name:</b> contractingPartners</p> <p><b>Title:</b> Contracting partners</p> <p><b>CSS Class:</b> ubsRepeatPanel</p>                                                                                                                                                                                 |
| 2 | <p>Add a “Content Panel” component inside your panel. This will represent the first element. Naming in the example:</p> <p><b>Name:</b> contractingPartner1</p> <p><b>Title:</b> Contracting partner 1</p>                                                                                                                                                                    |
| 3 | <p>Add another “Content Panel” inside the first element panel. This will be the header of the first element. Naming in the example:</p> <p><b>Name:</b> header</p> <p><b>Title:</b> Header</p>                                                                                                                                                                                |
| 4 | <p>Add the following two components to the header panel:</p> <ol style="list-style-type: none"> <li>1. Static title (Name for the first element. Example: Contracting partner 1)</li> <li>2. Removebutton</li> </ol> <p>Align those two components on one line with the layout editor. Also check the alignment for the mobile view (Select iPhone 5 in the layout mode).</p> |
| 5 | <p>Add all components needed to the “Contracting partner 1” panel.</p> <p>Note: The naming of the components need to be unique. Suggestion therefore is to add the element number to the name. For example: partner1Company for the first element and partner2Company in the second element and so on.</p>                                                                    |

|   |                                                                                                                                                                                                                                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | <p>Your static repeating panel should now look like this:</p>                                                                                                                                                                                            |
| 7 | <p>You can copy paste the first element to achieve the number of desired elements. After copying an element, adjust the numbers on the components, the static title and the content panel. Afterwards you should have a structure similar to this:</p>  |
| 8 | <p>Add a tertiary button the main panel and name it as you wish. For example: Add partner. The final structure will look similar to this:</p>                                                                                                          |
| 9 | <p>Optional: If you want separators between the elements, add a “Separator” on all elements except the first one. Place it inside the “Header” panel.</p>                                                                                                                                                                                 |

## 6.3.2 Rules

In order to show and hide the elements dynamically, the following rules need to be applied:

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>Add the following rule on the main panel (“Contracting partners” in the example).</p> <p><b>Event:</b> Initialize</p> <p><b>Purpose:</b> The idea is to hide and exclude every element except the first.</p> <pre>contractingPartner2.visible = false;<br/>contractingPartner2.dorExclusion = true;<br/>contractingPartner3.visible = false;<br/>contractingPartner3.dorExclusion = true;<br/>contractingPartner4.visible = false;<br/>contractingPartner4.dorExclusion = true;</pre> |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2

Add the following two rules to the Tertiary Button which should add another instance.

**Event:** Click

**Purpose:** Displays the next object when clicking add and handles the removebutton so that it's only visible on the latest element.

```
if (contractingPartner1.visible === false) {
    contractingPartner1.visible = true;
    contractingPartner1.dorExclusion = false;
} else if (contractingPartner2.visible === false) {
    contractingPartner2.visible = true;
    contractingPartner2.dorExclusion = false;
    contractingPartner1.header.removebuttonPartner1.visible = false;
} else if (contractingPartner3.visible === false) {
    contractingPartner3.visible = true;
    contractingPartner3.dorExclusion = false;
    contractingPartner2.header.removebuttonPartner2.visible = false;
} else if (contractingPartner4.visible === false) {
    contractingPartner4.visible = true;
    contractingPartner4.dorExclusion = false;
    contractingPartner3.header.removebuttonPartner3.visible = false;
}
```

**Event:** Click

**Purpose:** Hide the tertiary button when the last possible element is shown

```
if (contractingPartners.contractingPartner4.visible === true) {
    this.visible = false;
} else {
    this.visible = true;
}
```

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <pre>}</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3 | <p>Add the following rule on each of the remove buttons. Note: Remove button on the first element is slightly different.</p> <p><b>Event:</b> Click</p> <p><b>Purpose:</b> Hide the element and reset its data. Also show the remove button on the previous element:</p> <p><b>Element 1:</b></p> <pre>contractingPartner1.visible = false; contractingPartner1.dorExclusion = true; contractingPartner1.resetData();</pre> <p><b>Element 2-x:</b></p> <pre>contractingPartner2.visible = false; contractingPartner2.dorExclusion = true; contractingPartner2.resetData(); contractingPartner1.header.removebuttonPartner1.visible = true;</pre> |

### 6.3.3 Example

In the “Ajila Demos” folder there is a form called: Contracting partner. The configuration can be looked at there in a working example.

## 6.4 Performance Optimizations

When you are adding 10 or more instances of repeating panels you may encounter performance issues. To get rid of these issues, extract the content of the repeating panel into a fragment.

Steps for creating fragment of content of a repeating panel:



1. Group the elements of the panel (use the "Group" icon to activate the function and select all the elements)

2. Group selected objects in panel by clicking the icon "Group Objects in Panel"

3. Save the created panel as a fragment with the icon "Save as Fragment"

#### 4. Enter all the necessary details for the fragment

 Contracting Partner

---

Company

Last name

First name

Actual address of domicile / registered office

Date of birth

Country

## 7 7) PDF generation

- [UBS\\_Blank\\_BankingRelationship\\_DoR.xdp](#) (see page 82)

When the form is submitted, the “UBS Submit” action will be executed. Part of this submit action is the generation of one or multiple PDFs for the user to download. Every PDF produced will be in the PDF/A format. The compliance level of the PDF/A (1B, 2B or 3B) can be configured as part of the [OSGI configuration](#) (see page 140). The following chapters describe the different types of PDF generation an author can configure.

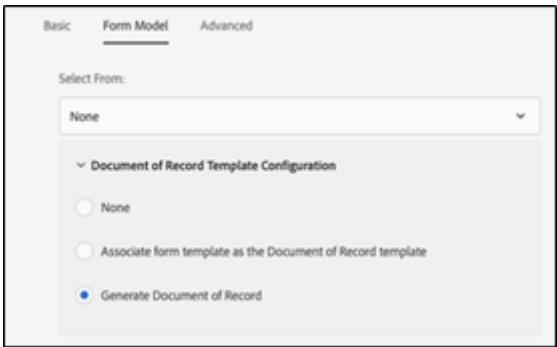
The naming convention of the PDF's and ZIP is documented in a dedicated subpage.

### 7.1 Document of Record

A Document of Record (short: DoR) is a PDF document rendered according to the definitions of the adaptive form. Every adaptive form with this configuration has the same template assigned and each component will be rendered into it. This is a generic approach for forms, where the PDF doesn't need to be pixel perfect.

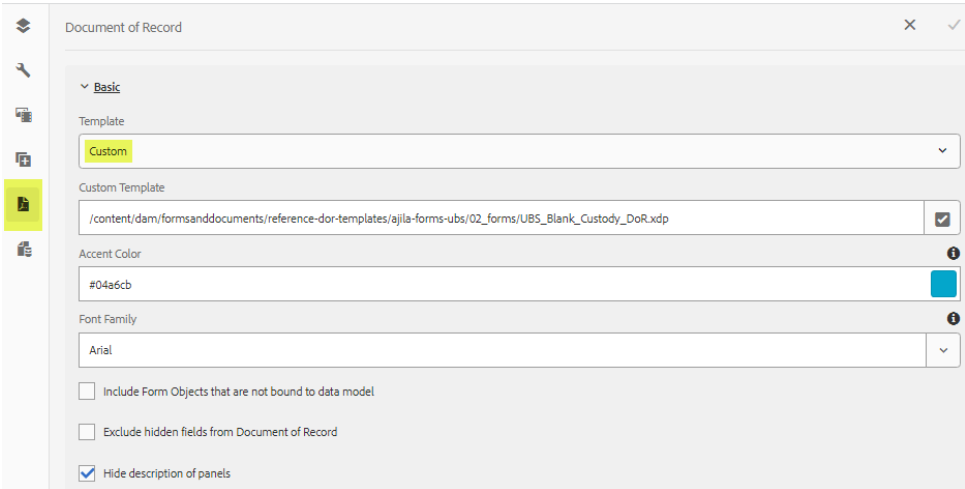
#### 7.1.1 Enable the DoR in the forms config

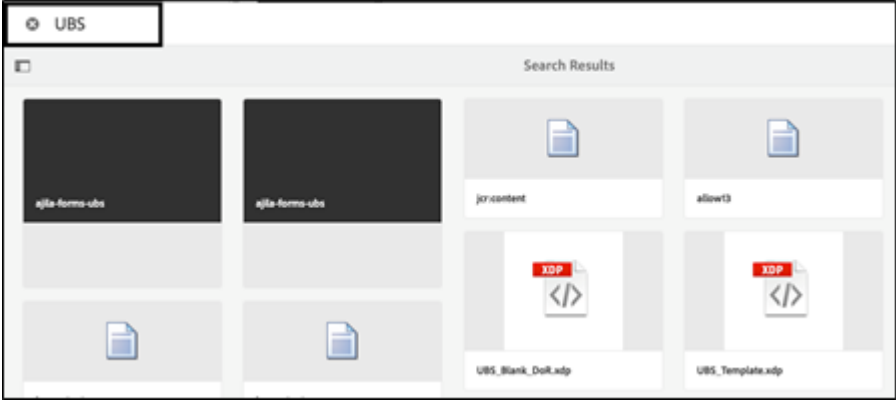
|   |                                           |
|---|-------------------------------------------|
| 1 | Open the properties of the adaptive form. |
| 2 | <p>Select the tab “Form Model”:</p>       |

|   |                                                                                                                                                  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <p>Select the radio button “Generate Document of Record”:</p>  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------|

You can configure the generic DoR during the creation of the adaptive form or afterwards!

### 7.1.2 Select the DoR template

|   |                                                                                                                                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Open the adaptive form in the “Edit” mode.                                                                                                   |
| 2 | <p>Click the “Document of Record” icon in the menu:</p>  |
| 3 | Switch the Template option to “Custom”.                                                                                                      |

|   |                                                                                                                                                                                                                                                                                                                                 |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Select the Custom Template, for example “UBS_Blank_DoR.xdp”.</p> <p><b>Info:</b> It can happen that the wizard does not show the XDP. In that case, search for your XDP with the input field on the top left-hand side of the wizard:</p>  |
| 5 | <p><b>Important:</b> Deselect the option “For Check Box and Radio Button components, show only the selected value(s).” and the bottom of the DoR configuration. Otherwise the checkboxes and radiobuttons are not displayed properly in the DoR or not displayed at all.</p>                                                    |

## 7.2 DoR templates

The following section covers the existing DoR templates available in UBS. The main difference is in the header section.

### 7.2.1 UBS\_Blank\_DoR.xdp

|                                                                                     |   |                      |
|-------------------------------------------------------------------------------------|---|----------------------|
|  | K | Banking relationship |
| FORM TITLE                                                                          |   |                      |

| Field    | Example Value | Description                                         |
|----------|---------------|-----------------------------------------------------|
| Logo     | -             | Fixed UBS logo                                      |
| FormType | K             | Can be configured in the DoR configuration per form |

|                         |                      |                                                                                                                 |
|-------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| senderAddressTitle      | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtBankingRelationship  | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime. |
| showBankingRelationship |                      | Checkbox to show / hide the txtBankingRelationship. If selected the text field will be displayed in the DoR.    |

### 7.2.2 UBS\_Blank\_Account\_Suffix\_DoR.xdp

|                                                                                                        |          |                                                            |
|--------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | Banking relationship<br><hr/> Custody account no.<br><hr/> |
|--------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------|

| Field                  | Example Value        | Description                                                                                                                                   |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                   | -                    | Fixed UBS logo                                                                                                                                |
| FormType               | K                    | Can be configured in the DoR configuration per form                                                                                           |
| senderAddressTitle     | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                                                 |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number). |
| CustodyAccountNo       | Custody account no.  | This can be configured in the DoR config per form.                                                                                            |
| Company                | UBS Switzerland AG   | This can be configured in the DoR config per form.                                                                                            |

|                  |                                    |                                                                                                                                                  |
|------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| P.O. Box         | P.O. Box                           | This can be configured in the DoR config per form.                                                                                               |
| txtPoBox         |                                    | The field “txtPoBox” will be populated here. This field can be dynamically filled during runtime.                                                |
| AccountSuffix    | Numbered account supplement/Suffix | This can be configured in the DoR config per form.                                                                                               |
| txtAccountSuffix |                                    | The field “txtAccountSuffix” will be populated here. This field can be dynamically filled during runtime. If it is left empty it will be hidden. |

### 7.2.3 UBS\_Blank\_Address\_DoR.xdp

| Field                    | Example Value        | Description                                                                                                                                  |
|--------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                     | -                    | Fixed UBS logo                                                                                                                               |
| FormType                 | K                    | Can be configured in the DoR configuration per form                                                                                          |
| senderAddressTitle       | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtBankingRelationship   | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number) |
| senderAddressCompanyName | UBS Switzerland AG   | Fixed value                                                                                                                                  |

|            |   |                                                                                                                                                                                                                                                                             |
|------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| txtAddress | - | <p>The address printed below the UBS Switzerland AG can be either be preconfigured in the DoR config or added at runtime by adding the “txtAddressHeader” field.</p> <p>It can have multiple lines. The page break can be enforced by adding the “&amp;#x2029;” symbol.</p> |
|------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.2.4 UBS\_Blank\_Address\_Fax\_DoR.xdp

|                                                                                                        |          |                                                  |
|--------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | Banking relationship<br><hr/> UBS Switzerland AG |
|--------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------|

| Field                    | Example Value        | Description                                                                                                                                  |
|--------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                     | -                    | Fixed UBS logo                                                                                                                               |
| FormType                 | K                    | Can be configured in the DoR configuration per form                                                                                          |
| senderAddressTitle       | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtBankingRelationship   | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number) |
| senderAddressCompanyName | UBS Switzerland AG   | Fixed value                                                                                                                                  |
| txtPostalCodeCity        |                      | The field “txtPostalCodeCity” will be populated here. This field can be dynamically filled during runtime.                                   |



|                     |   |                                                                                                              |
|---------------------|---|--------------------------------------------------------------------------------------------------------------|
| txtReceptionPhoneNo |   | The field “txtReceptionPhoneNo” will be populated here. This field can be dynamically filled during runtime. |
| txtBusinessArea     |   | The field “txtBusinessArea” will be populated here. This field can be dynamically filled during runtime.     |
| txtDepartment       |   | The field “txtDepartment” will be populated here. This field can be dynamically filled during runtime.       |
| txtFirstLastName    | - | The field “txtFirstLastName” will be populated here. This field can be dynamically filled during runtime.    |
| txtInternalCode     | - | The field “txtInternalCode” will be populated here. This field can be dynamically filled during runtime.     |
| txtStreetNo         | - | The field “txtStreetNo” will be populated here. This field can be dynamically filled during runtime.         |
| txtPhoneNo          | - | The field “txtPhoneNo” will be populated here. This field can be dynamically filled during runtime.          |
| txtFaxNo            | - | The field “txtFaxNo” will be populated here. This field can be dynamically filled during runtime.            |
| txtEmail            | - | The field “txtEmail” will be populated here. This field can be dynamically filled during runtime.            |

### 7.2.5 UBS\_Blank\_BankingRelationship\_DoR.xdp

|                                                                                                        |          |                                                                                                                         |
|--------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Banking relationship</b> <hr/> UBS<br>City Branch<br>PO Box<br>CC-12345 City<br>Tel. +41 12 345 67 89<br><br>ubs.com |
|--------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|

| Field                    | Example Value         | Description                                                                                                                                  |
|--------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                     | -                     | Fixed UBS logo                                                                                                                               |
| FormType                 | K                     | Can be configured in the DoR configuration per form                                                                                          |
| senderAddressTitle       | Banking relationship  | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtBankingRelationship   | -                     | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number) |
| senderAddressCompanyName | UBS                   | The company name.                                                                                                                            |
| senderAddressCityBranch  | City branch           | The city branch for this form.                                                                                                               |
| senderAddressPoBox       | PO Box                | The post office box information.                                                                                                             |
| senderAddressCity        | CC-12345 City         | The city information with zip information.                                                                                                   |
| senderAddressPhonenumber | Tel. +41 12 345 67 89 | The phonenumber information.                                                                                                                 |
| senderAddressSpacer      |                       | Spacing element between the phonenumber and the website.                                                                                     |

|                      |                                                   |                  |
|----------------------|---------------------------------------------------|------------------|
| senderAddressWebsite | <a href="http://ubs.com">ubs.com</a> <sup>2</sup> | The website url. |
|----------------------|---------------------------------------------------|------------------|

### 7.2.6 UBS\_Blank\_Company\_Account\_DoR.xdp

|                                                                                   |   |                                                              |
|-----------------------------------------------------------------------------------|---|--------------------------------------------------------------|
|  | K | Fisca Pension Fund Foundation of UBS AG<br>Account no. _____ |
|-----------------------------------------------------------------------------------|---|--------------------------------------------------------------|

| Field        | Example Value                           | Description                                                                                           |
|--------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| Logo         | -                                       | Fixed UBS logo                                                                                        |
| FormType     | K                                       | Can be configured in the DoR configuration per form                                                   |
| CompanyTitle | Fisca Pension Fund Foundation of UBS AG | Can be configured in the DoR configuration per form                                                   |
| AccountNo    | Account no.                             | The title of the account number. This title can be overwritten in the DoR configuration.              |
| txtAccountNo | -                                       | The field "txtAccountNo" will be populated here. This field can be dynamically filled during runtime. |

### 7.2.7 UBS\_Blank\_BankingRelationship\_Account\_Custody\_DoR.xdp

|                                                                                                       |   |                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><br>FORM TITLE | K | Banking relationship<br>_____<br>Account no.<br>_____<br>Custody account no.<br>_____<br>Fisca Pension Fund Foundation of UBS AG<br>Tel. +41-61-288 50 40<br><a href="http://www.ubs.com/fisca">www.ubs.com/fisca</a> |
|-------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

<sup>2</sup> <http://ubs.com>

| Field                  | Example Value                              | Description                                                                                                                                  |
|------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                   | -                                          | Fixed UBS logo                                                                                                                               |
| FormType               | K                                          | Can be configured in the DoR configuration per form                                                                                          |
| BankingRelationship    | Banking relationship                       | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtBankingRelationship | -                                          | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number) |
| AccountNo              | Account No                                 | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtAccountNo           | City branch                                | The field “txtAccountNo” will be populated here. This field can be dynamically filled during runtime (banking relationship number)           |
| CustodyAccountNo       | Custody Account No                         | This title can be configured / overwritten in the DoR configuration per form.                                                                |
| txtCustodyAccountNo    | CC-12345 City                              | The field “txtCustodyAccountNo” will be populated here. This field can be dynamically filled during runtime (banking relationship number)    |
| AddressOne             | Fisca Pension Fund<br>Foundation of UBS AG | This title can be configured / overwritten in the DoR configuration per form. The configured value is displayed bold.                        |
| AddressTwo             | Tel. +41-61-288 50 40                      | This title can be configured / overwritten in the DoR configuration per form.                                                                |

### 7.2.8 UBS\_Blank\_Master\_Account\_DoR.xdp

|                                                                                                        |          |                                                                                               |
|--------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | Master no.<br>-<br><hr/> Account no.<br>-<br><hr/> BC no.    Master no.    Object    Currency |
| UBS Switzerland AG<br>Portefeuille<br>P.O. Box<br>8098 Zurich                                          |          |                                                                                               |

| Field          | Example Value       | Description                                                                                                                                   |
|----------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo           | -                   | Fixed UBS logo                                                                                                                                |
| FormType       | K                   | Can be configured in the DoR configuration per form                                                                                           |
| MasterNo       | Master no.          | This title can be configured / overwritten in the DoR configuration per form.                                                                 |
| txtMasterNoOne | -                   | The field "txtBankingRelationship" will be populated here. This field can be dynamically filled during runtime (banking relationship number). |
| txtMasterNoTwo | Custody account no. | This can be configured in the DoR config per form.                                                                                            |
| AccountNo      | Account no.         | This title can be configured / overwritten in the DoR configuration per form.                                                                 |
| txtBCNo        |                     | The field "txtBCNo" will be populated here. This field can be dynamically filled during runtime (banking relationship number).                |
| txtMasterMo    |                     | The field "txtMasterMo" will be populated here. This field can be dynamically filled during runtime (banking relationship number).            |

|             |                                                            |                                                                                                                                    |
|-------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| txtObject   |                                                            | The field “txtObject” will be populated here. This field can be dynamically filled during runtime (banking relationship number).   |
| txtCurrency |                                                            | The field “txtCurrency” will be populated here. This field can be dynamically filled during runtime (banking relationship number). |
| BCNo        | BC no.                                                     | Can be configured / overwritten in the DoR configuration per form.                                                                 |
| MasterNo    | Master no.                                                 | Can be configured / overwritten in the DoR configuration per form.                                                                 |
| Object      | Object                                                     | Can be configured / overwritten in the DoR configuration per form.                                                                 |
| Currency    | Currency                                                   | Can be configured / overwritten in the DoR configuration per form.                                                                 |
| Address     | UBS Switzerland<br>Portefeuille<br>P.O. Box<br>8098 Zürich | Can be configured / overwritten in the DoR configuration per form.                                                                 |

### 7.2.9 UBS\_Blank\_Custody\_DoR.xdp

|                                                                                                                  |          |                                                            |
|------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------|
|  <b>UBS</b><br><br>FORM TITLE | <b>K</b> | Banking relationship<br><hr/> Custody account no.<br><hr/> |
|------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------|

| Field | Example Value | Description    |
|-------|---------------|----------------|
| Logo  | -             | Fixed UBS logo |

|                        |                      |                                                                                                                                               |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| FormType               | K                    | Can be configured in the DoR configuration per form                                                                                           |
| senderAddressTitle     | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                                                 |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number). |
| CustodyAccountNo       | Custody account no.  | This can be configured in the DoR config per form.                                                                                            |
| txtCustodyAccountNo    | Account no. value    | The field “txtCustodyAccountNo” will be populated here. This field can be dynamically filled during runtime.                                  |

### 7.2.10 UBS\_Blank\_Letter\_DoR.xdp

|                                                                                     |          |                               |
|-------------------------------------------------------------------------------------|----------|-------------------------------|
|  | <b>K</b> | Banking relationship<br><hr/> |
|-------------------------------------------------------------------------------------|----------|-------------------------------|

This template does not have the “FORM TITLE” in the header section.

| Field                  | Example Value        | Description                                                                                                                                   |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Logo                   | -                    | Fixed UBS logo                                                                                                                                |
| FormType               | K                    | Can be configured in the DoR configuration per form                                                                                           |
| senderAddressTitle     | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                                                 |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime (banking relationship number). |

### 7.2.11 UBS\_Blank\_Po\_Box\_DoR.xdp

|                                                                                                 |          |                                                                                                                             |
|-------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------|
| <br>FORM TITLE | <b>K</b> | Banking relationship<br><hr style="border: 0; border-top: 1px solid black; margin: 5px 0;"/> UBS Switzerland AG<br>P.O. Box |
|-------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------|

| Field                  | Example Value        | Description                                                                                                     |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| Logo                   | -                    | Fixed UBS logo                                                                                                  |
| FormType               | K                    | Can be configured in the DoR configuration per form                                                             |
| senderAddressTitle     | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime. |
| UbsSwitzerland         | UBS Switzerland AG   | This title can be configured in the DoR config of each form.                                                    |
| PoBox                  | P.O. Box             | This title can be configured in the DoR config of each form.                                                    |
| txtPoBox               | -                    | The field “txtPoBox” will be populated here. This field can be dynamically filled during runtime.               |



## 7.2.12 UBS\_Blank\_Safe\_Deposit\_Box\_DoR.xdp

|                                                                                                        |          |                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Banking relationship</b><br><hr/> <b>Product no.</b><br><hr/> <b>Safe deposit box</b><br><hr/> BC-no. / Agency      Safe deposit box no. |
|--------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|

| Field                  | Example Value        | Description                                                                                                     |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| Logo                   | -                    | Fixed UBS logo                                                                                                  |
| FormType               | K                    | Can be configured in the DoR configuration per form                                                             |
| senderAddressTitle     | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime. |
| ProductNo              | Product no.          | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtCustodyAccountNo    | -                    | The field “txtCustodyAccountNo” will be populated here. This field can be dynamically filled during runtime.    |
| SafeDepositBox         | Safe deposit box     | This title can be configured / overwritten in the DoR configuration per form.                                   |
| BcNo                   | BC-no.               | This title can be configured / overwritten in the DoR configuration per form.                                   |
| BcNo.                  |                      | The field “txtBcNo” will be populated here. This field can be dynamically filled during runtime.                |

|                  |                      |                                                                                                              |
|------------------|----------------------|--------------------------------------------------------------------------------------------------------------|
| Agency           | Agency               | This title can be configured / overwritten in the DoR configuration per form.                                |
| Agency           |                      | The field “txtAgency” will be populated here. This field can be dynamically filled during runtime.           |
| SafeDepositBoxNo | Safe deposit box no. | This title can be configured / overwritten in the DoR configuration per form.                                |
| SafeDepositBoxNo |                      | The field “txtSafeDepositBoxNo” will be populated here. This field can be dynamically filled during runtime. |

### 7.2.13 UBS\_Blank\_Safe\_Deposit\_Box\_No\_DoR.xdp

|                                                                                                         |          |                                           |
|---------------------------------------------------------------------------------------------------------|----------|-------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Banking relationship</b>               |
|                                                                                                         |          | <b>Safe deposit box</b>                   |
|                                                                                                         |          | BC-no. / Agency      Safe deposit box no. |

This is the same as the “UBS\_Blank\_Safe\_Deposit\_Box.xdp” just without the “Product no”.

### 7.2.14 UBS\_Blank\_Safe\_Deposit\_Box\_Size\_DoR.xdp

|                                                                                                          |          |                             |
|----------------------------------------------------------------------------------------------------------|----------|-----------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Banking relationship</b> |
|                                                                                                          |          | <b>BC no. / Agency code</b> |
|                                                                                                          |          | Safe deposit box no.        |
|                                                                                                          |          | Size                        |

| Field    | Example Value | Description                                         |
|----------|---------------|-----------------------------------------------------|
| Logo     | -             | Fixed UBS logo                                      |
| FormType | K             | Can be configured in the DoR configuration per form |

|                        |                      |                                                                                                                 |
|------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| Banking relationship   | Banking relationship | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtBankingRelationship | -                    | The field “txtBankingRelationship” will be populated here. This field can be dynamically filled during runtime. |
| CustodyAccountNo       | BC no. / Agency code | This title can be configured / overwritten in the DoR configuration per form.                                   |
| BcNo.                  |                      | The field “txtBcNo” will be populated here. This field can be dynamically filled during runtime.                |
| Agency                 | Agency               | The field “txtAgency” will be populated here. This field can be dynamically filled during runtime.              |
| SafeDepositBoxNo       | Safe deposit box no. | This title can be configured / overwritten in the DoR configuration per form.                                   |
| SafeDepositBoxNo       |                      | The field “txtSafeDepositBoxNo” will be populated here. This field can be dynamically filled during runtime.    |
| Size                   | Size                 | This title can be configured / overwritten in the DoR configuration per form.                                   |
| Size                   |                      | The field “txtSize” will be populated here. This field can be dynamically filled during runtime.                |

### 7.2.15 UBS\_Blank\_Sender\_Information\_DoR.xdp

|                                                                                                        |          |                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Banking relationship</b><br><hr/> <b>Account no.</b><br><hr/> Fisca<br>Pension Fund Foundation of UBS AG<br>UBS Switzerland AG<br>Basel (PV9V)<br>Your branch |
|--------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Field                  | Example Value          | Description                                                                                                     |
|------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------|
| Logo                   | -                      | Fixed UBS logo                                                                                                  |
| FormType               | K                      | Can be configured in the DoR configuration per form                                                             |
| Banking relationship   | Banking relationship   | This title can be configured / overwritten in the DoR configuration per form.                                   |
| txtBankingRelationship | -                      | The field "txtBankingRelationship" will be populated here. This field can be dynamically filled during runtime. |
| CustodyAccountNo       | Custody account no.    | This can be configured in the DoR config per form.                                                              |
| txtCustodyAccountNo    | Account no. value      | The field "txtCustodyAccountNo" will be populated here. This field can be dynamically filled during runtime.    |
| Address one            | Fisca                  | This can be configured in the DoR config per form.                                                              |
| Address two            | Pension Fund Founda... | This can be configured in the DoR config per form.                                                              |

|               |                   |                                                    |
|---------------|-------------------|----------------------------------------------------|
| Address three | UBS Switzerland A | This can be configured in the DoR config per form. |
| Address four  | Basel (PV9V)      | This can be configured in the DoR config per form. |
| YourBranch    | Your branch       | This can be configured in the DoR config per form. |

### 7.2.16 UBS\_Blank\_VestedBenefits\_DoR.xdp

|                                                                                                        |          |                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>FORM TITLE</b> | <b>K</b> | <b>Vested Benefits Foundation of UBS AG</b><br>P.O. Box, CH-4002 Basel<br>Tel. +41-61-226 75 75<br><br><a href="http://www.ubs.com/vb">www.ubs.com/vb</a><br>Vested Benefits Account<br><hr/> |
|--------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Field                 | Example Value                                                   | Description                                                                                 |
|-----------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Logo                  | -                                                               | Fixed UBS logo                                                                              |
| FormType              | K                                                               | Can be configured in the DoR configuration per form                                         |
| AddressOne            | Vested Benefits Foudation of UBS AG                             | This can be configured in the DoR config per form.. The configured value is displayed bold. |
| AddressTwo            | P.O. Box, CH-4002 Basel                                         | This can be configured in the DoR config per form.                                          |
| AddressThree          | <a href="http://www.ubs.com/vb">www.ubs.com/vb</a> <sup>3</sup> | This can be configured in the DoR config per form.                                          |
| VestedBenefitsAccount | Vested Benefits Account                                         | This title can be configured / overwritten in the DoR configuration per form.               |

<sup>3</sup> <http://www.ubs.com/vb>

|                          |   |                                                                                                                   |
|--------------------------|---|-------------------------------------------------------------------------------------------------------------------|
| txtVestedBenefitsAccount | - | The field “txtVestedBenefitsAccount” will be populated here. This field can be dynamically filled during runtime. |
|--------------------------|---|-------------------------------------------------------------------------------------------------------------------|

## 7.2.17 Footer

The footer is the same across all DoR templates:

|            |                    |            |          |
|------------|--------------------|------------|----------|
| 62308 E V1 | ABJC 26.07.2017 AF | 05.02.2021 | Page 1/1 |
|------------|--------------------|------------|----------|

| Option         | Description                                                                |
|----------------|----------------------------------------------------------------------------|
| AppCode        | The app code (fix AF).                                                     |
| footerFreeText | The free text in the footer (can be overwritten, name is txtMiddleFormId). |
| footerPage     | The naming of the page indicator.                                          |

**Note:** The other fields that are presented in the DoR configuration or the footer will not be reflected (like formcode, version, release date, language or mandator). Those information appear because we need to input it dynamically to the DoR. But the value is evaluated during the rendering of the form.

For more information, please read the following section: [AF Metadata Handling \(see page 148\)](#)

## 7.2.18 Configure your components

There are “Document of Record” options on most components. With these options, you can define what should be displayed in the generated PDF. Open up the components configuration and look for the “Document of Record” section. For example on a panel component:

Document of Record

☐ Exclude from Document of Record ⓘ

☐ Exclude title from Document of Record ⓘ

☐ Exclude description from Document of Record ⓘ

### 7.2.19 Multi-column DoR

The document of record can also have multi columns. This cannot be achieved with the layout mode. For the DoR multi column configuration, the component must be opened in the “Edit” mode. In order to enable multiple columns, open the parent panel and set the “Number of Columns in Document of Record” property to the desired number. Each component inside the panel will then have the property “Colspan for Document of Record”.

**Hint:** Before the parent panel has the property set, the components inside won’t have theirs.

### 7.2.20 DoR formtitle with linebreak

When the formtitle has a line break in the adaptive form, it doesn’t automatically have it in the DoR. If you want to achieve a line break in the DoR, you have to configure it as follows:

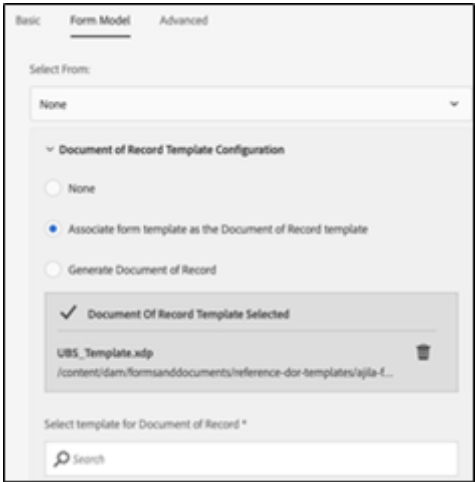
|   |                                                                                                                                                                                                                  |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Open the “Document of Record” configuration in your adaptive form.                                                                                                                                               |
| 2 | Under “Form Title” select “Enter Custom”                                                                                                                                                                         |
| 3 | <p>Enter your formtitle and add the following hex code to the point where the line break should be:</p> <p>&amp;#x2029;</p>  |

## 7.3 Custom XDP

If a form needs a customized PDF, a custom XDP template is created and assigned to the form. With this approach, a designer is able to define the generated PDF pixel perfect.

### 7.3.1 Enable the Custom XDP in the forms config

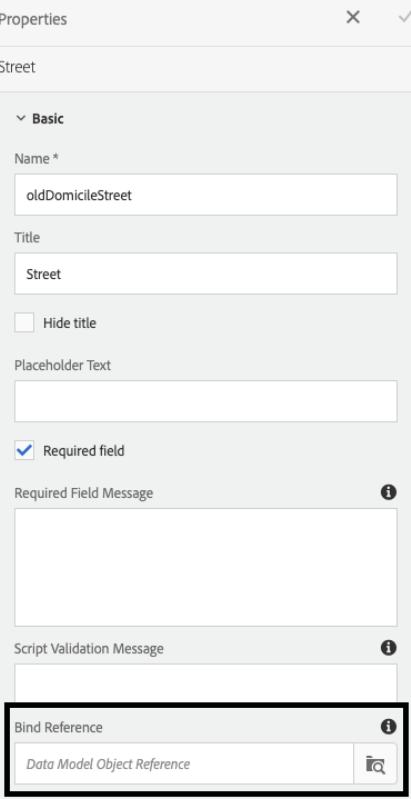
|   |                                           |
|---|-------------------------------------------|
| 1 | Open the properties of the adaptive form. |
|---|-------------------------------------------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | Select the tab “Form Model”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3 | Open the dropdown “Select From” and select the option “Schema”. Upload your schema or select it from the repository and define your Root Element.                                                                                                                                                                                                                                                                                                                                              |
| 4 | <p>Open the accordion “Document of Record Template Configuration”. Select the radio button “Associate form template as the Document of Record template”. Go ahead and select your desired template. Choose the template in the default language.</p> <p><b>Info:</b> Be aware of the multilingualism approach of the Custom XDP! Read the further instructions in the chapter “Additional information”.</p>  |
| 5 | Click “Save & Close”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### 7.3.2 Bind form components to the schema

|   |                                                  |
|---|--------------------------------------------------|
| 1 | Open the adaptive form in the “Edit” mode.       |
| 2 | Click on a component and open its configuration. |



|   |                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <p>In the configuration is a field “Bind reference”</p>  |
| 4 | Open the search dialog and a tree representing your selected schema will be displayed.                                                     |
| 5 | Select the XSD node you want to bind your component to.                                                                                    |
| 6 | Repeat the steps for every relevant component.                                                                                             |

### 7.3.3 Additional information

#### Multilingualism

In case of the Custom XDP, each language has its own XDP template. Hence every adaptive form with a Custom XDP has multiple XDP's. To keep the level of configuration low, the business logic looks up the proper XDP template according to the users locale (for example en\_US). But in order for this business logic to work, some rules need to be followed.

We assume that we have an adaptive form called “Change of address”. This adaptive form needs a Custom XDP. Therefore, the following XDP templates were created:

- ChangeOfAddress\_en\_us.xdp
- ChangeOfAddress\_en\_gb.xdp

- ChangeOfAddress\_de\_de.xdp
- ChangeOfAddress\_de\_ch.xdp

### 1) Template file naming

The template files need to have the following structure and each section of the filename is explained below:

TEMPLATENAME\_LOCALE.xdp

| Section      | Description / Rules                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEMPLATENAME | <p>The TEMPLATENAME needs to be the same for all languages.</p> <p><b>Restrictions:</b></p> <p>The template name section cannot have an underscore ( ) in it. The reason is that the logic assumes the locale (like en_US) after the first underscore in the template name.</p>                                                                                                                                                                                                                                                                                               |
| LOCALE       | <p>The locale for which the template is designed. A locale can consist of two parts: The language and the variant. In case of en_US, “en” is the language and “US” the variant.</p> <p>In the locale part you can define the language and the variant (en_US) or just the language (en). This has an impact of the prioritization!</p> <p>Examples:</p> <p>ChangeOfAddress_en_us.xdp</p> <p>ChangeOfAddress_en.xdp</p> <p><b>Restrictions:</b></p> <p>Always use lower case in the LOCALE section.</p> <p>Use an underscore ( ) to separate the language and the variant.</p> |

### Prioritization

If the naming is properly done, the business logic looks for templates in the following prioritization:

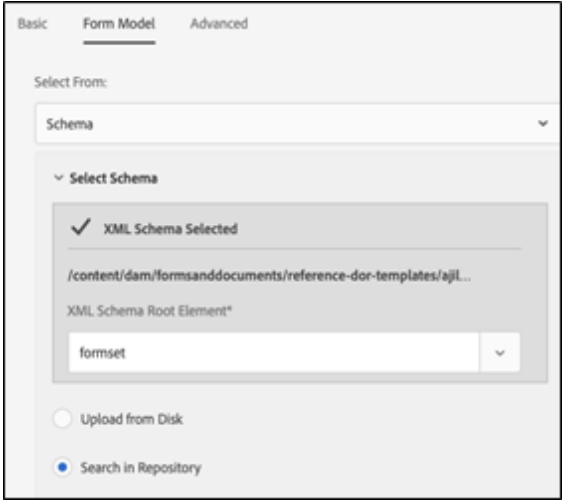
1. A template for the given language AND variant (for example en\_US).
2. A template for the given language (for example en).
3. The template referenced in the configuration.

## 7.4 Custom XDP in a FormSet

A Form Set has the option to generate every subforms PDF individually (generation per subform). In this case, a subform can be configured with a Custom XDP. In order for this to work within a Form Set, the following configurations need to be made.

**Info:** If the Form Set generates a default PDF with all subforms included, no configuration needs to be done.

### 7.4.1 Configure the form set schema

|   |                                                                                                                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Open the properties of the Form Set adaptive form.                                                                                                                                                                                  |
| 2 | Select the tab "Form Model".                                                                                                                                                                                                        |
| 3 | <p>Open the dropdown "Select From" and select the option "Schema". Select the predefined "formset.xsd" and select "formset" as RootElement.</p>  |
| 4 | Click "Save & Close".                                                                                                                                                                                                               |

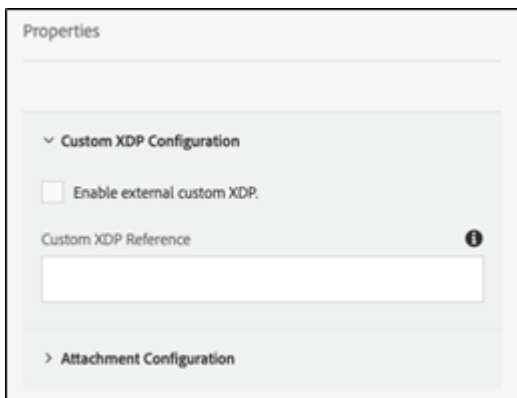
### 7.4.2 Add the binding reference to your adaptive form

|   |                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Open the form set in the "Edit" mode.                                                                                                                                                                                                                                                                                                                                                               |
| 2 | Select the panel which has the adaptive form referenced with a Custom XDP and open its configuration.                                                                                                                                                                                                                                                                                               |
| 3 | <p>Add the RootElement to the "Bind Reference". It needs to have a "/" as prefix. This RootElement needs to be the same as the one from the adaptive form configuration (of the referenced form).</p> <p><b>Info:</b> The RootElement cannot be found with the searching wizard. The reason is, that the schema is referenced on another adaptive form. You have to write the RootElement down!</p> |

|   |                         |
|---|-------------------------|
| 4 | Save the configuration. |
|---|-------------------------|

## 7.5 External custom XDP

In a case where the custom XDP is not on the AEM instance, the external custom XDP can be configured. In that case, the schema needs to be added the same way as in the normal custom XDP. In order to retrieve the XDP externally, the “Document of Record Configurations” component can be used:



For the external XDP's we update the XML Datastream with the following attributes:

| Attribute | Value | Description                                          |
|-----------|-------|------------------------------------------------------|
| APPCode   | AF    | Indicates that it's an adaptive form for the DCG XDP |

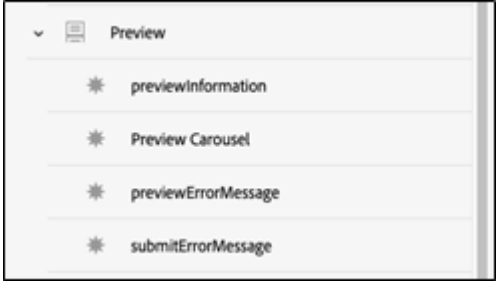
Please visit the component configuration chapter for more information.

## 7.6 Preview

The basic template provides a preview step. The configuration for the preview is embedded and no further configurations need to be done. The user will have the chance to generate a preview by clicking the “Preview” button on this step. The displayed preview has a “Preview” watermark to indicate that it's not the final document of record. It is possible to regenerate the preview, also with changed input data (when the user decides to apply changes to the form). The preview step needs to be the last step.

If you need to add a preview step to a form which doesn't have one, please complete the following steps:

|   |                                                                                             |
|---|---------------------------------------------------------------------------------------------|
| 1 | Add a new step “Preview” to your form.                                                      |
| 2 | Move the “submitErrorMessage” error box from the previous step to new created preview step. |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <p>Add the following components:</p> <ul style="list-style-type: none"> <li>• Information box (the hint for the user with the preview).</li> <li>• The “Carousel” component for displaying the preview with the name “carouselPreview”.</li> <li>• An error box with the name “previewErrorMessage”.</li> </ul> <p>You can compare your components with the ones generated by a template. Your preview step should now look like this:</p>  |
| 4 | <p>Add a “Preview Button” to your toolbar and place it after the “Submit” Button in the configuration. Add the following rules to the “Preview Button”:</p>  <p>Here you find the code for rule 2 and 3:</p> <pre>com.ajila.forms.control.carousel.initializeForPreview(carouselPreview, undefined, previewErrorMessage);</pre> <pre>this.visible=(!this.panel.navigationContext.hasNextItem);</pre>                                      |
| 5 | <p>Add the following rule to the next button:</p>                                                                                                                                                                                                                                                                                                                                                                                        |

**Hint:** The resolution of the preview can be configured is default 150 dpi. Please visit the “System component” chapter for further information.

## 7.7 7.1) PDF File Naming Convention

There are two possible file types that can result after a submit:

| File type | Detail                                                                                                                                                                           |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PDF       | Adaptive form produces only one document. Either a single adaptive form without individual attachments or a form0 that has the "Merged" approach without individual attachments. |
| ZIP       | Adaptive form produces multiple documents. Either a single adaptive form with individual attachments or a form0 with the "Archived" approach (most common one).                  |

### 7.7.1 PDF

When only one PDF is generated, the user can directly download this file. The naming pattern would be:

| Pattern                            | Example                              |
|------------------------------------|--------------------------------------|
| MasterCdok_MAIN_FinalCdok_Formcode | 64960_MAIN_64961_ABCQ.pdf<br>(FormK) |

### 7.7.2 ZIP

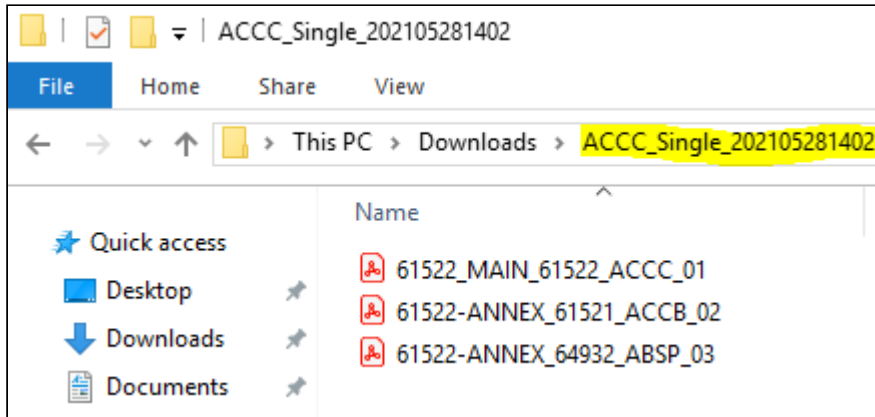
When multiple PDF's are generated, the user can download a ZIP file. The naming patterns would be:

| File Type   | Pattern                                       | Example                                                                                                                                                                                                                            |
|-------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZIP         | Formcode_Type_Timestamp                       | ACCC_Single_202105281402<br>→ Timestamp for uniqueness. Windows would merge folders when ZIP with same name is present.<br>→ Type would be either "Single" for Single Adaptive Forms and Form0 or "Formset" for a Business Formset |
| PDF (MAIN)  | MasterCdok_MAIN_FinalCdok_Formcode_Numbering  | 61522_MAIN_61522_ACCC_01.pdf                                                                                                                                                                                                       |
| PDF (ANNEX) | MasterCdok-ANNEX_FinalCdok_Formcode_Numbering | 61522-ANNEX_61521_ACCB_02.pdf<br>• 61521/ACCB are the CDOK/Formcode of the attachment                                                                                                                                              |

Every PDF file has a number for uniqueness.

### 7.7.2.1 Single Adaptive Form with Attachments

In this case, the user downloads a ZIP archive which includes multiple PDF's. In case of a single adaptive form with individual attachments, it would look like this:

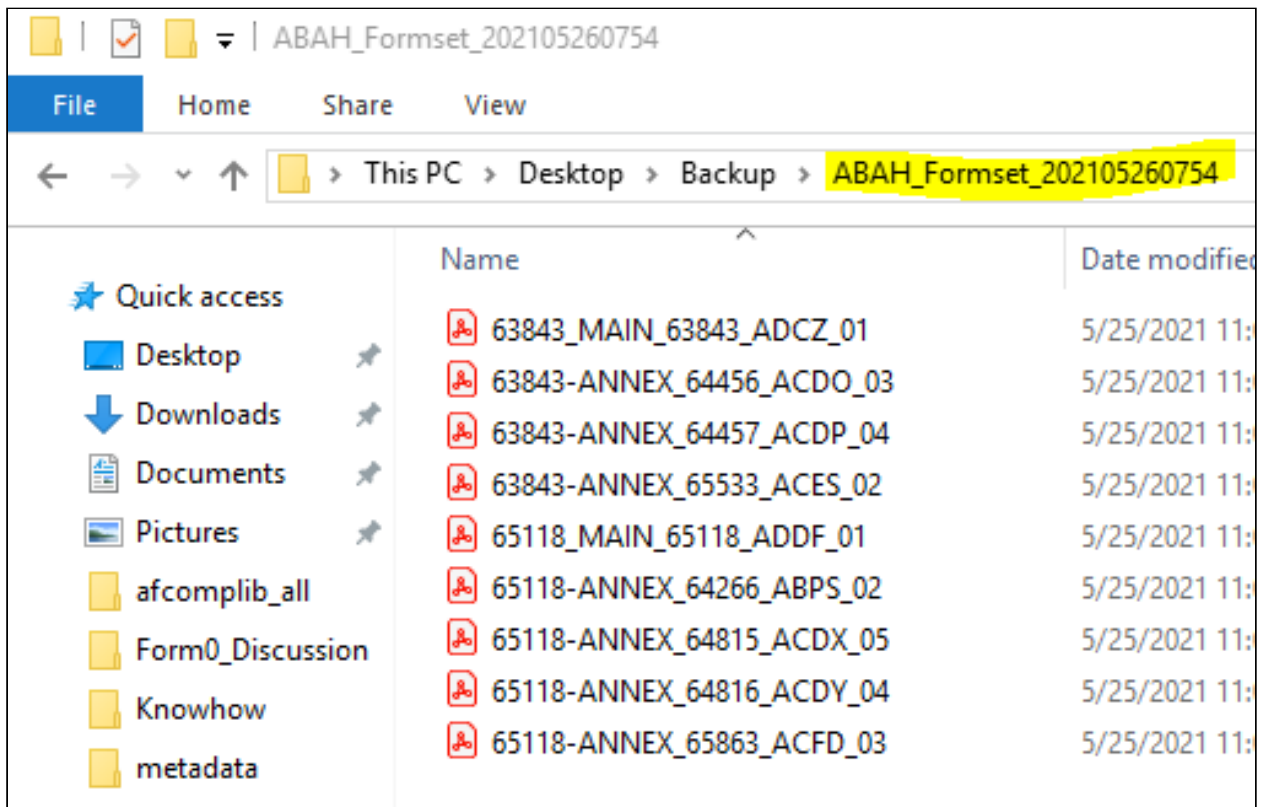


The ZIP has the Formcode and the AF Type (Single or Form0) in its name. Inside the ZIP, the PDF's follow the same structure as before.

Note: There is only one PDF of type MAIN. Every attachment is marked with the ANNEX type.

### 7.7.2.2 Business Formset

A Business Formset combines multiple adaptive forms. In that case, a business formset with two subforms (which have individual attachments) would look like this:



The user can quickly identify which documents belong to the same adaptive form because they all have the same master CDOK at the beginning. The numbering per master CDOK starts at 01.

## 7.8 7.2) Additional Zeppo Translation

In APAC, some of the forms need an additional output in Traditional Chinese (TC) or Simplified Chinese (SC). In order to enable the functionality for the user, the form designer must complete the following steps:

### 7.8.1 Translate the Adaptive Form

There are two possible documents that are generated on submit. The following table shows those two and describes, how the translation can be achieved:



| Type                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generated DoR - Template based     | <p>In this case, the Adaptive Form needs to be translated like always:</p> <ul style="list-style-type: none"> <li>• Generating the dictionary (for TC and/or SC)</li> <li>• Translation by ITT</li> <li>• Import of translated XLIFF</li> </ul> <p>Even though the user will never fill out the adaptive form in TC or SC, the Document of Record (DoR) uses this information to get a Chinese copy of the form. With that approach we achieve a similar layout between the English and Chinese DoR (easier for the customer).</p> |
| Custom (XDP) - Externally rendered | <p>In that case, the external XDP needs to be accessible in TC and SC and have a file with the translation in it. The form designer doesn't need to add anything to the adaptive form. It is assumed that for the same formcode, translations to the supported languages are available.</p>                                                                                                                                                                                                                                        |

### 7.8.2 Add the user selection

The form designer can add radio button component to the adaptive form. By selecting the option, the user can select which language copy he needs. Example:

## Declaration of status of the account holder as a Non-US or a US Person for income from a US source

---

### Preview

#### Please Choose

For the selected language, an additional translated document will be attached. Both language versions will be combined into one output document.

☐ Traditional Chinese ☐ Simplified Chinese ☐ No Translation

**i** By clicking on 'Preview', you can review your document before submission.  
Upon clicking onto 'Submit', you can no longer edit the document and the PDF will be created for signing.

Back

Preview

Submit

There needs to be a rule that adds the selected language to the metadata. For example:

```
if (this.value === "none") {
  window.forms.ubs.translation.setAdditionalOutputCopy("");
} else {
  window.forms.ubs.translation.setAdditionalOutputCopy(this.value);
}
```

If "this.value" is used, it must be ensured that the key of the option matches the locale:

- tc for "Traditional Chinese"
- sc for "Simplified Chinese"

If this is not the case, the mapping needs to be done in the rule. Example configuration:

The screenshot shows a panel titled 'Items' with a list of three items. Each item is in a white box with a label and a trash icon to its right. The items are:

- tc=Traditional Chinese
- sc=Simplified Chinese
- none=No Translation

Below the list is an 'Add' button. To the right of each item is a swap icon (two arrows pointing in opposite directions).

### 7.8.3 Output

When an additional translation is selected, the translated documents will be merged together with the original document. For CLP, we add the additional language to the object metadata. For example:

```
{
  "pages":4,
  "form_metadata":[
    {
      "formcode":"ACBP",
      "cdok":"61391",
      "version":"V2",
      "entity":"046",
      "language":"EN"
    },
    {
      "formcode":"ACBP",
      "cdok":"61391",
      "version":"V2",
      "entity":"046",
      "language":"TC"
    }
  ],
  "pass_through_dta":{
    "case-id":"",
    "filled-by-user":""
  }
}
```

## 7.9 Automatic Preview

The Online Forms are restyled to follow the official E-Banking styleguide. Part of these changes is the handling of the "Preview" functionality. Before, a user had to actively click on the "Preview" button to generate it. Now, the "Preview" should automatically be executed.

The implementation of this is done in two steps:

| Step 1                        | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual Change per Form        | Since we need to fulfill this requirement for the first E-Banking Forms, this procedure will be done manually by a form designer. Given the time, we cannot attempt to change this on all forms and gurantee, that everything works properly.                                                                                                                                                              |
| Migration of all Online Forms | When we have more time, the migration of all Online Forms needs to be planned and executed. This will require some additional conceptual thoughts. To be included: <ul style="list-style-type: none"> <li>▪ Automatic generation of the Preview</li> <li>▪ Handling in FormSets (Subform and FormSet)</li> <li>▪ Removal of the Preview Button (or hide in a first case for fallback scenario).</li> </ul> |

### 7.9.1 Manual Change per Form - Step by Step

1. Add the following rule to the "Next" Step of the form - Event "Click":

```
if (!this.panel.navigationContext.hasNextItem) {
    com.ajila.forms.control.carousel.initializeForPreview(carouselPreview,
        undefined, previewErrorMessage);
}
```

2. Hide the Object of the "Preview" Infobox in the last step

## 7.10 DoR Styling Options

### 7.10.1 General information

Under all components there is now a new dropdown under their respective document of record properties (*Panel properties > Document of Record > Component Style*) in the inspector :

This dropdown contains certain stylings which we have defined. This will effect the look of the specified panel on the Document of record side of things as well as the form level in some select cases.

You do not need to do anything to the direct setup of the Document of record on a form level, the change is component only.

## 7.10.2 Styling options

### 7.10.2.1 Numeration of repeating panels - [GitLab Issue 1301](#)<sup>4</sup>,[1330](#)<sup>5</sup>,[169](#)<sup>6</sup>

**Prerequisites** -The repeating panel should be created in following manner as most of existing form utilize the show and hide functionality for adding instances instead of addInstance on panel.

- Create a panel on any AF step and add components as needed.
- Under panel properties configure the minimum and maximum instance for panel.

<sup>4</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1301>

<sup>5</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1330>

<sup>6</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/169>

The screenshot shows a 'Properties' dialog box with a close button (X) and a checkmark button (✓). The dialog is divided into several sections:

- Section:** A large empty rectangular area.
- Form or Fragment:** A text input field with a checkmark icon to its right.
- Repeat Settings:** A section containing two input fields:
  - Minimum:** A text input field with a yellow highlight containing the number '1'.
  - Maximum:** A text input field with a yellow highlight containing the letter 'S'.

c) Add a button outside the panel from step 1 and under button action use following script (This can be customized for remove button too)

```
window.forms.ubs.addInstance(nameOfPanel);
```

d) From this title you will need to **not** exclude the title from the document of record for this panel.

Properties

✕

✓

My title

Document of Record

☐ Exclude from Document of Record

☐ Exclude title from Document of Record

☐ Exclude description from Document of Record

Pagination

Place

Following Previous

✓

After

Continue Filling Parent

✓

Overflow

None

✓

Layout for Document of Record

Column

▼

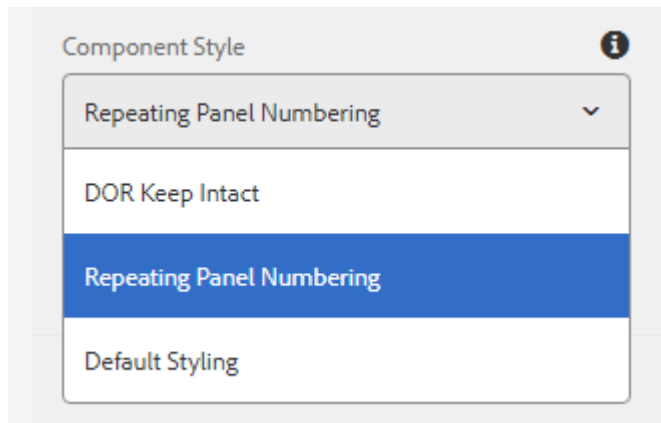
Component Style

Repeating Panel Numbering

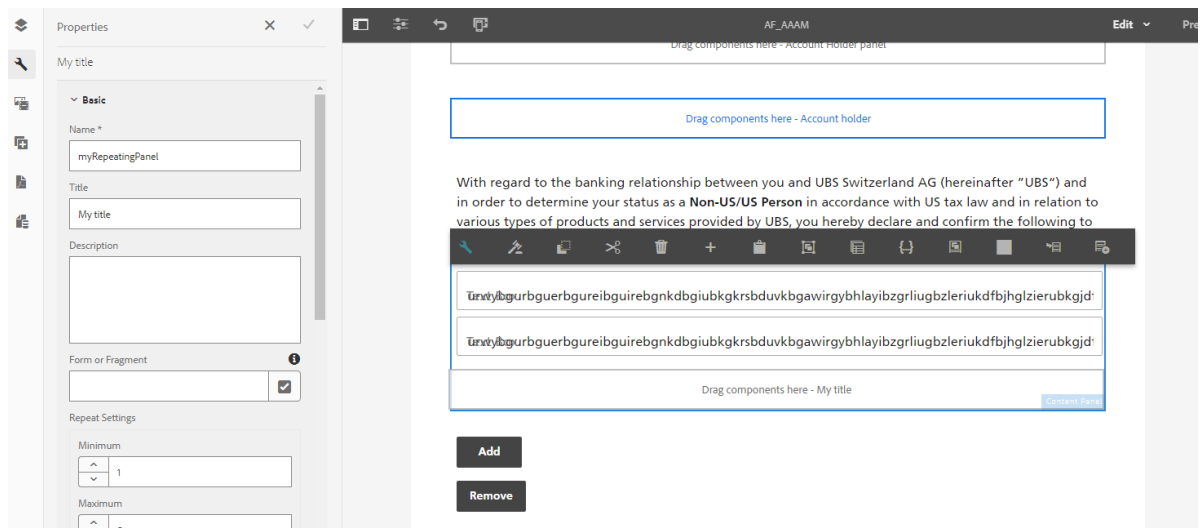
▼

To have numeration in the title of repeating panels on a front form side as the user inputs as well as on the Document of Record you need to follow the steps outlined below.

1. On the panel you will be repeating and the title will show you must select from the dropdown the repeating panel numbering which is under the document of record properties.

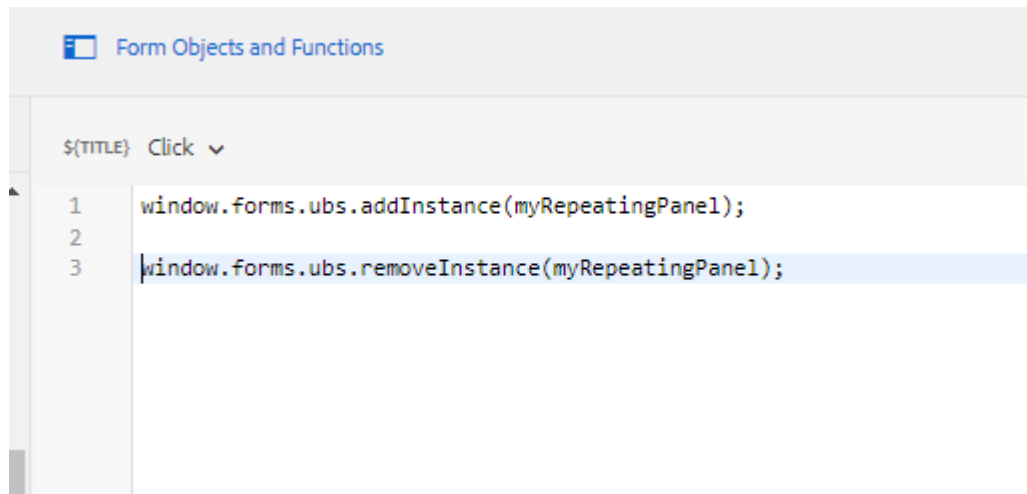


- The panel you have chosen it will display the title assigned to the component on the front side and DoR so please ensure this is the correct text.



- To add or remove instances you must use a specific call and not the standard Adobe call to add or remove instance. If you use other the numbers will not update correctly. Please attach the following to the add instance or remove instance rule editor:





Then the output of a form with multiple instances will look like so on the form (the title property of panel will be used with numbering) :

The screenshot shows a form with three repeating panels. Each panel has a title and two text boxes. The titles are 'My title 1', 'My title 2', and 'My title 3'. The text boxes contain placeholder text: 'urwybgurbguerbgureibguirebgnkdbgiubkgkrdbuvkgawirgybhlayibzgrliugbzleriukdfbjhglzierubkgjd!'. At the bottom of the form are two buttons: 'Add' and 'Remove'.

And then the DoR :

**non-UBS Person** in accordance with UBS tax law and in relation to various types of products and services provided by UBS, you hereby declare and confirm the following to UBS:

**My title 1**

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

**My title 2**

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

**My title 3**

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

urwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfburwybgurbguerbgureibguirebgnkdbgiubkgkrsbduvkgawirgybhlaybzgriugbzleriukdfbjhglzierubkgjdfb;zoieurtkjgndfb;oeizjrgndfb

Text Box

### 7.10.2.2 Keep Content Together - [Gitlab Issue 1411](https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1411)<sup>7</sup>

This feature is designed to allow groups of components within a panel to not split pages and will be grouped together under one page where applicable size dependant.

Select the panel you wish to keep grouped then select from the document of record propertied dropdown DOR Keep Intact.

Component Style

DOR Keep Intact

DOR Keep Intact

Repeating Panel Numbering

Default Styling

<sup>7</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1411>

Before - Content is split over two pages

**PREVIEW**

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List  
This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

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Master no.

-

Account no.

-

| BC no.                                                                                                        | Master no. | Object |
|---------------------------------------------------------------------------------------------------------------|------------|--------|
| <div> <div>UBS Switzerland AG</div> <div>Portefeuille</div> <div>P.O. Box</div> <div>8098 Zürich</div> </div> |            |        |

Radio Button

☐ Item 1
 ☐ Item 2

Dropdown List

After - Content is not split and all on the same page to view

This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List

This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List

This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List

This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List

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15.05.2024
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**Master no.**  
 \_\_\_\_\_

**Account no.**  
 \_\_\_\_\_

| BC no.                                                    | Master no. | Object |
|-----------------------------------------------------------|------------|--------|
| UBS Switzerland AG<br>Postfach<br>P.O. Box<br>8098 Zürich |            |        |

This is an instance of repeatable panel

Radio Button  
☐ Item 1  
☐ Item 2

---

Dropdown List

### 7.10.2.3 Signature layout without static text - 1372<sup>8</sup>

Request was to have the position of the signature component without including the empty static text to have it placed to the right of the DoR page.

To do so you select the DoR styling option on the signature component like the other dor styling options.

Step 1

Old fragment with placeholder

Mandator 001

Signature of client advisor

a

Place Date (dd.mm.yyyy) Name of client advisor

Mandator 019

Signature of client

a

Place Date (dd.mm.yyyy) Name of client

Mandator 033

Signature of client

Place Date (dd.mm.yyyy) Name of client

New fragment, without placeholder

Signature of client

Place Date (dd.mm.yyyy) Name of client

### 7.10.2.4 Adjust height for DOR template - 1375<sup>9</sup>

Request was to reduce the height in the DoR template content area by 3.2mm, so there will be enough space between the content and Footer line. This requirement is applicable for all the DoR templates we have in our AEM environment except Letter templates where the gap is already sufficient.

This applies to case where there was previously little gap to the footer of the page and we have reduced the length of content area to allow additional spacing at the end of the page.

BAKF had this problem. Before:

<sup>8</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1372>

<sup>9</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1375>

https://a302-6989-2131.gddz.ubs.net/app/LS4/content/forms/af/afforms\_apac\_all/af\_ba/AF\_BAKF.html?wcmmode=disabled&mandator=030

FormLinksProd Documentation General Close Dev Cloud Machine Iteration Board DCD machines ddg-wm16-nch-pro... 62046 - AAA

We acknowledge and agree that the Attorney shall be responsible for ensuring that it has the necessary (and permitted) authority and investment power(s) in order to fully comply with all applicable laws and regulations and/or any agreements or arrangements it has entered into with me/us together, the "Obligations". We further acknowledge that:

(a) UBS is not party to such agreements or arrangements between the Attorney and me/us and shall have no responsibility under any circumstances for monitoring compliance with the Obligations (and that UBS solely provides custody and transaction execution services to me/us) and

(b) it is the sole duty of the Attorney to ensure full compliance with such Obligations.

I am/We are aware that the Attorney may have an interest in certain financial instruments used by the Attorney when providing me/us with the Asset Management Services. A potential conflict of interest on the part of the Attorney may arise, for example, if the Attorney invests my/our assets in financial instruments (e.g. investment funds) which are established, administered, managed and/or directly or indirectly controlled by the Attorney. We acknowledge that the Attorney is solely responsible for disclosing any potential conflicts of interest to me/us. In addition, we hereby confirm that the Attorney shall be entitled to invest any or all of my/our assets in such financial instruments.

**I/We hereby accept responsibility for any obligation that arises as a result of any activity undertaken by the Attorney and furthermore, I/we accept responsibility as a debtor(s) to UBS.**

For the avoidance of doubt, I/we confirm that the Attorney is my/our agent and UBS shall not be responsible or liable to me/us for any breach.

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## After :

https://a302-6989-2131.gddz.ubs.net/app/LS4/content/forms/af/afforms\_apac\_all/af\_ba/AF\_BAKF.html?wcmmode=disabled&mandator=030

FormLinksProd Documentation General Close Dev Cloud Machine Iteration Board DCD machines ddg-wm16-nch-pro... 62046

1. Comprehensive Asset Management

We authorise the above named attorney (the "Attorney") - without power of substitution - to manage on my/our behalf, - all assets deposited in any account at any time for my/our account with UBS AG (hereinafter called "UBS"). In particular, the Attorney is authorised to issue instructions for the purchasing, sale of securities, foreign exchange and precious metals, to convert and exchange securities, to engage in arbitrage operations and subscriptions, to exercise or sell subordinated rights, to purchase shares of non-UBS investment funds or other limited securities, to carry out over-the-counter or a fiduciary basis - in UBS's own name but at my/our expense and risk - to utilise electronic banking services provided by UBS in order to carry out any of the activities permitted under this Power of Attorney, to obtain information on my/our accounts from UBS via such means and on such terms and conditions as set out in this Power of Attorney (in addition to regular mail), to choose a place of jurisdiction, to change the currency in which the performance of my/our investments is measured (i.e. reference currency), to agree on the price or fee payable by me/us to UBS for the products and/or services provided by UBS to the accounts managed by the Attorney (including subsequent changes of such price or fee), as well as take all other measures which the Attorney may deem appropriate in connection with the management of my/our assets and ensuring investment advice on my/our accounts to me/us. "Asset Management Services". The Attorney is also authorised to give legally binding instructions to UBS in my/our name for the management of assets and to conclude securities lending agreements on my/our account(s) concerning securities in custody.

We acknowledge and agree that the Attorney shall be responsible for ensuring that it has the necessary (and permitted) authority and investment power(s) in order to fully comply with all applicable laws and regulations and/or any agreements or arrangements it has entered into with me/us together, the "Obligations". We further acknowledge that:

(a) UBS is not party to such agreements or arrangements between the Attorney and me/us and shall have no responsibility under any circumstances for monitoring compliance with the Obligations (and that UBS solely provides custody and transaction execution services to me/us) and

(b) it is the sole duty of the Attorney to ensure full compliance with such Obligations.

I am/We are aware that the Attorney may have an interest in certain financial instruments used by the Attorney when providing me/us with the Asset Management Services. A potential conflict of interest on the part of the Attorney may arise, for example, if the Attorney invests my/our assets in financial instruments (e.g. investment funds) which are established, administered, managed and/or directly or indirectly controlled by the Attorney. We acknowledge that the Attorney is solely responsible for disclosing any potential conflicts of interest to me/us. In addition, we hereby confirm that the Attorney shall be entitled to invest any or all of my/our assets in such financial instruments.

**I/We hereby accept responsibility for any obligation that arises as a result of any activity undertaken by the Attorney and furthermore, I/we accept responsibility as a debtor(s) to UBS.**

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### 7.10.2.5 Red seal on signature block - 1891<sup>10</sup>

Request was to create a red seal on the signature block for signatures to be legally binding in certain territories.

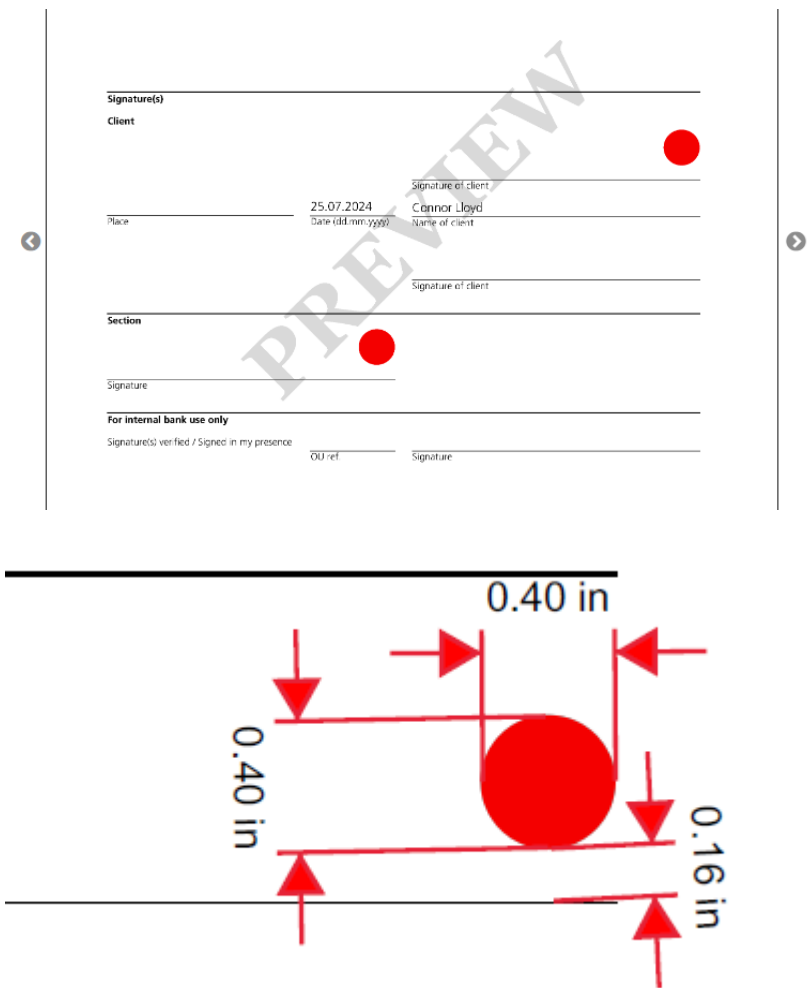
There is some prerequisites and mild limitations for usage in the current version but it is available on all versions of the DoR.

To change the signature block you select the signature scribble and then change the DoR styling in the dropdown to 'Red seal version' like all the previous DoR stylings.

<sup>10</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1891>

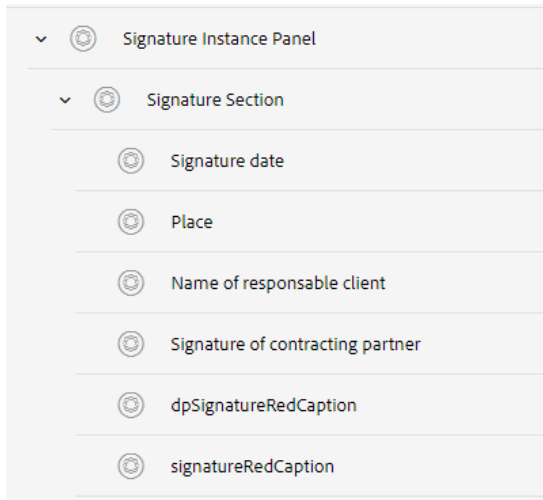
The current limitation with the design is with the aspect ratio and for the image on the signature field to maintain the correct dimensions we require the setup to be in line with the current signature settings on existing forms of 3 cols for the signature in the document of record over a 6 col panel (which is what the current APAC and other form layout is). Different sizes and layouts may require different images to be implemented.

Final result when applied to a signature section



### 7.10.3 Wet signature panel - [GitLab Issue 2091](#)<sup>11</sup>, [1672](#)<sup>12</sup>, [1857](#)<sup>13</sup>

**Prerequisites** - This will give the signature section a new look and feel on the DoR for AF that are given this option in the dropdown.



For the following items the setup will be as follows

#### Signature Instance Panel

This is where the panels instance should be added for creating multiples when using the instance manager

#### Signature Section

This will be where the styling is added so the dropdown for DoR styling will need to be selected as 'Wet Signature Section'. For additional signature sections we offer up to 6 around the form so we will need to select the appropriate one Wet Signature Section Two etc.

For everything below X is the corresponding number to whichever DoR Styling option you have selected. E.G if you select Wet Signature Section it will be 1, then Wet Signature Section Three will be 'dpSignatureDate3' and so on.

#### Signature Date

This is the datepicker and will need to be named 'dpSignatureDateX' with the title Signature date (as per the original PDF)

#### Place

<sup>11</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/2091>

<sup>12</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1672>

<sup>13</sup> <https://devcloud.ubs.net/ubs/wmpc/client-doc-and-records-mgmt/info-capturing-processing/1-information-capture-core/commons/1-information-capture-core-home/-/issues/1857>



This is the place text field and will need to be named 'txtSignaturePlaceX' and with the title Place (as per the original PDF)

Name of responsible client

This is the name text field and will need to be named 'txtSignatureNameX' and with the title Name of responsible client (as per the original PDF)

Signature of contracting partner

This is a placeholder text field which will be hidden from the user to populate the wet signature signature block text. The name will need to be 'txtSignatureSectionX' and with the title 'Please write inside the corners'

dpSignatureRedCaption

This is a placeholder static text field which will be hidden from the user to populate the date underneath red text. The name will need to be 'dpSignatureRedCaptionX' and the text 'Please write inside the fields' (as per original PDF)

signatureRedCaption

This is a placeholder static text field which will be hidden from the user to populate the date underneath red text. The name will need to be 'signatureRedCaptionX' and the text 'Please write inside the borders' (as per original PDF)

Example images

The image displays two examples of a signature form layout. Each example consists of three main components: a date field, a signature field, and a name field. The date field is a grid of boxes for DDMMYY, with labels 'D', 'D', 'M', 'M', 'Y', 'Y', 'Y', 'Y' below each box. The signature field is a large rectangular box with a placeholder line and a green checkmark icon. The name field is a single-line text box. A large 'REVIEW' watermark is overlaid on the right side of the examples.

**Example 1:**

- Date field: 01 01 2001 (D D M M Y Y Y Y)
- Signature field: Placeholder line, green checkmark icon.
- Name field: 1

**Example 2:**

- Date field: 01 01 2002 (D D M M Y Y Y Y)
- Signature field: Placeholder line, green checkmark icon.
- Name field: 2

## 8 8) Rule Editor

The rule editor offers authors to extend the functionality of adaptive form objects. Adobe has documented the rule editor in detail here:

<https://docs.adobe.com/content/help/en/experience-manager-65/forms/adaptive-forms-advanced-authoring/rule-editor.html>

Under the following link you can find the Javascript API of AEM:

<https://helpx.adobe.com/experience-manager/6-5/forms/javascript-api/index.html>

### 8.1 8.1) Coding Guidelines Rule Editor

The AEM Forms Rule Editor offers the possibility to add custom JavaScript code to properly handle all Adaptive Form cases. This section summarizes Best-Practices, how to write code in the Rule Editor.

#### 8.1.1 ClientLib (afcomplib\_all) and Rule Editor

The afcomplib\_all ClientLib offers generic JavaScript code, which can be executed from the Rule Editor. This offers the possibility, to reuse code across multiple forms and avoid inconsistencies.

**Note:** Whenever Rules have to be rewritten or copied around, consider reaching out to the Form Developers. A generic JavaScript function can be checked and planned for implementation!

| Rule Editor                                     | afcomplib_all ClientLib                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> window.forms.ubs.getFormMetadata() </div> | <pre> /**  * Returns a metadata object with the  * information about the language and the mandator.  * With this function, the content can be handled  * dynamically.  */ context.getFormMetadata = function () {     var metadata = {};     var browserUrl = new     URL(window.location);     var mandator =     browserUrl.searchParams.get("mandator");     metadata.mandator = mandator ?     mandator : "001";     var language =     browserUrl.searchParams.get("afAcceptLang");     var defaultLanguage = "en";     if (language) {         metadata.language = language.length         &gt; 1 ? language.substring(0, 2) : defaultLanguage;     } else {         metadata.language =         defaultLanguage;     }     return metadata; } </pre> |

## 8.1.2 Coding Practices

### 8.1.2.1 Avoid multiple rules with same conditions

For one component, add the code for one condition to the same rule. Try to avoid things like the example below. This leads to:

- Harder to read rules
- More effort on maintenance

| <input type="checkbox"/> | Title                                     | Content                                                           |   | Status    |
|--------------------------|-------------------------------------------|-------------------------------------------------------------------|---|-----------|
| <input type="checkbox"/> | Additional address details - Value Commit | <pre> if (this.value === "1") {     // Do something } </pre>      | ▼ | ✓ Enabled |
| <input type="checkbox"/> | Additional address details - Value Commit | <pre> if (this.value === "1") {     // Do something else } </pre> | ▼ | ✓ Enabled |

### 8.1.2.2 Use descriptive and meaningful variable names

Use variable names that make it clear, what the purpose is. This makes the code more readable and maintainable:

| Don't                             | Do                                                  |
|-----------------------------------|-----------------------------------------------------|
| <pre>var k = company.value;</pre> | <pre>var externalCompanyName = company.value;</pre> |

## 9 9) Cross-Topic Information

This chapter provides information about topics that are affecting multiple topics described earlier in the documentation.

### 9.1 Hide components

Components can be hidden on multiple levels and for multiple use cases. They can be combined to achieve different goals. See the following table to gain an overview:

| Level              | Description                                                                                                                                                              | Impact                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component          | As mentioned earlier, a component can be hidden in the adaptive form by enabling the option “Hide object”.                                                               | The component is hidden in the adaptive form but will be shown in the Document of Record.                                                                                                                                                 |
| Component          | There is a “Document of Record” section when authoring a component in the editor mode. You can exclude a component from the DoR here and add other configurations.       | The configuration will be applied to the DoR. For example excluded and therefore not shown.                                                                                                                                               |
| Rule editor        | Creating a rule in the rule editor which hides the component.                                                                                                            | The component will be hidden in the Adaptive Form but still be shown as empty value in the DoR.                                                                                                                                           |
| Document of Record | In the Document of Record section of an adaptive form (where you can select the DoR Template) you can enable the option “Exclude hidden fields from Document of Record”. | When this option is enabled:<br><br>If a component is hidden on component level via “Hide Object”, it will still be shown in the DoR as empty value.<br><br>If a component is hidden by the rule editor, it will not be shown in the DoR. |

The description above can result in different combinations depending on what the goal is. For example: If you want a component hidden in the adaptive form but shown in the DoR you can select “Hide Object” and not exclude it from the DoR on a component level. But if you hide the component via the Rule Editor, other configurations need to be considered. If that’s the case, you cannot select “Exclude hidden fields from

Document of Record". Because when the component is hidden by the rule editor and this option is enabled, nothing will appear in the DoR.

## 9.2 Change list styles

On the static text component you are able to use 2 different list by default; bullet list and numbered list. In case you need other types as well then you can use the html "list-style-type" option.

Create a bullet list and click on the source edit icon:



You should have now a text like this:

```

  B  /  U  T1  T2  ✂
  <ul>
  <li>Text</li>
  <li>text</li>
  <li>text</li>
  <li>text</li>
  </ul>
```

Add one of the style-codes listed below to the <ol> tag. Example:

```

  B  /  U  T1  T2
  ☰ ☱ 🖼️ ⏏️ ★ ▼
  <ul class="dashed">
  <li>a</li>
  <li>b</li>
  <li>c</li>
  <li>d</li>
  <li>e</li>
  </ul>
  <ol style="list-style-type: lower-roman;">
  <li>a</li>
  <li>b</li>
  <li>c</li>
  <li>d</li>
  <li>e</li>
  </ol>
```

| Name | Style-Code |
|------|------------|
|------|------------|

|         |                                                                                      |
|---------|--------------------------------------------------------------------------------------|
| Dash    | class="dashed"                                                                       |
| Roman   | style="list-style-type: upper-roman;"<br>Or<br>style="list-style-type: upper-roman;" |
| Letters | style="list-style-type: lower-roman;"<br>Or<br>style="list-style-type: upper-roman;" |

**NOTE:** Do not use the "list-style" option. Then the listing is not properly displayed in the DoR.

## 9.3 Show / Hide options

If some options need to be dynamically shown or hidden, a centralized function can be called via the rule editor. This functions can be used for:

- Dropdowns
- Radiobuttons
- Checkboxes

### 9.3.1 Show options

```
window.forms.ubs.components.showOptions(["ARG", "CHE", "DEU"], countryList);
```

The showOptions method needs two parameters:

| Parameter     | Description                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| Array of keys | Every option meeting one of the keys provided in the array will be shown to the user. Every other option will be hidden. |
| Component     | Reference to the component where the provided keys should be shown.                                                      |

### 9.3.2 Hide options

```
window.forms.ubs.components.hideOptions(["ARG", "CHE", "DEU"], countryList);
```

The showOptions method needs two parameters:

| Parameter     | Description                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------|
| Array of keys | Every option meeting one of the keys provided in the array will be hidden. Every other option will be shown. |
| Component     | Reference to the component where the provided keys should be hidden.                                         |

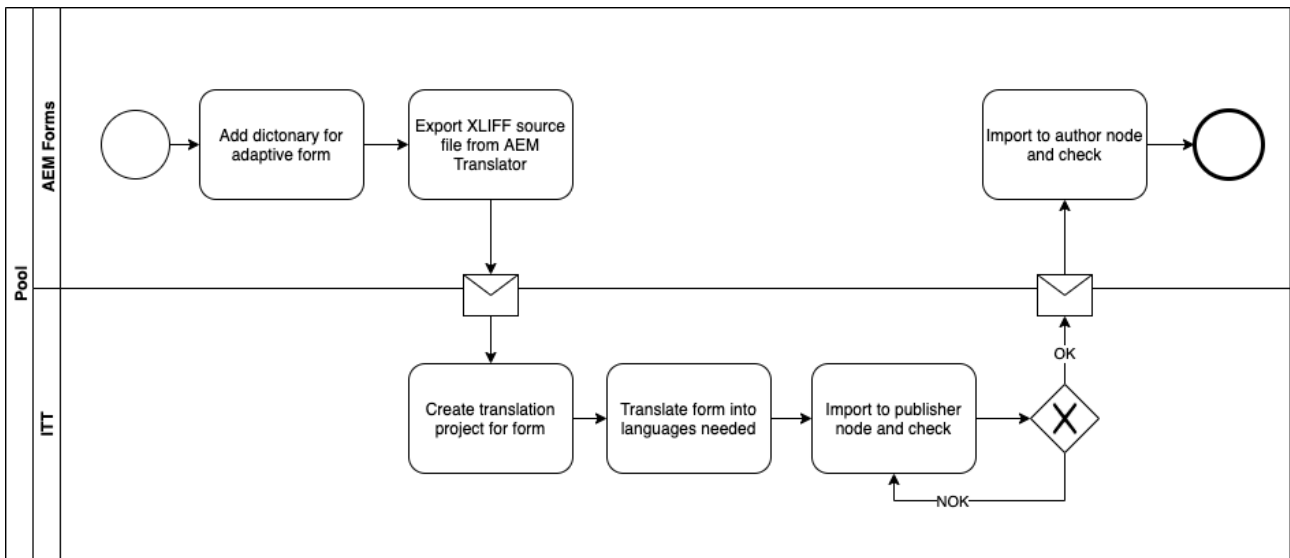


## 10 10) Translations

This chapter describes how an adaptive form can be translated. A detailed documentation by Adobe can be found here:

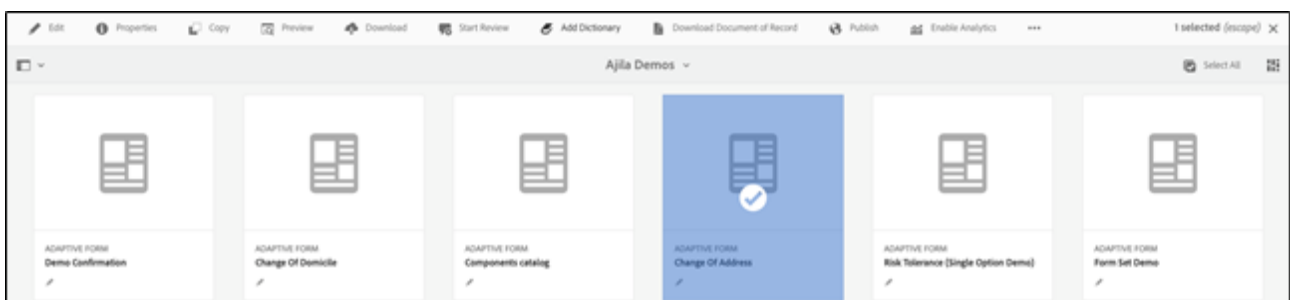
<https://docs.adobe.com/content/help/en/experience-manager-65/forms/adaptive-forms-advanced-authoring/using-aem-translation-workflow-to-localize-adaptive-forms.html>

The process for the translation looks as follows. Each action required on the AEM side is documented in a separate subchapter afterwards.



### 10.1 Adding a dictionary

A dictionary can be added in the AEM authoring section by selecting the adaptive form and clicking “Add Dictionary”:



In the following wizard, define a project title and the languages in which the adaptive form should be translated. Click “Done” to complete the dictionary creation.

**Note:** When the language and country are the same (example: es-ES, de-DE, it-IT) we go only with the language. Therefore the examples mentioned are not selectable. Select es, de it instead.

## 10.2 Export the dictionary

1. Open the following URL to access the AEM translator: `HOST:PORT/libs/cq/i18n/translator.html`.
2. The previously created dictionary is available in the “Dictionaries” dropdown list with its complete path.
3. Export the XLIFF source file by clicking *Export / Export selection as xliiff*. **Note:** Make sure that no rows are selected. Otherwise AEM only exports them.
4. AEM opens a new tab with the XLIFF file.
5. Save the file with the UTF-8 encoding and name it: `FORMCODE_source_file.xliiff`. **Important:** Do not copy it manually into a new file! Make a right click in the new tab and select "Save as..."!
6. **Note:** If the form contains a fragment that’s not yet translated, this fragment needs a dictionary as well. Complete the steps for the fragment as well.

The XLIFF source file can be sent to ITT for translation.

## 10.3 Import translation files

URL for ITT Team: <https://a302-6990-9482.gdzd.ubs.net/app/LS4/content/forms/af/translation/xliff-preprocessor.html>

This section is the same for the ITT team to check their translations on the publisher node as well as for the form designer to finally import them to the author node. **Important:** The import on the publisher node will be overwritten from the author every night. So the final step of importing it to author by the form designer is crucial, otherwise it will be not checked into GIT.

The import can be done with a dedicated form. The URL for the author and publisher will be sent individually. The form looks like this:

Complete the following steps to import a translation:

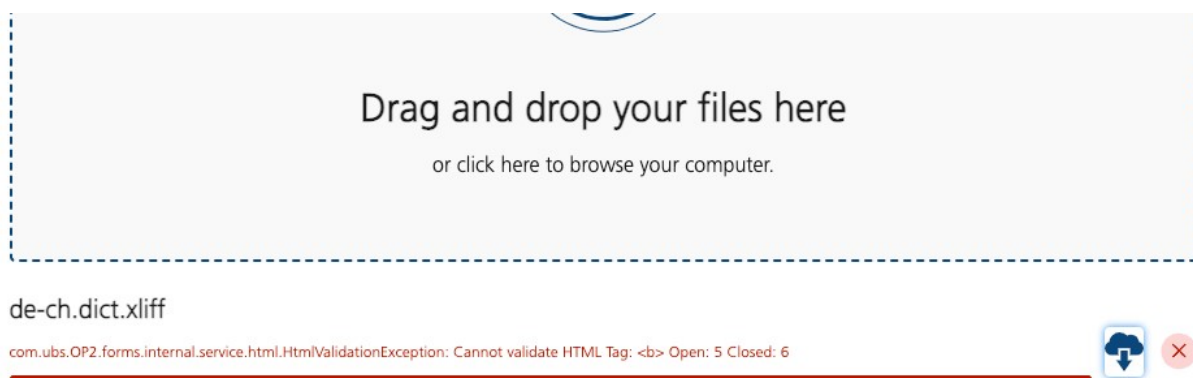
1. Select the form / fragment with the first dropdown. You can filter the entries.
2. Upload all the target files either by drag and drop or by browsing your local computer.
3. Click the button below in order to preprocess and upload the translations. A browser alert will appear with the information if it was successful or not.

The import can be done multiple times for a form.

### 10.3.1 HTML Validation

While uploading the new XLIFF file, a html validator is triggered to check the counts of open and closed html tags.

If you as example forgot to open or close a HTML-Tag then an error is shown with the referencing tag and the counts of open and closed tag.



## 10.4 Translation scope

The translations made with the previously described procedure will have an effect on:

- The adaptive form
- The document of record

If an adaptive form is translated with a custom XDP, the translation won't take effect. In case of custom XDP's, an XDP in each language needs to be described. See further information in the chapter "Custom XDP".

## 10.5 AF Manual Translations

For the Adaptive forms we will create the form only in the source language and target languages will be implemented using AEM Translator tool, unlike the Classic forms design in all languages.

To implement the Target languages, the form must be approved in the source language. Reason being, if there are any changes made in the source language content/style, it will also impact the target languages.

In order to use the AEM Translator tool (<https://a302-6991-0117.gdzd.ubs.net/app/LS4/libs/cq/i18n/translator.html>), we need to get the Access from POD 1 (IC Core). Send an email to [dl-cdrm-icp-informationcapturecore@ubs.com](mailto:dl-cdrm-icp-informationcapturecore@ubs.com)<sup>14</sup>, asking the for the access to AEM Translator.

<sup>14</sup> <mailto:dl-cdrm-icp-informationcapturecore@ubs.com>



## Additional text translations for DoR & AF

### Text box

Default values configured on Text Box component are excluded when generating a dictionary in AEM translator.

As a workaround, please add a hidden Text Box component and set the **Title** equivalent to the default value of the Text Box.

Switzerland

▼ Basic

Name \*

TXT\_Switzerland

Title

Switzerland

☒ Hide title

Placeholder Text

Then add a script to the Text Box component which requires a default value.

\$(TITLE) Initialize ▼

```
1 this.value = TXT_Switzerland.title;
```

### TextMiddleFooter

The footerFreeText on Document of Record are excluded when generating a dictionary in AEM translation. The workaround for TextMiddleFooter is the same with the Text Box. The only difference is by setting the property name to "txtMiddleFormId".

▼ Basic

Name \*

txtMiddleFormId

Title

One copy for each party

☒ Hide title

Placeholder Text

Tick the Hide Object and Exclude from Document of Record.

☒ Hide Object

☐ Disable Object

Field Layout

CSS Class

widget\_ajila-forms-ubs-textbox

Document of Record

☒ Exclude from Document of Record

Colspan for Document of Record

Then, add the script in the Textbox component. Set the execution to Initialize.

\$ {TITLE} Initialize

1

this.value = this.title;

Banking relationship

The Banking relationship text on the AF will not be added to the Dictionary, so we have to add the "Banking relationship" text in the Document of Record configuration under the **senderAddressTitle** as shown in below screenshot.

Then this text will be added in Dictionary and we can translate the text in Target languages.

Document of Record

Header Text

Use Template Header Text

Form Title

Use Adaptive Form Title

senderAddressTitle

Use Custom

Banking relationship

| String                     | Comment | DE | ES | FR | IT | PT | SC | TC | PSBR | ZH-CN | ZH-TW | JA | KO-KR | EN-US | EN-GB | DE-CH                    | FR-CH                    | IT-CH                    | Z |
|----------------------------|---------|----|----|----|----|----|----|----|------|-------|-------|----|-------|-------|-------|--------------------------|--------------------------|--------------------------|---|
| At_Banking_relationship    |         |    |    |    |    |    |    |    |      |       |       |    |       |       |       | Banking relationship     | Banking relationship     | Banking relationship     |   |
| It_sendBankingRelationship |         |    |    |    |    |    |    |    |      |       |       |    |       |       |       | ObankbankingRelationship | ObankbankingRelationship | ObankbankingRelationship |   |
| It_sendBankingRelationship |         |    |    |    |    |    |    |    |      |       |       |    |       |       |       | bbBankingRelationship    | bbBankingRelationship    | bbBankingRelationship    |   |

## 10.6 Chinese forms Translations

Please refer to this page: [7.2\) Additional Zeppo Translation \(see page 104\)](#)

If any translations required please refer to this page [AF Translations](#)<sup>15</sup>. If you find any new translation for any text please update the page.

## 10.7 Invalid Translation Servlet

With the Adobe Ticket EFORMS-20680, the invalid tags issue in the DoR was fixed by AEM raising an exception when an invalid tag is found in the translation. In that case, the Preview and Submit don't work for that Online Form. In the development phase, this is not a problem, since the form designer sees it during testing and can fix the issue. However, existing forms would now have an issue. That's why the following servlets were introduced to support the identification of Online Forms with an issue as well as detailed information about the wrong tags.

Background information: Before the fix, those texts would not appear in the DoR which could potentially lead to legal issues.

### 10.7.1 Usage

The servlets are intended to be used via Postman. The following Postman Collection can be used:



The collection checks against the 1316 Author. This can be adjusted to whatever instance desired.  
Important:

- In the "Authorization" section, a user with enough privileges must be provided

---

<sup>15</sup> <https://central-confluence.ubs.net/confluence/display/GFS/AF+Translations>



- The instance must open up the endpoint for not being MU9 relevant (if the Authentication is enabled):

| Component <b>com.ubs.OP2.forms.internal.web.security.OnlineFormsAuthenticationConfiguration</b> |                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Online Forms Authentication Configuration                                                       |                                                                                                                                                                                                                                           |
| Enabled                                                                                         | <input checked="" type="checkbox"/>                                                                                                                                                                                                       |
| Enable the Authentication Handler and Servlet Filter (authenticationEnabled)                    |                                                                                                                                                                                                                                           |
| URL Exclusion List                                                                              | <div> <div>/content/forms/af/translation/xliff-preprocessor.html</div> <div>/bin/com/ajila/forms/ubs/translation</div> <div>/bin/com/ajila/forms/ubs/fileupload</div> <div>/bin/com/ajila/forms/ubs/translations/htmlchecker</div> </div> |
| List all Endpoints that should not be handled by the Authentication Module (urlExclusionList)   |                                                                                                                                                                                                                                           |
| Configuration Information                                                                       |                                                                                                                                                                                                                                           |
| Persistent Identity (PID)                                                                       | com.ubs.OP2.forms.internal.web.security.OnlineFormsAuthenticationConfigura                                                                                                                                                                |
| Configuration Binding                                                                           | Unbound or new configuration                                                                                                                                                                                                              |

### 10.7.1.1 GET Invalid Online Forms

- Returns a list of Online Form Paths which have an issue with invalid tags
- Doesn't provide detail information

**Example:**

```
[
  "/content/forms/af/affforms_global_all/af_ba/AF_BAFQ",
  "/content/forms/af/affforms_global_all/af_ba/AF_BAAQ",
  "/content/forms/af/affforms_global_all/af_ba/AF_BAFR",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACFE",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACEM",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACBF",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACFG",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACBW",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACBG",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACZT",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACBM",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACAZ",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACDQ",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACEH",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACBX",
  "/content/forms/af/affforms_global_all/af_ac/AF_ACCT",
  "/content/forms/af/affforms_global_all/af_ad/AF_ADDO",
  "/content/forms/af/affforms_global_all/af_ad/AF_ADDP",
  "/content/forms/af/affforms_global_all/af_ad/AF_ADDN",
  "/content/forms/af/affforms_global_all/af_aa/AF_AAZY",
  "/content/forms/af/affforms_global_all/af_aa/AF_AAVN",
  "/content/forms/af/affforms_global_all/af_aa/AF_AAVI",
  "/content/forms/af/affforms_global_all/af_aa/AF_AABC",
  "/content/forms/af/affforms_global_all/af_ab/AF_ABMO",

```

```

"/content/forms/af/affforms_global_all/af_ab/AF_ABBE",
"/content/forms/af/affforms_global_all/af_ab/AF_ABFG",
"/content/forms/af/affforms_global_all/af_ab/AF_ABQU",
"/content/forms/af/affforms_global_all/af_ab/AF_ABFP",
"/content/forms/af/affforms_global_all/af_ab/AF_ABZE",
"/content/forms/af/affforms_global_all/af_ab/AF_ABJV",
"/content/forms/af/affforms_global_all/af_ab/AF_ABQP",
"/content/forms/af/affforms_global_all/af_ab/AF_ABJS",
"/content/forms/af/affforms_global_all/af_ab/AF_ABBS",
"/content/forms/af/affforms_global_all/af_ab/AF_ABSD",
"/content/forms/af/invalidtag"
]

```

### 10.7.1.2 GET Detail Information For Form

With the response of the first endpoint, detailed information about the issues in the Online Form can be requested. The parameter "formReference" needs to be provided containing the form path (as returned in the first request). In addition, a second parameter "language" can be provided to get the detailed information per language. Without this, all issues are shown per language.

#### Example:

```

{
  "de-ch": [
    {
      "translation": {
        "key": "fd_<p><b>I/We confirm</b> that I/we am/are not an entity that
issues or has the ability to issue bearer shares.</p>\n",
        "value": "<p><b>Ich/Wir bestätige(n),</b> dass ich/wir keine Firma
bin/sind, die Inhaberaktien ausgibt oder ausgeben kann.<p>\n"
      },
      "validation": "The element type \"p\" must be terminated by the matching
end-tag \"</p>\"."
    }
  ]
}

```

The servlet can also be access directly from browser on dev author instance after logging in :

1) List of all forms with xliiff issues - <http://a302-4937-1316.gdzd.ubs.net:4502/bin/com/ajila/forms/ubs/translations/htmlchecker>

2)Xliiff issue summary for a specific form path with specific language - [https://a302-6991-0117.gdzd.ubs.net/bin/com/ajila/forms/ubs/translations/htmlchecker?formReference=/content/forms/af/affforms\\_global\\_all/af\\_ba/AF\\_BAHF&language=de-ch](https://a302-6991-0117.gdzd.ubs.net/bin/com/ajila/forms/ubs/translations/htmlchecker?formReference=/content/forms/af/affforms_global_all/af_ba/AF_BAHF&language=de-ch)

← → ↻ 🏠 [https://a302-6991-0117.gdzd.ubs.net/bin/com/ajila/forms/ubs/translations/htmlchecker?formReference=/content/forms/af/afforms\\_global\\_all/af\\_ba/AF\\_BAHF&language=de-ch](https://a302-6991-0117.gdzd.ubs.net/bin/com/ajila/forms/ubs/translations/htmlchecker?formReference=/content/forms/af/afforms_global_all/af_ba/AF_BAHF&language=de-ch)

📁 UBS 📁 ForDev 📁 FormLinksProd 📁 Documentation 📁 General 📁 DevCloudMachine 🔍 INFORMATION CAP...

---

```
[{"translation":{"key":"fd_4","value":"<4"},"validation":"The content of elements must consist of well-formed character data or markup."}]
```

## 11 11) Technical documentation

### 11.1 Embedding Adaptive Forms

#### 11.1.1 Embedding via iFrame (recommended)

iFrames can be used to display a web page within a web page. The HTML code is simple and makes the integration easy (example):

```
<iframe src="http://localhost:4503/content/forms/af/affforms_global_all/AF_64/AF_64647/AF_64647_en.html" title="SomeTitle" height="1000px" width="1000px">
```

The browser includes all CSS / JS files as part of the iFrame and ensures the encapsulation of the other parts of the web page.

| PRO                                  | CON                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Easy to include in existing web page | Responsiveness / Mobile View needs additional libraries in order to work properly. By default, the iFrame keeps the defined height and width. |
| No configuration of a reverse proxy  |                                                                                                                                               |

##### 11.1.1.1 Prerequisites

In order for the PoC to work, the default response header “X-Frame-Origin” needed to be updated. This OSGI config can be found in “Apache Sling Main Servlet”. The default value there is: X-Frame-Origin=SAMEORIGIN. This leads to a CORS error on the web page that includes a form via iFrame.

Therefore, every application which is allowed to integrate adaptive form needs to be listed. For example:

| Apache Sling Main Servlet                                                                                |                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main processor of the Sling framework controlling all aspects of processing requests inside of Sling, na |                                                                                                                                                                                                                 |
| Number of Calls per Request                                                                              | 1500<br>Defines the maximum number of Servlet and Script calls while processing a request. The default value is 1500. The resources required to process a request in case of errors. The default value is 1500. |
| Recursion Depth                                                                                          | 50<br>The maximum number of recursive Servlet and Script calls while processing a request. The default value is 50. (sling.max.inclusions)                                                                      |
| Allow the HTTP TRACE method                                                                              | <input type="checkbox"/><br>If set to true, the HTTP TRACE method will be enabled. By default the HTTP TRACE method is disabled.                                                                                |
| Number of Requests to Record                                                                             | 20<br>Defines the number of requests that internally recorded for display on the "Recent Requests" page.                                                                                                        |
| Recorded Request Path Patterns                                                                           | <br>One or more regular expressions which limit the requests which are stored by the recorder.                                                                                                                  |
| Server Info                                                                                              | Apache Sling<br>The server info returned by Sling. If this field is left empty, Sling generates a default value.                                                                                                |
| Additional response headers                                                                              | X-Content-Type-Options=nosniff<br>X-Frame-Options=ALLOW-FROM https://someapplication.ubs.com<br>Provides mappings for additional response headers. Each entry is of the form: header=value                      |
| Servlet Name                                                                                             | <br>Optional name for the Sling main servlet registered by this component (servletName)                                                                                                                         |
| Configuration Information                                                                                |                                                                                                                                                                                                                 |
| Persistent Identity (PID)                                                                                | org.apache.sling.servlets.core.RequestDispatcherServlet                                                                                                                                                         |
| Configuration Binding                                                                                    | org.apache.sling.servlets.core.RequestDispatcherServlet                                                                                                                                                         |

UPDATE: The X-Frame-Options=ALLOW-FROM is not supported by Chrome and Firefox. SAMEORIGIN on the other hand is. That means it is still an open question, on how to allow specific hosts...

### 11.1.2 Embedding via loading resources

The second approach is documented in detail by Adobe: <https://experienceleague.adobe.com/docs/experience-manager-65/forms/adaptive-forms-basic-authoring/embed-adaptive-form-external-web-page.html?lang=en#sample-topology>

This approach result is much more configuration. The CSS / JS for the frontend is loaded directly into the web page. This means it can conflict with existing CSS / JS paths. The paths provided by Adobe are given (apps/...).

In addition, the communication with the AEM Publisher instance needs to be configured. Therefore, a reverse proxy is needed for every request that should go to the AEM publisher.

| PRO                                                 | CON                   |
|-----------------------------------------------------|-----------------------|
| Reposiveness / Mobile View given with that approach | Configuration effort. |

|  |                                                                             |
|--|-----------------------------------------------------------------------------|
|  | Risk of corrupting the web page that includes the adaptive form (CSS, ...). |
|--|-----------------------------------------------------------------------------|

### 11.1.3 Proposal

At the current state of the project, we suggest to use iFrames for integrating adaptive forms into other web application. The implementation for an iFrame is rather easy and a lot of configuration work doesn't need to be done.

The iFrame automatically handles the request to the AEM Publisher and everything form related is there. The control for the forms stays where it should be.

## 11.2 Languages

AEM Forms supports the following languages out-of-the-box (see chapter "Add localization support for non-supported locales"):

<https://experienceleague.adobe.com/docs/experience-manager-65/forms/manage-administer-aem-forms/supporting-new-language-localization.html?lang=en-US#manage-administer-aem-forms>

**In addition to those language**, we introduced the following languages:

- de-ch (German Switzerland)
- en-us (English US)
- en-gb (Englisch Great Britain)
- fr-ch (French Switzerland)
- it-ch (Italian Switzerland)
- ja-jp (Japanese Japan)
- nl (Dutch)
- pt (Portugese)
- sc (Simplified Chinese)
- tc (Traditional Chinese)

The languages have an impact on the translation of the adaptive forms and on the patterns used. For example there is a differentiation between "de" and "de-ch" when it comes to the display pattern of numbers:

- de → 1.000
- de-ch → 1'000

### 11.2.1 Language mapping

The language mapping service was introduced to address two topics:

| Topic                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display Language on DoR | <p>When calling an adaptive form, a language code with two characters is provided. But not all mandators can handle two characters in the DoR (001 / 101 for example). Therefore we need to have the possibility to configure which language is displayed according to the given language and mandator.</p> <p>Examples:</p> <p>001 and DE → D<br/>001 and ES → S<br/>019 and DE → DE<br/>019 and ES → ES</p> <p><b>Note:</b> The goal in the future is to have two characters across all mandators. But that's not yet possible because the scanning center would need up to 100'000 CHF to do the change.</p> |
| AEM Language            | <p>AEM Forms supports not only the language but also the country. This means there is not just "DE" but also "DE-CH". Same goes for French, Italian, etc. When an adaptive form is called by Kiosk for example, we always get the language ("DE" for example). Depending on the mandator, this can either result in just "DE" or in "DE-CH". The redirect service will take up the configuration and redirect to the concrete AEM language.</p> <p>Examples:</p> <p>001 and DE → DE-CH<br/>019 and DE → DE</p>                                                                                                  |

In order to solve this, the "MandatorLanguageMappingConfiguration" config was introduced:

| Component com.ubs.OP2.forms.internal.service.language.MandatorLanguageMappingConfiguration       |                |
|--------------------------------------------------------------------------------------------------|----------------|
| Description for com.ubs.OP2.forms.internal.service.language.MandatorLanguageMappingConfiguration |                |
| Mandator / Language Mapping                                                                      | 001;en;en-us;E |
|                                                                                                  | 001;de;de-ch;D |
|                                                                                                  | 001;fr;fr-ch;F |
|                                                                                                  | 001;it;it-ch;I |
|                                                                                                  | 001;es;es;S    |

In this configuration, the various combinations can be added. They always have the following pattern which is described in detail below:

*Mandator;IncomingLanguage;AemLanguage;DisplayLanguage*

- Mandator: The mandator provided in the URL while opening the form
- IncomingLanguage: The language provided in the URL while opening the form
- AemLanguage: Final language in which the form is opened in (ex. de-ch, ...)
- DisplayLanguage: Language that will be shown in the DoR (stored in the formdata data as "txtDisplayLanguage")

**Note:** The language code that is sent to CLP is always two characters!

## 11.2.2 Additional information

In AEM Forms we can't handle languages where the language and country is the same. For example de-de, es-es, .... This leads to errors when importing the dictionary. AEM Forms removes the duplicate country information and looks for the language only (ex. de, es, ...).

## 11.3 OSGI configuration

The following table describes UBS specific system components, their goal and their configuration possibilities. You can find them under the following URL: `HOST:PORT/system/console/configMgr`.

| Component                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JcrCleanupScheduler       | <p>This scheduler removes files from the UBS temp folder according the configurations given. Its intend is to periodically clean up generated Document of Records so disk space can be optimized. The following configurations can be made:</p> <ul style="list-style-type: none"> <li>• <code>scheduler.expression</code>: Provide the CRON pattern to define, when the service should be executed.<br/>Default: <code>0 0 1 * * ?</code></li> <li>• <code>keep.minutes</code>: Number of minutes a generated Document of Record should be persisted. Older documents will be deleted from CRX when the service is executed.<br/>Default: 60 (minutes)</li> </ul> |
| DorPreviewService         | <p>This service creates the preview of document of records. The following configurations can be made:</p> <ul style="list-style-type: none"> <li>• <code>preview.imageResolutionDpi</code>: Provide the dpi resolution with whom the preview should be rendered. The default value is 150 dpi.</li> </ul>                                                                                                                                                                                                                                                                                                                                                          |
| Document NumeratorService | <p>This services is responsible for the numeration of the document of records (for attachments, etc.). The following configurations can be made:</p> <ul style="list-style-type: none"> <li>• <code>Font size</code>: The size of the numeration</li> <li>• <code>Margin right</code>: Position of the page numbers from the right side</li> <li>• <code>Margin bottom</code>: Position of the page numbers from the bottom</li> <li>• <code>Page translation</code>: Define the translations per language for the word "Page"</li> </ul>                                                                                                                          |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UbsAttachmentAdapterDefaultImpl | <p>This service fetches the attachments from the UBS form api service. The following configurations can be made:</p> <ul style="list-style-type: none"> <li>• Enabled: Enable or disable the attachment fetching</li> <li>• Service Location: URL of the host incl. endpoint</li> <li>• <del>Mandator: Mandator code</del></li> </ul> <div style="border: 1px solid black; padding: 10px; margin-top: 10px;"> <p>Obsolete → Use the one in <i>DorAttachmentResolver</i>. This default mandator config will disappear once every form must be opened with a mandator</p> </div> |
| PdfConfiguration                | <p>This configuration handles all PDF relevant configurations for adaptive forms.</p> <ul style="list-style-type: none"> <li>• Compliance level: Select the compliance level used for the PDF/A conversion. Every document of record or attachment will be converted with that level of compliance.</li> </ul>                                                                                                                                                                                                                                                                 |

## 11.4 System user

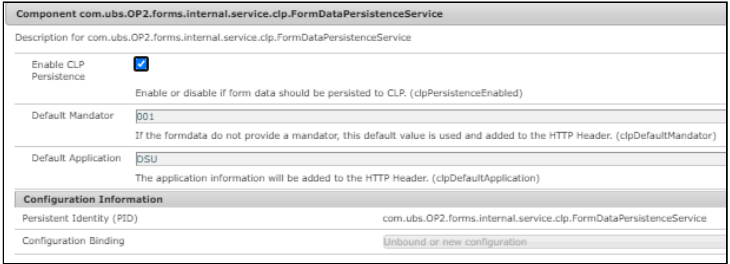
During the installation, a system user with the name “ubs-forms-tmp-writer” is created and has the following permissions. Those permissions are also applied to the subfolders.

| Folder (CRX)                   | Permission                  | Reason                                                                                                                      |
|--------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| /content/dam/formsanddocuments | Read                        | We need to access this folder for the pdf generation per subform (the meta data of the referenced forms).                   |
| /content/forms/af              | Read                        | We need to access this folder for every pdf generation. Mainly to determine the relevant configurations on how to generate. |
| /tmp/ubsdocs                   | Read, Write, Modify, Delete | The generated pdf's will be stored in this folder. We also need to have delete permission for the clean up service.         |

## 11.5 CLP integration

The CLP integration is used to persist the form data into CLP. To achieve this, we first need to authorize via the ISGA service and then PUT to the CLP. The requests are handled through the OP2 bundle provided by the platform team. For the “formData” section of the CLP API we currently follow the XML structure V3.1. Additional information can be found here: [AF Metadata Handling \(see page 148\)](#)

With the retrieved access token we put the form data to the CLP.

| Component                  | Description                                                                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FormDataPersistenceService | <p>Handles the persistence of the form data (retrieving the access token, put to CLP). This can be activated and deactivated:</p> <ul style="list-style-type: none"> <li>• clpPersistenceEnabled: true / false</li> </ul>  |

### 11.5.1 Fire and Forget approach

Form submission cycle now creates logs under **logs/ubsdata-clp.log**. Upon adaptive form submission we will capture the status specific to the form submitted in case:

- The adaptive form submission is successful.
- CLP integration fails.

The form submission is not blocked if CLP integration fails for any reason. With that approach, the user can continue when the CLP integration fails. The DoR has the following differences when the CLP integration fails:

- No QR-Code with the UUID in it (visually)
- The UUID in the custom property is empty

If the CLP integration is mandatory for a specific adaptive form and mandator, this can be configured in the metadata component. In this case, the submit will fail when the integration with CLP is not successful.

**Entities**

Entity Number \*  
001

Language Subset \*  
EN,DE,IT,FR

CLP Mandatory  
true

In addition, there is also the possibility to decide at runtime, whether the integration is required or not. The following function can be used in the rule editor by the form designer:

```
window.forms.ubs.setClpMandatory("AAAA", true);
```

The first parameter is needed to identify the adaptive form. This is important in case of a formset, where multiple adaptive forms are handled at the same time. The second parameter can either be true (mandatory) or false (not mandatory).

## 11.5.2 Health check

In order to monitor the CLP availability, health checks will be executed. Adaptive form that are used for such a health check can be marked as such. If an adaptive form is marked, the status for the CLP XML will be set to "healthcheck". With this marker, CLP can easily identify the health check forms and reorganize them accordingly.

In order to mark an adaptive form as a health check form, open the properties and switch to the "UBS" tab:

components-catalog

Basic Form Model Advanced UBS

Healthcheck Form  
Yes

Components catalog

Generate Preview Upload Image

**Note:** By default, the value is empty which means the adaptive form is no health check form.

## 11.6 AF Caching

### 11.6.1 JS / CSS Caching

When the afcomplib is released, we want the clients browser to reload all the JS / CSS files. After reloading, the caching should still be made. In order to achieve this, the following approach is implemented:

One all publishers, there is a filter which updates the HTML for the adaptive form that is returned to the client. With every build of the afcomplib, we add a timestamp of that build to the HTML:

```
<html lang="enUS">
<head>
  <title>Change Of Address</title>
  <meta http-equiv="content-type" content="text/html; charset=UTF-8">
  <link rel="stylesheet" href="/etc.clientlibs/fd/af/runtime/clientlibs/guidetheme2/core.min.css?20210706080057" type="text/css">
  <link rel="stylesheet" href="/libs/wcm/foundation/components/page/responsive.min.css?20210706080057" type="text/css">
  <title>Change Of Address</title>
  <meta http-equiv="X-UA-Compatible" content="IE=edge,chrome=1">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="version" content="6.5.0.0-SNAPSHOT">
  <meta name="build.timestamp" content="20210706080057" == $0
  <link rel="shortcut icon" href="/apps/ajila-forms-customers/ajila-forms-ubs/clientlibs/themes/standard-theme/assets/favicon.ico" type="image/x-icon">
  <script src="https://polyfill.io/v3/polyfill.min.js?20210706080057"></script>
</head>
```

The "CacheManagementFilter" then applies this timestamp to all CSS and JS links in the HTML:

```
<link rel="stylesheet" href="/apps/ajila-forms-customers/ajila-forms-ubs/clientlibs/components/ajila-forms-ubs-checkbox/common.min.css?20210706080057" type="text/css">
<link rel="stylesheet" href="/apps/ajila-forms-customers/ajila-forms-ubs/clientlibs/components/ajila-forms-ubs-carousel/common.min.css?20210706080057" type="text/css">
<script src="/etc.clientlibs/fd/af/runtime/clientlibs/widgets.min.js?20210706080057"></script>
<script src="/apps/ajila-forms-customers/ajila-forms-ubs/clientlibs/components/ajila-forms-ubs-radiobutton/common.min.js?20210706080057"></script>
<script src="/apps/ajila-forms-customers/ajila-forms-ubs/clientlibs/components/ajila-forms-ubs-fileupload/common.min.js?20210706080057"></script>
```

With that extra query param, the browser knows that he has to reload the files after a deployment. Until the next deployment, the caching is applied as before.

## 11.7 AF Metadata Handling

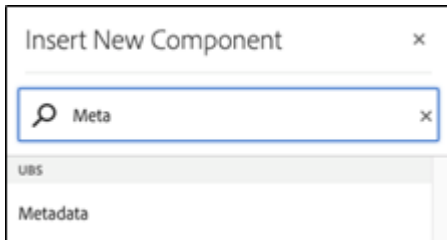
The basis for the metadata handling is the dedicated component that needs to be added to each adaptive form. The metadata component is part of the template and automatically added on the preview step for new adaptive forms.

### 11.7.1 Metadata component

The metadata component is used to store information about the adaptive form that are relevant to Kiosk. These information will be used for:

- The data dump to provide Kiosk with the information which adaptive form's are available
- To provide a service which can retrieve an adaptive form URL by the business triple
- To get the information for the DoR Footer
- To get the information for the CLP integration

The metadata component is part of the basic and formset template. The component is placed on the “Preview” step and won’t be shown in the adaptive form and DoR. If the component is not yet part of the adaptive form, add as normal component:



The configuration of the component looks as follows:

|  |                                                                                                                                                                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <h3>Formcode</h3>                                                                                                                                                                                                                                                               |
|  | <p>The formcode here is the relevant information that is used for the integration with Kiosk.</p>                                                                                                                                                                               |
|  | <p>The formcode entered here is automatically populated to the formcode in the DoR configuration section. The DoR formcode configuration is disabled and therefore readonly.</p>                                                                                                |
|  | <h3>Master Language</h3>                                                                                                                                                                                                                                                        |
|  | <p>The language in which the adaptive form was initially created (two characters).</p>                                                                                                                                                                                          |
|  | <h3>Languages (XLIFF translations)</h3>                                                                                                                                                                                                                                         |
|  | <p>In this field, all available languages are listed. Please be aware, that for example the languages “de-ch” and “de” will result in one entry “DE”. This applies to all other combinations as well.</p>                                                                       |
|  | <h3>Type</h3>                                                                                                                                                                                                                                                                   |
|  | <p>The type of the adaptive form. Currently we have the following types:</p>                                                                                                                                                                                                    |
|  | <ul style="list-style-type: none"> <li>• Adaptive Form</li> <li>• Form0</li> <li>• Business Formset</li> </ul>                                                                                                                                                                  |
|  | <p>Types "Adaptive Form" and "Form0" will be exported to Kiosk as "normal" adaptive forms. The "Business Formset" will be exported as true formset in the future.</p>                                                                                                           |
|  | <h3>Entities</h3>                                                                                                                                                                                                                                                               |
|  | <p>Multiple entities can be added here. In the Language subset, please add the languages necessary according to the information in the “Master Language” and the “Languages (XLIFF translations)”. The languages are comma separated and should always have two characters.</p> |
|  | <p>Within an entity, multiple integrated CDOK’s can be added. The combination of CDOK and mandator is relevant for the DoR footer and the CLP integration (see next chapter). The "Release Date" needs to be added in the format: <b>dd.MM.yyyy</b>.</p>                        |
|  | <p>The first entry underneath each mandator is considered the <i>Master CDOK</i>. This is the fallback when no explicit final CDOK is defined (see priority below).</p>                                                                                                         |

## 11.7.2 DoR Footer / CLP Integration

### 11.7.2.1 Final CDOK Definition

For each CDOK, the following information are provided in the metadata component:

- CDOK
- Partner Level
- Release Date
- Version

These information are needed for the DoR footer and the CLP XML. Therefore, every adaptive form needs to provide the information on this final CDOK. The final CDOK is defined by the following priority:

| Priority | Field                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | finalCdokList         | <p>When an adaptive form has a dynamic CDOK which can change during runtime, the following Javascript function can be used in the Rule Editor:</p> <div> <p><b>Dynmic CDOK function</b></p> <pre> window.forms.ubs.setFinalCdok("AAAB", "62589"); </pre> </div> <p>The first parameter is the formcode of the current adaptive form. The second one is the final CDOK. Whenever this function is executed for a formcode that already exists, the final CDOK is updated.</p> <p>Upon preview or submission, this information is read out and if an entry is found for the adaptive form, this will be considered the final CDOK. The mapping of the formcode is also done via the metadata component. This means that the first parameter of the function needs to match the value of the formcode in the metadata component. This function can be used regardless of a single adaptive form or a formset.</p> |
| 2        | txtFormId (Form Data) | <p>The txtFormId is part of the AF formdata. When a txtFormId is present with a value, this is considered the final CDOK when previewing or submitting the adaptive form.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Priority | Field       | Description                                                                                                                                                                                          |
|----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3        | Master CDOK | When no explicit final CDOK is defined, the first entry for the given mandator in the metadata component is considered the final CDOK.<br>→ If no entry is present, the preview or submit will fail. |

**Deprecated priority 2:** The txtFormId is deprecated. The form designers should not manipulate them directly as it can conflict with a formset. Once everything is migrated, this can be removed.

### 11.7.2.2 Metadata extraction

With the final CDOK present and the mandator provided when opening an adaptive form, the metadata information can be extracted from the component. When the ABSD form is opened with mandator 001 and the final CDOK is 64877, the following information will be extracted:

The screenshot shows a form titled "Entities" with the following fields:

- Entity Number \***: 001
- Language Subset \***: EN,DE,FR,IT
- CDOK's** (highlighted with a red box):
  - CDOK Number \***: 64877
  - Partner Level**: ☐
  - Release Date \***: 2021-04-01
  - Version \***: V14

Those information will then be populated to the DoR footer and to CLP. If there is no entry for the final CDOK and the mandator, the form preview and submit will fail.

#### 11.7.2.2.1 DoR Footer

The information shown above are part of the DoR footer.

### 11.7.2.2.2 CLP Integration

The same information is used to complete the CLP definition for further DPE processing. The example below is part of the CLP XML definition: [CLP Metadata](#)<sup>16</sup>.

```
{
  "pages":2,
  "form_metadata":[
    {
      "formcode":"ABSD",
      "cdok":"64877",
      "version":"V14",
      "entity":"001",
      "language":"EN"
    }
  ],
  "pass_through_dta":{
    "case-id":"",
    "filled-by-user":""
  }
}
```

## 11.8 Referenzdata (RDS)

<sup>16</sup> <https://central-confluence.ubs.net/confluence/display/IDM/CLP+Metadata>



## 12 12) API

This chapter describes the API's that are offered by adaptive form library bundle. For each API a separate chapter is provided.

**Important:** In order to be able to access those services, the requestor needs to authenticate via BasicAuth.

### 12.1 RenderDor

The renderDoR servlet can be used to directly create a PDF Document. This service can be used by LOB apps which have all the necessary data to render the PDF Document.

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL             | [SERVER]/bin/api/bdr/renderdor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Method          | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Request Headers | <ul style="list-style-type: none"> <li>• X-Mandator: The mandator to generated the PDF for.</li> <li>• X-Application-Id: Application ID of the calling application.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              |
| Request Body    | form-data <ul style="list-style-type: none"> <li>▪ formcode: Formcode of the form to be rendered</li> <li>▪ mandator: The mandator to generate the PDF for.</li> <li>▪ language: The 2 characters for the language (ex. EN, DE, FR, IT, ES, ...)</li> <li>▪ data: Base64 encoded XML Data stream individual to each form.</li> </ul>                                                                                                                                                                                                                                                        |
| Response Type   | <p><b>200 OK</b><br/>When the DoR Rendering was successful, a 200 response is returned.</p> <p><b>400 BAD REQUEST</b><br/>This is returned, when one of the mandatory Headers or Form Data Parameters are missing. The issue can also occur, when the internal submit call was successful, but the reponse did not contain any document. Please provide the submitTransactionId for analysis by the UBS Forms Team.</p> <p><b>500 INTERNAL SERVER ERROR</b><br/>The request failed due to an internal error. Please provide the submitTransactionId for analysis by the UBS Forms Team.</p> |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response Body | <p><b>Content-Type:</b> application/json</p> <p>The response body is always a JSON (Success or Error Case):</p> <div data-bbox="360 394 1423 1059"> <p><b>renderDor Response Body</b></p> <pre> {   "submitTransactionUuid": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",   "exception": false,   "exceptionMessage": "",   "listSize": 1,   "forms": [     {       "formUuid": "423cd7e3-5e60-414b-b022-acbee45d04cf",       "formcode": "BAEW",       "finalCdok": "65091",       "releaseDate": "30.12.2022",       "version": "V2",       "file": "BASE_64 File"     }   ] }</pre> </div> <p>The individual attributes are described here:</p> <ul style="list-style-type: none"> <li>exception: Indicates if there was an exception (also a given by the HTTP Response Code)</li> <li>listSize: Indicates the number of documents generated</li> <li>submitTransactionId: Internal transactionId of UBS Forms for tracing (provided in success and failed case)</li> <li>forms: Array of documents containing the clpUuid and the base64 PDF Document.</li> <li>exceptionMessage: Additional information in error case.</li> </ul> |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.2 Flattened XSD Schema

Returns the flattened XSD Schema for the given Formcode.

|                   |                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------|
| URL               | [SERVER]/bin/api/xsd/getAdaptiveFormSchema                                                                           |
| Method            | GET                                                                                                                  |
| Request Parameter | <ul style="list-style-type: none"> <li>formcode: Formcode for which the flattened XSD Schema is requested</li> </ul> |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response Type | <p><b>200 OK</b><br/>The servlet always returns a 200. The body defines, if the flattening was successful or not. In case of a non successful flattening, the returned JSON contains the errorMessage.</p> <p><b>500 Internal Server Error</b><br/>Unexpected error while creating JSON Response</p>                                                                                                                                                                                                |
| Response Body | <div> <div>Successful Flattening</div> <pre>{   "formcode": "AAAZ",   "flattenedXsd": "BASE64_XSD",   "xsdFileReference": "/content/dam/formsanddocuments/data-dictionary/forms/AAAM.xsd",   "exception": false }</pre> </div> <div> <div>Failed Flattening</div> <pre>{   "formcode": "ABSD",   "exception": true,   "exceptionMessage": "Online Form has no referenced XSD. Reference Path: /content/dam/formsanddocuments/affforms_global_all/af_ab/AF_ABSD/jcr:content/metadata" }</pre> </div> |

## 12.3 Redirect service

The redirect service is used to have a centralized entry point to the adaptive forms by providing the business triple (formcode, mandator, language).

|        |                                                                                         |
|--------|-----------------------------------------------------------------------------------------|
| URL    | [SERVER]/bin/api/affforms?<br>formcode=FORMCODE&mandator=MANDATOR&afAcceptLang=LANGUAGE |
| Method | GET                                                                                     |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Request information | <p>Form triple as part of the query params:</p> <ul style="list-style-type: none"> <li>• formcode: The 4 character formcode (ex. ABSD)</li> <li>• mandator: The 3 digits for the mandator (ex. 001)</li> <li>• language: The 2 characters for the language (ex. EN, DE, FR, IT, ES, ...)</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| Response Type       | The servlet returns a redirect status when the adaptive form was found and redirects automatically to it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Validation          | <p><b>Triple validation</b><br/>When one of the provided triple information is missing, the servlet returns an error.</p> <p><b>Formcode validation</b><br/>When multiple adaptive forms with the same formcode are found in the repository, the servlet returns an error.</p> <p><b>Language validation</b><br/>The redirect service checks if the provided language is present for the given formcode. When the given language is not present and it also doesn't match the "Master Language", the servlet returns an error.<br/>The service evaluates the relevant language. When 001 and DE is provided, a de-ch translations needs to be present.</p> |

## 12.4 Consistency check

The consistency check was implemented to easily identify adaptive forms with inconsistencies including:

- Fragment / Subform references that point to a non-existing fragment / subform
- Document of record templates that point to a non-existing template

|                     |                                                   |
|---------------------|---------------------------------------------------|
| URL                 | [SERVER]/bin/com/ajila/forms/ubs/consistencycheck |
| Method              | GET                                               |
| Request information | No extra parameter needed                         |
| Response Type       | application/json                                  |

|               |                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response body | <p>A string array with inconsistent adaptive forms</p> <p>Example:</p> <pre>[ "/content/forms/af/standalone/64/64647/64647_EN", "/content/forms/af/standalone/62/62308/62308_EN", "/content/forms/af/standalone/64/64279/64279_en", "/content/forms/af/affforms_global_integrated/AF_64/ AF_64470_en" ]</pre> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.5 Reference resolver

The reference resolver can be used to retrieve all forms that have a reference on one or more paths provided. The following paths can be searched for:

- Fragment paths (ex. /content/forms/af/affforms\_global\_fragmentlib/affrg\_global\_BankingRelationship)
- Document of record template paths (ex. /content/dam/formsanddocuments/reference-dor-templates/ajila-forms-ubs/02\_forms/UBS\_Blank\_Address\_DoR.xdp)
- Component paths – **Attention:** The referenced path in the forms is not the complete path. Example:  
*Complete CRX Path (will deliver no results):* /apps/ajila-forms-customers/ajila-forms-ubs/components/controls/datepicker  
*Referenced paths (to be used):* ajila-forms-customers/ajila-forms-ubs/components/controls/datepicker

|                     |                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL                 | [SERVER]/bin/com/ajila/forms/ubs/referenceresolver                                                                                                               |
| Method              | GET                                                                                                                                                              |
| Request information | <p>Query Parameter: referencePaths</p> <p>The referencePaths query parameter can include multiple reference paths which need to be separated by a comma (,).</p> |
| Response Type       | application/json                                                                                                                                                 |

|               |                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response body | <p>A string array with inconsistent adaptive forms</p> <p>Example:</p> <pre>[ "/content/forms/af/standalone/64/64647/64647_EN", "/content/forms/af/standalone/62/62308/62308_EN", "/content/forms/af/standalone/64/64279/64279_en", "/content/forms/af/affforms_global_integrated/AF_64/ AF_64470_en" ]</pre> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.6 Form component metadata service

The form component metadata service can be used, to retrieve all components used within one form. For each component, the configured metadata will be returned. The components that are taken into account are:

- guideTextBox
- guideNumericBox
- guideTelephone
- guideDatePicker
- guideDropDownList
- guideCheckBox
- guideRadioButton

|                     |                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URL                 | [SERVER]/bin/com/ajila/forms/ubs/componentmetadata                                                                                      |
| Method              | GET                                                                                                                                     |
| Request information | <p>Query Parameter: form</p> <p>The complete path to the form, for example:<br/>/content/forms/af/ajila-forms-ubs/change-of-address</p> |
| Response Type       | application/json                                                                                                                        |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response body | <p>An array of form components with the following JSON structure:</p> <pre>[   {     "autoFieldKeyWord": "String",     "guideNodeClass": "String",     "jcrPrimaryType": "String",     "jcrTitle": "String",     "name": "String",     "options": [       {         "key": "String",         "value": "String"       }     ],     "slingResourceType": "String",     "textIsRich": "String",     "validatePictureClause": "String",     "validatePictureClauseMessage": "String",     "validationPatternType": "String"   } ]</pre> |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.7 RetrieveDor

This servlet checks against MOF, if the current user has access to the final document requested. If the access is granted, the servlet returns the PDF/A file as PDF in the response.

|        |                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------|
| URL    | [SERVER]/bin/api/forms/retrievedor?<br>clpUuid=CLPUUID&mandator=MANDATOR&applicationId=APPLICATION |
| Method | GET                                                                                                |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Request information | <p><b>Request Parameters</b></p> <ul style="list-style-type: none"> <li>• clpUuid: The CLP UUID of the document that is requested.</li> <li>• mandator: The 3 digits for the mandator (ex. 001).</li> <li>• applicationId: applicationID of the caller.</li> </ul> <p><b>Headers</b></p> <ul style="list-style-type: none"> <li>▪ CSIV2: Token of the user. Normally added by the ISSO provider. But can also be provided explicitly as header.</li> </ul> |
| Response Type       | <p><b>200 OK</b><br/>When the access is granted by MOF, the response is the PDF/A document as application/pdf.</p> <p><b>401 UNAUTHORIZED</b><br/>The CSIV2 token is not provided</p> <p><b>409 CONFLICT</b><br/>The request failed because the user has no access to the document or any other technical issue.</p>                                                                                                                                       |

## 12.8 RetrieveDorData

The retrieveDorData servlet can be used to retrieve a dor. This service can be used by LOB apps (through OP3).

|                                                         |                                                                                                                                                                                                             |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL                                                     | [SERVER]/bin/api/bdr/retrievedordata                                                                                                                                                                        |
| Method                                                  | GET                                                                                                                                                                                                         |
| Request Headers<br><i>*All parameters are mandatory</i> | <ul style="list-style-type: none"> <li>• X-Mandator: The mandator to retrieve the draft for</li> <li>• X-Application-Id: Application ID of the calling application.</li> <li>• ASN1: ASN.1 token</li> </ul> |
| Request Parameters<br><i>*All fields are mandatory</i>  | <ul style="list-style-type: none"> <li>• clpUuid: CLP UUID of DoR</li> </ul>                                                                                                                                |



|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response Type | <p><b>200 OK</b><br/>When the retrieveDor is successful, a 200 response is returned.</p> <p><b>400 BAD REQUEST</b><br/>This is returned, when one of the mandatory Headers or Form Data Parameters are missing or there is an error in the validation. Please provide the retrieveTransactionId for analysis by the UBS Forms Team.</p> <p><b>500 INTERNAL SERVER ERROR</b><br/>The request failed due to an internal error. Please provide the retrieveTransactionId for analysis by the UBS Forms Team.</p> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Response  
Body**Content-Type:** application/json

The response body is always a JSON (Success or Error Case):

Success Case:

**retrieveDorData Response Body**

```
{
  "retrieveTransactionId": "a97c206a-2233-4672-96bf-3aaa6468bbe3",
  "exception": "false",
  "exceptionMessage": "",
  "contentType": "application/pdf"
  "formData": "JVBERi0xLjcKJfbk/
N8KMSAwIG9iago8PAovTGFuZyAoZW4tVVMpCi9NYXJrSW5mbyA8PA...."
}
```

**retrieveDorData Response Body**

```
{
  "retrieveTransactionId": "433c8c1d-be66-4f8b-8904-0388b20b2caa",
  "exception": "false",
  "exceptionMessage": "",
  "contentType": "application/zip"
  "formData":
"UESDBBQACAgIALtC2VoAAAAAAAAAAAAAAAAAgAAAANjU4MDJfTUFJTl9ET1JfNjU4MDJf
QUJXWl8wMS...."
}
```

Error Case:

**retrieveDorData Response Body**

```
{
  "retrieveTransactionId": "795e6a45-5725-4b86-b70b-78ba2074b3a4",
  "exception": "true",
  "exceptionMessage": "Unauthorized: User has no access to the final
document",
  "contentType" : ""
  "formData": ""
}
```

```
}

```

The individual attributes are described here:

- `retrieveTransactionId`: Internal `transactionId` of UBS Forms for tracing (provided in success and failed case)
- `exception`: Indicates if there was an exception (also a given by the HTTP Response Code)
- `exceptionMessage`: Additional information in error case.
- `contentType`: "application/pdf" or "application/zip" or empty if exception
- `formData`: Base64 encoded string of Byte Array.

## 12.9 OpenDraft

The Opendraft servlet can be used to open an adaptive form with prefilled data. This service can be used by LOB apps to display forms with prefilled information.

|                  |                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL              | [SERVER]/app/LS4/bin/api/afforms/[LOCATION]?<br>formcode=[FORMCODE]&afAcceptLang=[LANG]&mandator=[MANDATOR]&clpUuid=[CLP<br>UUID]                                                                                                                                                                                                          |
| Method           | GET                                                                                                                                                                                                                                                                                                                                        |
| Request<br>Body  | form-data <ul style="list-style-type: none"> <li>• FORMCODE: Formcode of the form to be rendered</li> <li>• MANDATOR: The mandator to generate the PDF for.</li> <li>• LANG: The 2 characters for the language (ex. EN, DE, FR, IT, ES, ...)</li> <li>• CLPUUID: Clpuuid of the draft</li> <li>• LOCATION: Internal or External</li> </ul> |
| Response<br>Type | <p><b>OK</b><br/>Form loads with prefilled data.</p> <p><b>UUID Validation error</b><br/>This is returned, when the clpuuid is invalid number of characters.</p> <p><b>Form loads with no data error</b><br/>This is returned, when the clpuuid lookup finds no data, the form still loads but will be empty</p>                           |

|               |                         |
|---------------|-------------------------|
| Response Body | <b>N/A - Form Loads</b> |
|---------------|-------------------------|

## 12.10 CreateDraft

The createdraft servlet can be used to persist a draft by providing a data stream. This service can be used by LOB apps (through OP3) which have all the necessary data to save draft.

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL                                                     | [SERVER]/bin/api/bdr/createdraft                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Method                                                  | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Request Headers<br><i>*All parameters are mandatory</i> | <ul style="list-style-type: none"> <li>• X-Mandator: The mandator to generated the PDF for.</li> <li>• X-Application-Id: Application ID of the calling application.</li> <li>• ASN1: ASN.1 token</li> </ul>                                                                                                                                                                                                                                                                                             |
| Request Body<br><i>*All fields are mandatory</i>        | form-data <ul style="list-style-type: none"> <li>• formcode: Formcode of the form to be rendered</li> <li>• mandator: The mandator to generate the PDF for.</li> <li>• language: The 2 characters for the language (ex. EN, DE, FR, IT, ES, ...)</li> <li>• data: Base64 encoded XML Data stream individual to each form.</li> </ul>                                                                                                                                                                    |
| Response Type                                           | <p><b>200 OK</b><br/>When the createdraft is successful, a 200 response is returned.</p> <p><b>400 BAD REQUEST</b><br/>This is returned, when one of the mandatory Headers or Form Data Parameters are missing or there is an error in the validation. Please provide the draftTransactionId for analysis by the UBS Forms Team.</p> <p><b>500 INTERNAL SERVER ERROR</b><br/>The request failed due to an internal error. Please provide the draftTransactionId for analysis by the UBS Forms Team.</p> |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response Body | <p><b>Content-Type:</b> application/json</p> <p>The response body is always a JSON (Success or Error Case):</p> <p>Success Case:</p> <div data-bbox="360 439 1423 784"> <p><b>createDraft Response Body</b></p> <pre>{   "draftTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",   "exception": "false",   "exceptionMessage": "",   "clpUuid": "6591bcd7-37b1-447b-9ea5-dfc08385c97b" }</pre> </div> <p>Error Case:</p> <div data-bbox="360 855 1423 1232"> <p><b>createDraft Response Body</b></p> <pre>{   "draftTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",   "exception": "true",   "exceptionMessage": "Exception while validating inputs: Language must be provided in the header or as parameter.",   "clpUuid": "" }</pre> </div> <p>The individual attributes are described here:</p> <ul style="list-style-type: none"> <li>• draftTransactionId: Internal transactionId of UBS Forms for tracing (provided in success and failed case)</li> <li>• exception: boolean. Indicates if there was an exception (also a given by the HTTP Response Code)</li> <li>• exceptionMessage: Additional information in error case. &lt;blank&gt; if no exception.</li> <li>• clpUuid: clpUuid of a successful save draft case, &lt;blank&gt; if exception.</li> </ul> |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 12.11 OpenAPI Specifications

### 12.11.1 UBS Forms afcomplib

This is a list of APIs offered by the adaptive form library bundle  
 File is created with goto/swagger and best viewed using it. (Yaml attachment can be downloaded)  
 Version: 1.0.0  
 BasePath:/app/LS4

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### 12.11.1.1 Access

#### 1. HTTP Basic Authentication

### 12.11.1.2 Methods

[ Jump to [Models](#) (see page 0) ]

#### 12.11.1.2.1 Table of Contents

##### 12.11.1.2.1.1 DoR

- GET /bin/api/afforms/retrievedor (see page 0)
- GET /bin/api/bdr/retrievedordata (see page 0)
- POST /bin/com/ajila/forms/ubs/dor (see page 0)
- POST /bin/com/ajila/forms/ubs/printpartial (see page 0)

##### 12.11.1.2.1.2 FileUploadDownload

- GET /bin/com/ajila/forms/ubs/download (see page 0)
- DELETE /bin/com/ajila/forms/ubs/fileupload (see page 0)
- GET /bin/com/ajila/forms/ubs/fileupload (see page 0)
- HEAD /bin/com/ajila/forms/ubs/fileupload (see page 0)
- POST /bin/com/ajila/forms/ubs/fileupload (see page 0)

##### 12.11.1.2.1.3 Process

- POST /bin/api/bdr/createdraft (see page 0)
- POST /bin/api/bdr/renderdor (see page 0)
- POST /bin/com/ajila/forms/ubs/saveasdraft (see page 0)

##### 12.11.1.2.1.4 RDS

- GET /app/LS4/bin/com/ajila/forms/ubs/rds/siap-subcategory (see page 0)
- GET /bin/com/ajila/forms/ubs/rds/country (see page 0)
- GET /bin/com/ajila/forms/ubs/rds/elsig (see page 0)

- GET /bin/com/ajila/forms/ubs/rds/nci-helperparams (see page 0)
- GET /bin/com/ajila/forms/ubs/rds/nci-identifier (see page 0)

#### 12.11.1.2.1.5 Redirection

- GET /bin/api/afforms/external (see page 0)
- GET /bin/api/afforms (see page 0)
- GET /bin/api/afforms/internal (see page 0)

#### 12.11.1.2.1.6 Retrieval

- GET /bin/api/xsd/getAdaptiveFormSchema (see page 0)
- GET /bin/com/ajila/forms/ubs/authoringmetadata (see page 0)
- GET /bin/com/ajila/forms/ubs/componentmetadata (see page 0)
- GET /bin/com/ajila/forms/ubs/consistencycheck (see page 0)
- GET /bin/com/ajila/forms/ubs/referenceresolver (see page 0)
- GET /bin/com/ajila/forms/ubs/translation (see page 0)
- GET /bin/com/ajila/forms/ubs/translations/htmlchecker (see page 0)
- GET /bin/ubsforms/formfragment (see page 0)

#### 12.11.1.2.1.7 SessionCheck

- POST /bin/com/ajila/forms/ubs/keepalive (see page 0)
- GET /bin/com/ajila/forms/ubs/ping (see page 0)

### 12.11.2 DoR

(see page 165)

GET /bin/api/afforms/retrievedor

Up (see page 0)

Returns the PDF/A file (**binApiAfformsRetrievedorGet**)

This servlet checks against MOF, if the current user has access to the final document requested. If the access is granted, the servlet returns the PDF/A file as PDF in the response

#### 12.11.2.1 Query parameters

##### **clpUuid (required)**

*Query Parameter* – The CLP UUID of the document that is requested.

##### **mandator (required)**

*Query Parameter* – The 3 digits for the mandator

**formcode (required)**

*Query Parameter* – Form Code (identifier) of form

**afAcceptLang (required)**

*Query Parameter* – Language of form

**caseSwsc (optional)**

*Query Parameter* – caseSwsc

**caseUuid (optional)**

*Query Parameter* – caseUuid

## 12.11.2.2 Responses

### 12.11.2.2.1 200

When the access is granted by MOF, the response is the PDF/A document application/pdf. ([see page 165](#))

### 12.11.2.2.2 400

Bad Request - e.g. Missing/Invalid mandatory attributes ([see page 165](#))

### 12.11.2.2.3 401

Unauthorized - e.g. missing CSIV2 token ([see page 165](#))

### 12.11.2.2.4 404

Not Found ([see page 165](#))

### 12.11.2.2.5 409

Conflict - The request failed because the user has no access to the document or any other technical issue. ([see page 165](#))

([see page 165](#))

GET /bin/api/bdr/retrievedordata

Up ([see page 0](#))

Returns the PDF/A file (**binApiBdrRetrievedordataGet**)

The retrieveDorData servlet can be used to retrieve a dor. This service can be used by LOB apps (through OP3



### 12.11.2.3 Request headers

### 12.11.2.4 Query parameters

**clpUuid (required)**

*Query Parameter* – The CLP UUID of the document that is requested.

### 12.11.2.5 Return type

[RetrieveDorDataResponse](#) (see page 0)

### 12.11.2.6 Example data

Content-Type: application/json

```
{
  "exception" : "false",
  "retrieveTransactionId" : "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",
  "formData" : "6591bcd7-37b1-447b-9ea5-dfc08385c97a",
  "contentType" : "application/pdf",
  "exceptionMessage" : ""
}
```

### 12.11.2.7 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

### 12.11.2.8 Responses

#### 12.11.2.8.1 200

When the retrieveDor is successful, a 200 response is returned. [RetrieveDorDataResponse](#) (see page 0)

### 12.11.2.9 Example data

Content-Type: successExample

```
{
  "retrieveTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",
  "exception": "false",
  "exceptionMessage": "",
  "contentType": "application/pdf",
  "formData": "JVBERi0xLjcKJfbk/N8KMSAwIG9iago8PAovTGFuZyAoZW4tVVMpCi9NYXJrSW5mbmA8PA..."
}
```

#### 12.11.2.9.1 400

This is returned, when one of the mandatory Headers or Form Data Parameters are missing or there is an error in the validation. Please provide the retrieveTransactionId for analysis by the UBS Forms Team.

[RetrieveDorDataResponse](#) (see page 0)

### 12.11.2.10 Example data

Content-Type: badRequestExample

```
{
  "retrieveTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",
  "exception": "true",
  "exceptionMessage": "Exception while validating inputs: clpUuid must be provided in the header or as parameter.",
  "contentType": "",
  "formData": ""
}
```

#### 12.11.2.10.1 500

The request failed due to an internal error. Please provide the retrieveTransactionId for analysis by the UBS Forms Team. (see page 165)

(see page 165)

POST /bin/com/ajila/forms/ubs/dor

[Up](#) (see page 0)

Generate Document of Record (**binComAjilaFormsUbsDorPost**)

Generates DoR - turns into image and displays image

### 12.11.2.11 Consumes

This API call consumes the following media types via the Content-Type request header:

- multipart/form-data

### 12.11.2.12 Form parameters

**data (required)**

*Form Parameter –*

**locale (required)**

*Form Parameter –*

**resource (required)**

*Form Parameter –*

### 12.11.2.13 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `text/plain`

### 12.11.2.14 Responses

#### 12.11.2.14.1 200

Picture Links of DoR preview ([see page 165](#))

### 12.11.2.15 Example data

Content-Type: text/plain

```
/app/LS4/tmp/ubsdocs/b6ec3fb0-92da-4e0d-aaa8-a89b3077e8f4/
AF_AAAM-4-4-2025_Preview_Page_1.png,/app/LS4/tmp/ubsdocs/d2932a2b-
f173-4f35-9b7e-8aec9f0d4e8a/AF_AAAM-4-4-2025_Preview_Page_2.png
```

([see page 165](#))

POST `/bin/com/ajila/forms/ubs/printpartial`

[Up \(see page 0\)](#)

Returns incomplete PDFA file (**binComAjilaFormsUbsPrintpartialPost**)

Validates against MOF and allows for the download of an incomplete form.

### 12.11.2.16 Consumes

This API call consumes the following media types via the Content-Type request header:

- `multipart/form-data`

### 12.11.2.17 Form parameters

**data (optional)**

*Form Parameter —*

**locale (optional)**

*Form Parameter —*

**resource (optional)**

*Form Parameter —*

### 12.11.2.18 Responses

#### 12.11.2.18.1 200

When the access is granted by MOF, the response is the PDF/A document ([see page 165](#))

#### 12.11.2.18.2 500

Error while generating the print partial document ([see page 165](#))

## 12.11.3 FileUploadDownload

([see page 165](#))

```
GET /bin/com/ajila/forms/ubs/download
```

Up ([see page 0](#))

For users to download file (`binComAjilaFormsUbsDownloadGet`)

Downloading of file based on identification

### 12.11.3.1 Query parameters

**file identification (optional)**

*Query Parameter —*

## 12.11.3.2 Responses

### 12.11.3.2.1 200

File Downloaded ([see page 165](#))

([see page 165](#))

DELETE /bin/com/ajila/forms/ubs/fileupload

Up ([see page 0](#))

Delete File Upload (**binComAjilaFormsUbsFileuploadDelete**)

Delete File Upload

## 12.11.3.3 Query parameters

### identification (optional)

*Query Parameter* – identification from the upload service in POST

### formContainerPath (optional)

*Query Parameter* –

## 12.11.3.4 Responses

### 12.11.3.4.1 204

Deleted Successfully ([see page 165](#))

### 12.11.3.4.2 409

SC Conflict ([see page 165](#))

([see page 165](#))

GET /bin/com/ajila/forms/ubs/fileupload

Up ([see page 0](#))

Query Path of Upload (**binComAjilaFormsUbsFileuploadGet**)

Query Path of Upload

### 12.11.3.5 Query parameters

**identification (optional)**

*Query Parameter* – identification from the upload service in POST

**formContainerPath (optional)**

*Query Parameter* –

### 12.11.3.6 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `text/string`

### 12.11.3.7 Responses

#### 12.11.3.7.1 200

Path of file in identification ([see page 165](#))

### 12.11.3.8 Example data

Content-Type: text/string

```
@/C:/Users/T123456/Downloads/Test_textfile.txt
```

([see page 165](#))

```
HEAD /bin/com/ajila/forms/ubs/fileupload
```

Up ([see page 0](#))

Check Availability of Upload (**binComAjilaFormsUbsFileuploadHead**)

Check Availability of Upload

### 12.11.3.9 Query parameters

**identification (optional)**

*Query Parameter* – identification from the upload service in POST

**formContainerPath (optional)**

*Query Parameter* –

### 12.11.3.10 Responses

#### 12.11.3.10.1 204

Available [\(see page 165\)](#)

#### 12.11.3.10.2 404

Not Found [\(see page 165\)](#)

#### 12.11.3.10.3 409

SC Conflict [\(see page 165\)](#)

[\(see page 165\)](#)

POST /bin/com/ajila/forms/ubs/fileupload

Up [\(see page 0\)](#)

For Users to upload file (**binComAjilaFormsUbsFileuploadPost**)

Uploading of file

### 12.11.3.11 Consumes

This API call consumes the following media types via the Content-Type request header:

- `multipart/form-data`

### 12.11.3.12 Form parameters

#### **file (optional)**

*Form Parameter —*

#### **runtimeLocale (optional)**

*Form Parameter —*

#### **formContainerPath (optional)**

*Form Parameter —*

### 12.11.3.13 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

### 12.11.3.14 Responses

#### 12.11.3.14.1 201

File Created ([see page 165](#))

### 12.11.3.15 Example data

Content-Type: application/json

```
{ "id": "d00e95f3-a3c2-4f42-99ab-e21e07178596", "fileName": null, "contentType": null }
```

## 12.11.4 Process

([see page 165](#))

POST /bin/api/bdr/createdraft

[Up \(see page 0\)](#)

Create Draft from data stream (**binApiBdrCreatedraftPost**)

The createDraft servlet can be used to persist a draft by providing a data stream

### 12.11.4.1 Consumes

This API call consumes the following media types via the Content-Type request header:

- multipart/form-data

### 12.11.4.2 Request headers

### 12.11.4.3 Form parameters

**formcode (required)**

*Form Parameter –*

**mandator (required)**

*Form Parameter –*

**language (required)**



*Form Parameter –***data (required)***Form Parameter –*

#### 12.11.4.4 Return type

[CreateDraftResponse](#) (see page 0)

#### 12.11.4.5 Example data

Content-Type: application/json

```
{
  "draftTransactionId" : "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",
  "exception" : "false",
  "clpUuid" : "6591bcd7-37b1-447b-9ea5-dfc08385c97a",
  "exceptionMessage" : ""
}
```

#### 12.11.4.6 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

#### 12.11.4.7 Responses

##### 12.11.4.7.1 200

When the createDraft is successful, a 200 response is returned with the clpUuid [CreateDraftResponse](#) (see page 0)

#### 12.11.4.8 Example data

Content-Type: successExample

```
{"draftTransactionId":"6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610","exception":"false","exceptionMessage":"","clpUuid":"6591bcd7-37b1-447b-9ea5-dfc08385c97b"}
```

#### 12.11.4.8.1 400

This is returned, when one of the mandatory Headers or Form Data Parameters are missing or there is an error in the validation. Please provide the draftTransactionId for analysis by the UBS Forms Team.

[CreateDraftResponse](#) (see page 0)

#### 12.11.4.9 Example data

Content-Type: badRequestExample

```
{ "draftTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610", "exception": "true", "exceptionMessage": "Exception while validating inputs: Language must be provided in the header or as parameter.", "clpUuid": "" }
```

#### 12.11.4.9.1 500

The request failed due to an internal error. Please provide the draftTransactionId for analysis by the UBS Forms Team. [CreateDraftResponse](#) (see page 0)

#### 12.11.4.10 Example data

Content-Type: internalServerErrorExample

```
{ "draftTransactionId": "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610", "exception": "true", "exceptionMessage": "Exception while creating draft: Could not save document into MOF: {'type': 'https://httpstatus.es/500', 'status': 500}", "clpUuid": "" }
```

[\(see page 165\)](#)

POST /bin/api/bdr/renderdor

[Up](#) (see page 0)

#### Render PDF Document (**binApiBdrRenderdorPost**)

The renderDoR servlet can be used to directly create a PDF Document. This service can be used by LOB apps which have all the necessary data to render the PDF Document.

#### 12.11.4.11 Consumes

This API call consumes the following media types via the Content-Type request header:

- multipart/form-data

#### 12.11.4.12 Request headers

#### 12.11.4.13 Form parameters

**formcode (required)***Form Parameter –***mandator (required)***Form Parameter –***language (required)***Form Parameter –***data (required)***Form Parameter –*

#### 12.11.4.14 Return type

[RenderDorResponse](#) (see page 0)

#### 12.11.4.15 Example data

Content-Type: application/json

```
{
  "exception" : "false",
  "submitTransactionUuid" : "6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610",
  "listSize" : 1,
  "exceptionMessage" : "",
  "forms" : [ {
    "formcode" : "BAEW",
    "file" : "BASE_64 File",
    "formUuid" : "423cd7e3-5e60-414b-b022-acbee45d04cf",
    "releaseDate" : "30.12.2022",
    "finalCdok" : "65091",
    "version" : "V2"
  }, {
    "formcode" : "BAEW",
    "file" : "BASE_64 File",
```

```

    "formUUid" : "423cd7e3-5e60-414b-b022-acbee45d04cf",
    "releaseDate" : "30.12.2022",
    "finalCdok" : "65091",
    "version" : "V2"
  } ]
}

```

#### 12.11.4.16 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

#### 12.11.4.17 Responses

##### 12.11.4.17.1 200

When the DoR Rendering was successful, a 200 response is returned. [RenderDorResponse](#) (see page 0)

##### 12.11.4.17.2 400

This is returned, when one of the mandatory Headers or Form Data Parameters are missing. The issue can also occur, when the internal submit call was successful, but the reponse did not contain any document. Please provide the submitTransactionId for analysis by the UBS Forms Team. ([see page 165](#))

##### 12.11.4.17.3 500

The request failed due to an internal error. Please provide the submitTransactionId for analysis by the UBS Forms Team. ([see page 165](#))

([see page 165](#))

POST `/bin/com/ajila/forms/ubs/saveasdraft`

[Up](#) (see page 0)

Save Form Draft in CLP (**binComAjilaFormsUbsSaveasdraftPost**)

Save form data in CLP and returns CLP uuid when successful

#### 12.11.4.18 Consumes

This API call consumes the following media types via the Content-Type request header:

- `multipart/form-data`

#### 12.11.4.19 Form parameters

##### **formData (required)**

*Form Parameter –*

##### **formPath (required)**

*Form Parameter –*

#### 12.11.4.20 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `string`

#### 12.11.4.21 Responses

##### **12.11.4.21.1 200**

When draft is successfully saved, CLP UUID is returned. ([see page 165](#))

#### 12.11.4.22 Example data

Content-Type: string

```
9a5bc76f-48f4-4eec-9a36-953e5cda5e07
```

##### **12.11.4.22.1 403**

User not allowed to persist data as saveAsDraft flag is... ([see page 165](#))

##### **12.11.4.22.2 409**

Exception while saving or update the draft. ([see page 165](#))

### 12.11.5 RDS

([see page 165](#))

GET /app/LS4/bin/com/ajila/forms/ubs/rds/siap-subcategory

Up ([see page 0](#))

**SIAP Subcategory (appLS4BinComAjilaFormsUbsRdsSiapSubcategoryGet)**

This servlet calls RDS for a list of SIAP Sub-categories

**12.11.5.1 Query parameters****formMandator (required)**

*Query Parameter* – Mandator

**formLanguage (required)**

*Query Parameter* – Language of form

**12.11.5.2 Produces**

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

**12.11.5.3 Responses****12.11.5.3.1 200**

List of SIAP Sub-categories based on mandator and language ([see page 165](#))

**12.11.5.4 Example data**

Content-Type: application/json

```
[{"100=Casinos, betting or other gambling related activities","101=Defense,
arms or war materials related industries, private military services
providers","102=Precious metals professionals","103=Precious stones
professionals","104=Professional dealers in art, antiques and cultural
objects","105=Professional sports related industries","106=Extractive
industries: extraction in SCAP countries","107=Money remittance businesses
(regulated/licensed)","108=Bank note dealers outside the banking
sector","109=Digital assets related business","110=Cannabis related
business (CRB)","111=Religious persons/organizations","112=Sensitive
charities / Non-profit organizations (NPO)","116=Public construction
(government) in SCAP country","117=Commodity trading (energy, metals and
agricultural products)","118=Maritime shipping / transport"]
```

---

(see page 165)

GET /bin/com/ajila/forms/ubs/rds/country

Up (see page 0)

Returns list of countries populated by RDS (**binComAjilaFormsUbsRdsCountryGet**)

This servlet gets a list of countries from RDS based on the following parameters

### 12.11.5.5 Query parameters

**formMandator (required)**

*Query Parameter – Mandator*

**formLanguage (required)**

*Query Parameter – Language of form*

**countryType (required)**

*Query Parameter – Country Type specified for RDS to generate country list*

### 12.11.5.6 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

### 12.11.5.7 Responses

#### 12.11.5.7.1 200

List of countries (see page 165)

### 12.11.5.8 Example data

Content-Type: application/json

```
["104=Andorra","644=United Arab Emirates","601=Afghanistan","344=Antigua  
and Barbuda","343=Anguilla ..."]
```

---

(see page 165)

GET /bin/com/ajila/forms/ubs/rds/elsig

Up (see page 0)

Returns electronic signature eligibility of a form (**binComAjilaFormsUbsRdsElsigGet**)

This servlet makes a call to RDS to determine if a form is eligible for electronic signature

### 12.11.5.9 Query parameters

**formMandator (required)**

*Query Parameter* – Mandator

**formLanguage (required)**

*Query Parameter* – Language of form

**finalCdok (required)**

*Query Parameter* – CDOK of form. (Add a 0 to the front of the 5-digit CDOK in the request)

**formPath (required)**

*Query Parameter* – formPath of form

**subFormPath (required)**

*Query Parameter* – formPath of sub form (if available). if not, leave a '/' as the value of parameter

**isSubformActive (required)**

*Query Parameter* – boolean value to indicate if subform is active

### 12.11.5.10 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- plain text

### 12.11.5.11 Responses

#### 12.11.5.11.1 200

Text String to display electronic signature eligibility ([see page 165](#))

### 12.11.5.12 Example data

Content-Type: No Digital Signatures

Digital signatures are not possible for this document.

### 12.11.5.13 Example data

Content-Type: EACP



Help save paper by using eAcceptance+ to sign this document. The client can confirm this with the Access App or Access Card in their Digital Banking.

### 12.11.5.14 Example data

Content-Type: Default

We cannot currently determine whether this form can be signed digitally. Please try again later. Alternatively, print the form, sign it manually and send it to UBS by postal mail.

### 12.11.5.15 Example data

Content-Type: EAQS

Help save paper by using the qualified electronic signature (QES) for this document. The client can confirm this with the Access App or Access Card in their Digital Banking.

[\(see page 165\)](#)

GET /bin/com/ajila/forms/ubs/rds/nci-helperparams

[Up \(see page 0\)](#)

NCI Helper Params (**binComAjilaFormsUbsRdsNciHelperparamsGet**)

This servlet calls RDS for a list of NCI Helper Params

### 12.11.5.16 Query parameters

**formMandator (required)**

*Query Parameter* – Mandator

### 12.11.5.17 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

## 12.11.5.18 Responses

### 12.11.5.18.1 200

List of NCI Helper Params based on mandator [\(see page 165\)](#)

### 12.11.5.19 Example data

Content-Type: application/json

```
{
  "Nigéria": {
    "isoCode": "NG",
    "eeaMember": false
  },
  "Northern Cyprus": {
    "isoCode": "TR",
    "eeaMember": false
  },
  "Gana": {
    "isoCode": "GH",
    "eeaMember": false
  },
  "Solomon Islands": {
    "isoCode": "SB",
    "eeaMember": false
  }
}
```

[\(see page 165\)](#)

GET /bin/com/ajila/forms/ubs/rds/nci-identifier

[Up \(see page 0\)](#)

NCI Identifier (**binComAjilaFormsUbsRdsNciIdentifierGet**)

This servlet calls RDS for a list of NCI Identifier

## 12.11.5.20 Query parameters

### **formMandator (required)**

*Query Parameter – Mandator*

### **formLanguage (required)**

*Query Parameter – Language of form*

## 12.11.5.21 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

## 12.11.5.22 Responses

### 12.11.5.22.1 200

List of NCI Identifier based on mandator and language ([see page 165](#))

## 12.11.5.23 Example data

Content-Type: application/json

```
{
  "00": {
    "1": "National Passport Number",
    "2": "CONCAT"
  },
  "PS": {
    "1": "National Passport Number",
    "2": "CONCAT"
  },
  "PT": {
    "1": "Tax Number",
    "2": "National Passport Number",
    "3": "CONCAT"
  },
  "PY": {
    "1": "National Passport Number",
    "2": "CONCAT"
  },
  "QA": {
    "1": "National Passport Number",
    "2": "CONCAT"
  },
  "AD": {
    "1": "National Passport Number",
    "2": "CONCAT"
  }
}
```

## 12.11.6 Redirection

([see page 165](#))

GET /bin/api/affforms/external

Up ([see page 0](#))

External Redirection Resolver (**binApiAffformsExternalGet**)

Uses the redirect service for external

### 12.11.6.1 Query parameters

**mandator (required)**

*Query Parameter* – The 3 digits for the mandator

**formcode (required)**

*Query Parameter* – Form Code (identifier) of form

**afAcceptLang (required)**

*Query Parameter* – Language of form

### 12.11.6.2 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `text/html`

### 12.11.6.3 Responses

#### 12.11.6.3.1 302

Redirect ([see page 165](#))

#### 12.11.6.3.2 404

Not Found ([see page 165](#))

### 12.11.6.4 Example data

Content-Type: Multiple Forms

#### 12.11.6.5 Multiple Online Forms found for given Formcode.

For the given formcode AAAM, multiple adaptive forms are found in the repository.

### 12.11.6.6 Example data

Content-Type: No Online Form

#### 12.11.6.7 No Online Form found for given Formcode.

For the given formcode AMMM, no adaptive form was found in the repository.

([see page 165](#))

GET `/bin/api/afforms`

[Up \(see page 0\)](#)

#### Redirection Resolver (**binApiAfformsGet**)

The redirect service is used to have a centralized entry point to the adaptive forms by providing the business triple (formcode, mandator, language).

### 12.11.6.8 Query parameters

**mandator (required)**

*Query Parameter* – The 3 digits for the mandator

**formcode (required)**

*Query Parameter* – Form Code (identifier) of form

**afAcceptLang (required)**

*Query Parameter* – Language of form

### 12.11.6.9 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `text/html`

### 12.11.6.10 Responses

#### 12.11.6.10.1 302

Redirect ([see page 165](#))

#### 12.11.6.10.2 404

Not Found ([see page 165](#))

### 12.11.6.11 Example data

Content-Type: Multiple Forms

#### 12.11.6.12 Multiple Online Forms found for given Formcode.

For the given formcode AAAM, multiple adaptive forms are found in the repository.

### 12.11.6.13 Example data

Content-Type: No Online Form

#### 12.11.6.14 No Online Form found for given Formcode.

For the given formcode AMMM, no adaptive form was found in the repository.

---

([see page 165](#))

GET /bin/api/affforms/internal

[Up \(see page 0\)](#)**Internal Redirection Resolver (binApiAffformsInternalGet)**

Uses the redirect service but when it's internal, url is replaced with dev/cx/i4/r0/p0 - changed to the official name instead of the others

**12.11.6.15 Query parameters****mandator (required)**

*Query Parameter* – The 3 digits for the mandator

**formcode (required)**

*Query Parameter* – Form Code (identifier) of form

**afAcceptLang (required)**

*Query Parameter* – Language of form

**12.11.6.16 Produces**

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `text/html`

**12.11.6.17 Responses****12.11.6.17.1 302**

Redirect ([see page 165](#))

**12.11.6.17.2 404**

Not Found ([see page 165](#))

**12.11.6.18 Example data**

Content-Type: Multiple Forms

**12.11.6.19 Multiple Online Forms found for given Formcode.**

For the given formcode AAAM, multiple adaptive forms are found in the repository.

### 12.11.6.20 Example data

Content-Type: No Online Form

#### 12.11.6.21 No Online Form found for given Formcode.

For the given formcode AMMM, no adaptive form was found in the repository.

## 12.11.7 Retrieval

(see page 165)

GET /bin/api/xsd/getAdaptiveFormSchema

Up (see page 0)

Returns the flattened XSD Schema (**binApiXsdGetAdaptiveFormSchemaGet**)

Returns the flattened XSD Schema for the given Formcode

### 12.11.7.1 Query parameters

#### **formcode (required)**

*Query Parameter* – Form Code (identifier) of form

### 12.11.7.2 Return type

[FlattenedXSDFormResponse](#) (see page 0)

### 12.11.7.3 Example data

Content-Type: application/json

```
{
  "xsdFileReference" : "/content/dam/formsanddocuments/data-dictionary/
forms/AAAM.xsd",
  "exception" : false,
  "formcode" : "AAAZ",
```

```
"flattenedXsd" : "BASE64_XSD"
}
```

#### 12.11.7.4 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

#### 12.11.7.5 Responses

##### 12.11.7.5.1 200

The servlet always returns a 200. The body defines, if the flattening was successful or not. In case of a non successful flattening, the returned JSON contains the `errorMessage`.

[FlattenedXSDFormResponse](#) (see page 0)

#### 12.11.7.6 Example data

Content-Type: Failed Flattening

```
{ "formcode": "ABSD", "exception": true, "exceptionMessage": "Online Form has no
referenced XSD. Reference Path: /content/dam/formsanddocuments/
aforms_global_all/af_ab/AF_ABSD/jcr:content/metadata" }
```

#### 12.11.7.7 Example data

Content-Type: Form Code Not Found

```
{ "formcode": "ABSD", "exception": true, "exceptionMessage": "No Online Form
found for given Formcode." }
```

#### 12.11.7.8 Example data

Content-Type: Successful Flattening

```
{ "formcode": "AAAZ", "flattenedXsd": "BASE64_XSD", "xsdFileReference": "/
content/dam/formsanddocuments/data-dictionary/forms/
AAAM.xsd", "exception": false }
```



### 12.11.7.9 Example data

Content-Type: No Form Code Provided

```
{"exception":true,"exceptionMessage":"No Formcode was provided."}
```

#### 12.11.7.9.1 401

Unauthorized ([see page 165](#))

#### 12.11.7.9.2 500

Unexpected error while creating JSON Response ([see page 165](#))

([see page 165](#))

GET /bin/com/ajila/forms/ubs/authoringmetadata

Up ([see page 0](#))

Supported languages for forms (**binComAjilaFormsUbsAuthoringmetadataGet**)

Checks what languages form supports

### 12.11.7.10 Query parameters

#### **formPath (required)**

*Query Parameter* – formPath of form

### 12.11.7.11 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

### 12.11.7.12 Responses

#### 12.11.7.12.1 200

List of Supported Languages ([see page 165](#))

### 12.11.7.13 Example data

Content-Type: application/json

```
["DE", "EN", "IT", "FR", "ES"]
```

(see page 165)

GET /bin/com/ajila/forms/ubs/componentmetadata

Up (see page 0)

Retrieve all components used within one form (**binComAjilaFormsUbsComponentmetadataGet**)

The form component metadata service can be used, to retrieve all components used within one form. For each component, the configured metadata will be returned. The components that are taken into account are:

- guideTextBox
- guideNumericBox
- guideTelephone
- guideDatePicker
- guideDropDownList
- guideCheckBox
- guideRadioButton

### 12.11.7.14 Query parameters

#### **form (required)**

*Query Parameter* —

The complete path to the form, for example:

- /content/forms/af/ajila-forms-ubs/change-of-address

### 12.11.7.15 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

## 12.11.7.16 Responses

### 12.11.7.16.1 200

An array of form components with the following JSON structure: [\(see page 165\)](#)

### 12.11.7.17 Example data

Content-Type: application/json

```
[{"guideNodeClass":"guideNumericBox","jcrPrimaryType":"nt:unstructured","name":"NB_Counter_ACBY","slingResourceType":"ajila-forms-customers/ajila-forms-ubs/components/controls/numericbox","textIsRich":"true, true, true"}, {"autoFieldKeyWord":"name","guideNodeClass":"guideTextBox","jcrPrimaryType":"nt:unstructured","jcrTitle":"Format","name":"TXT_Format","slingResourceType":"ajila-forms-customers/ajila-forms-ubs/components/controls/textbox","textIsRich":"true, true, true"},"..."]
```

[\(see page 165\)](#)

GET /bin/com/ajila/forms/ubs/consistencycheck

Up (see page 0)

Identify adaptive forms with inconsistencies (**binComAjilaFormsUbsConsistencycheckGet**)

The consistency check was implemented to easily identify adaptive forms with inconsistencies including

- Fragment / Subform references that point to a non-existing fragment / subform
- Document of record templates that point to a non-existing template

## 12.11.7.18 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- application/json

## 12.11.7.19 Responses

### 12.11.7.19.1 200

A string array with inconsistent adaptive forms [\(see page 165\)](#)

### 12.11.7.20 Example data

Content-Type: application/json

```
["/content/forms/af/standalone/64/64647/64647_EN", "/content/forms/af/standalone/62/62308/62308_EN", "/content/forms/af/standalone/64/64279/64279_en", "/content/forms/af/affforms_global_integrated/AF_64/AF_64470_en"]
```

(see page 165)

GET /bin/com/ajila/forms/ubs/referenceresolver

Up (see page 0)

Retrieve all forms that have a reference on one or more paths provided

**(binComAjilaFormsUbsReferenceresolverGet)**

The reference resolver can be used to retrieve all forms that have a reference on one or more paths provided. The following paths can be searched for

- Fragment paths
  - (ex. /content/forms/af/affforms\_global\_fragmentlib/affrg\_global\_BankingRelationship)
- Document of record template paths
  - (ex. /content/dam/formsanddocuments/reference-dor-templates/ajila-forms-ubs/02\_forms/UBS\_Blank\_Address\_DoR.xdp)
- Component paths
  - Attention -The referenced path in the forms is not the complete path.
  - Example:
    - Complete CRX Path (will deliver no results): /apps/ajila-forms-customers/ajila-forms-ubs/components/controls/datepicker
    - Referenced paths (to be used) - ajila-forms-customers/ajila-forms-ubs/components/controls/datepicker"

### 12.11.7.21 Query parameters

#### **referencePaths (required)**

*Query Parameter* – The referencePaths query parameter can include multiple reference paths which need to be separated by a comma (,).

### 12.11.7.22 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

### 12.11.7.23 Responses

#### 12.11.7.23.1 200

A string array of forms with the references defined in the referencePaths [\(see page 165\)](#)

### 12.11.7.24 Example data

Content-Type: application/json

```
["/content/forms/af/affforms_global_all/af_ac/AF_ACAY/", "/content/forms/af/affforms_global_all/af_ac/AF_ACBK/", "/content/forms/af/affforms_global_all/af_ac/AF_ACBY/", "/content/forms/af/affforms_global_all/af_ac/AF_ACFF/"]
```

[\(see page 165\)](#)

GET `/bin/com/ajila/forms/ubs/translation`

[Up \(see page 0\)](#)

Returns list of all dictionaries (**binComAjilaFormsUbsTranslationGet**)

Get list of paths of dictionaries

### 12.11.7.25 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

### 12.11.7.26 Responses

#### 12.11.7.26.1 200

A string array of dictionary paths [\(see page 165\)](#)

### 12.11.7.27 Example data

Content-Type: application/json

```
["/content/forms/af/affforms_global_fragmentlib/
affrg_global_InternalBankUse_Dispatch_OUGPN_Date/jcr:content/
guideContainer/assets/dictionary, /content/forms/af/affforms_global_all/
af_ac/AF_ACZT/jcr:content/guideContainer/assets/dictionary, /content/forms/
af/affforms_global_all/af_ab/AF_ABB0/jcr:content/guideContainer/assets/
dictionary, /content/forms/af/affforms_global_all/af_ba/AF_BAJ0/jcr:content/
guideContainer/assets/dictionary, /content/forms/af/affforms_global_all/
af_aa/AF_AAVI/jcr:content/guideContainer/assets/dictionary, /content/forms/
af/affforms_global_all/af_ac/AF_ACCF/jcr:content/guideContainer/assets/
dictionary, ..."]
```

(see page 165)

GET /bin/com/ajila/forms/ubs/translations/htmlchecker

Up (see page 0)

Returns list of invalid HTML translations (**binComAjilaFormsUbsTranslationsHtmlcheckerGet**)

Returns list of invalid HTML translations

### 12.11.7.28 Query parameters

**formReference (required)**

*Query Parameter* – path reference of a form

**language (required)**

*Query Parameter* – language

### 12.11.7.29 Responses

#### 12.11.7.29.1 200

A string array of invalid HTML translations (see page 165)

(see page 165)

GET /bin/ubsforms/formfragment

[Up \(see page 0\)](#)List of fragments and xsds for form (**binUbsformsFormfragmentGet**)

Prints out a list of fragments and their xsds for the form given formcode and mandator

### 12.11.7.30 Query parameters

**mandator (required)***Query Parameter* – The 3 digits for the mandator**formcode (required)***Query Parameter* – Form Code (identifier) of form

### 12.11.7.31 Produces

This API call produces the following media types according to the Accept request header; the media type will be conveyed by the Content-Type response header.

- `application/json`

### 12.11.7.32 Responses

#### 12.11.7.32.1 200

Receive a list of fragments and xsds ([see page 165](#))

### 12.11.7.33 Example data

Content-Type: application/json

```
{
  "formCode": "ABZE",
  "formPath": "/content/forms/af/afforms_global_all/af_ab/AF_ABZE",
  "formXSDDef": "No XSD Assigned",
  "formXSDDefExists": false,
  "includedFragments": [
    {
      "componentName": "panel",
      "componentPath": "/content/forms/af/afforms_global_all/af_ab/AF_ABZE/jcr:content/guideContainer/rootPanel/items/panel1624433485924/items/panel",
      "componentFragment": "/content/dam/formsanddocuments/afforms_global_fragmentlib/affrg_global_Signature_AuthorizedRepresentative_CH",
      "componentFragmentExists": true,
      "componentXSDDef": "No XSD Assigned",
      "componentXSDDefExists": false,
      "fragmentInfo": []
    },
    {
      "componentName": "panel_copy_copy_copy",
      "componentPath": "/content/forms/af/
```

```

affforms_global_all/af_ab/AF_ABZE/jcr:content/guideContainer/rootPanel/
items/panel_1086389067/items/panel_copy_copy_copy","componentFragment":"/
content/dam/formsanddocuments/affforms_global_fragmentlib/
affrg_global_PersonalDetails_CompanyName_CH","componentFragmentExists":true
,"componentXSDDRef":"No XSD
Assigned","componentXSDDExists":false,"fragmentInfo":[]},
{"componentName":"panel","componentPath":"/content/forms/af/
affforms_global_all/af_ab/AF_ABZE/jcr:content/guideContainer/rootPanel/
items/panel_1086389067/items/panel","componentFragment":"/content/dam/
formsanddocuments/affforms_global_fragmentlib/
affrg_global_AddressBlock_CountryDD","componentFragmentExists":true,"compon
entXSDDRef":"No XSD Assigned","componentXSDDExists":false,"fragmentInfo":[]},
{"componentName":"panel_copy","componentPath":"/content/forms/af/
affforms_global_all/af_ab/AF_ABZE/jcr:content/guideContainer/rootPanel/
items/panel_copy","componentFragment":"/content/dam/formsanddocuments/
affforms_global_fragmentlib/
affrg_global_InternalBankUse_Text_OURef_Signature","componentFragmentExists
":true,"componentXSDDRef":"No XSD
Assigned","componentXSDDExists":false,"fragmentInfo":[]},
{"componentName":"panel_copy","componentPath":"/content/forms/af/
affforms_global_all/af_ab/AF_ABZE/jcr:content/guideContainer/rootPanel/
items/step_1/items/panel_copy","componentFragment":"/content/dam/
formsanddocuments/affforms_global_fragmentlib/
affrg_global_BankingRelationship","componentFragmentExists":true,"component
XSDDRef":"No XSD Assigned","componentXSDDExists":false,"fragmentInfo":[]}]}
```

#### 12.11.7.33.1 500

General Error Message:

- Example: can be due to no form found or multiple forms found

(see page 165)

#### 12.11.7.34 Example data

Content-Type: application/json

```
Error while fetching form information for inputted form code
```



## 12.11.8 SessionCheck

(see page 165)

POST /bin/com/ajila/forms/ubs/keepalive

Up (see page 0)

Keep Alive (**binComAjilaFormsUbsKeepalivePost**)

Checks if session is still active

### 12.11.8.1 Responses

#### 12.11.8.1.1 200

Receive an empty 200 response if session is still active (see page 165)

(see page 165)

GET /bin/com/ajila/forms/ubs/ping

Up (see page 0)

Session Check (**binComAjilaFormsUbsPingGet**)

Checks if session is still active

### 12.11.8.2 Responses

#### 12.11.8.2.1 200

Receive an empty 200 response if session is still active (see page 165)

## 12.11.8.3 Models

[ Jump to [Methods \(see page 0\)](#) ]

### 12.11.8.3.1 Table of Contents

1. CreateDraftResponse - [CreateDraftResponse \(see page 0\)](#)
2. FlattenedXSDFormResponse - [FlattenedXSDFormResponse \(see page 0\)](#)
3. FormComponents - [FormComponents \(see page 0\)](#)

4. `FormComponentsArray` - [FormComponentsArray](#) (see page 0)
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11. `fileUpload` - [FileUpload](#) (see page 0)
12. `printPartialFormData` - [Print Partial Form Data](#) (see page 0)
13. `ubs_dor_body` ([see page 0](#))

#### 12.11.8.3.2 `CreateDraftResponse` - [CreateDraftResponse](#)

Up

##### **draftTransactionId (optional)**

[String](#) (see page 0) Internal transactionId of UBS Forms for tracing (provided in success and failed case)  
*example: 6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610*

##### **exception (optional)**

[String](#) (see page 0) Indicates if there was an exception (also a given by the HTTP Response Code)  
*example: false*

##### **exceptionMessage (optional)**

[String](#) (see page 0) Additional information in error case.

##### **clpUuid (optional)**

[String](#) (see page 0) ClpUuid provided for draft that is successfully persisted in CLP & MOF  
*example: 6591bcd7-37b1-447b-9ea5-dfc08385c97a*

#### 12.11.8.3.3 `FlattenedXSDFormResponse` - [FlattenedXSDFormResponse](#)

Up

##### **formcode (optional)**

[String](#) (see page 0)  
*example: AAAZ*

##### **flattenedXsd (optional)**

[String](#) (see page 0)  
*example: BASE64\_XSD*

##### **xsdFileReference (optional)**

[String](#) (see page 0)  
*example: /content/dam/formsanddocuments/data-dictionary/forms/AAAM.xsd*

**exception (optional)**[Boolean](#) (see page 0)*example: false***12.11.8.3.4** FormComponents - [FormComponents](#)[Up](#)**autoFieldKeyword (optional)**[String](#) (see page 0)**guideNodeClass (optional)**[String](#) (see page 0)**jcrPrimaryType (optional)**[String](#) (see page 0)**jcrTitle (optional)**[String](#) (see page 0)**name (optional)**[String](#) (see page 0)**options (optional)**[array\[FormOptions\]](#) (see page 0)**slingResourceType (optional)**[String](#) (see page 0)**textIsRich (optional)**[String](#) (see page 0)**validatePictureClause (optional)**[String](#) (see page 0)**validatePictureClauseMessage (optional)**[String](#) (see page 0)**validationPatternType (optional)**[String](#) (see page 0)**12.11.8.3.5** FormComponentsArray - [FormComponentsArray](#)[Up](#)**12.11.8.3.6** FormData - [Form-Data](#)[Up](#)**formcode**[String](#) (see page 0)*example: AAAM*

**mandator**[String \(see page 0\)](#)*example: 1***language**[String \(see page 0\)](#)*example: EN***data**[String \(see page 0\)](#)*example: JVBERi0xLjUNCiW1tbW1DQoxIDAgb2JqDQo8PC9UeXBIL0NhdGFsb2cvUGFnZXMg***12.11.8.3.7** **FormObject** - [FormObject](#)[Up](#)**formUuid (optional)**[String \(see page 0\)](#)*example: 423cd7e3-5e60-414b-b022-acbee45d04cf***formcode (optional)**[String \(see page 0\)](#)*example: BAEW***finalCdok (optional)**[String \(see page 0\)](#)*example: 65091***releaseDate (optional)**[String \(see page 0\)](#)*example: 30.12.2022***version (optional)**[String \(see page 0\)](#)*example: V2***file (optional)**[String \(see page 0\)](#)*example: BASE\_64 File***12.11.8.3.8** **FormOptions** - [FormOptions](#)[Up](#)**key (optional)**[String \(see page 0\)](#)**value (optional)**[String \(see page 0\)](#)

### 12.11.8.3.9 `RenderDorResponse` - [RenderDorResponse](#)

Up

**submitTransactionUuid (optional)**

[String](#) (see page 0) Internal transactionId of UBS Forms for tracing (provided in success and failed case)  
*example: 6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610*

**exception (optional)**

[String](#) (see page 0) Indicates if there was an exception (also a given by the HTTP Response Code)  
*example: false*

**exceptionMessage (optional)**

[String](#) (see page 0) Additional information in error case.

**listSize (optional)**

[Integer](#) (see page 0) Indicates the number of documents generated  
*example: 1*

**forms (optional)**

[array\[FormObject\]](#) (see page 0) Array of documents containing the clpUuid and the base64 PDF Document

### 12.11.8.3.10 `RetrieveDorDataResponse` - [RetrieveDorDataResponse](#)

Up

**retrieveTransactionId (optional)**

[String](#) (see page 0) Internal transactionId of UBS Forms for tracing (provided in success and failed case)  
*example: 6fc4ed3d-dba9-4ae3-bf2c-7cde9c072610*

**exception (optional)**

[String](#) (see page 0) Indicates if there was an exception (also a given by the HTTP Response Code)  
*example: false*

**exceptionMessage (optional)**

[String](#) (see page 0) Additional information in error case.

**contentType (optional)**

[String](#) (see page 0) application/pdf or application/zip or empty if exception  
*example: application/pdf*

**formData (optional)**

[String](#) (see page 0) Base64 encoded string of Byte Array.  
*example: 6591bcd7-37b1-447b-9ea5-dfc08385c97a*

### 12.11.8.3.11 `SaveDraftFormData` - [Save Draft Form Data](#)

Up

**formData**

[String](#) (see page 0)  
*example: EEN*

**formPath**[String \(see page 0\)](#)*example: /content/forms/af/affforms\_global\_all/af\_ab/AF\_ABZE/jcr:content/guideContainer***12.11.8.3.12** `fileUpload` - [FileUpload](#)[Up](#)**file**[String \(see page 0\)](#)**runtimeLocale**[String \(see page 0\)](#)**formContainerPath**[String \(see page 0\)](#)**12.11.8.3.13** `printPartialFormData` - [Print Partial Form Data](#)[Up](#)**data**[String \(see page 0\)](#)*example: E...***locale**[String \(see page 0\)](#)*example: en-us***resource**[String \(see page 0\)](#)*example: /content/forms/af/affforms\_ch\_all/af\_aa/AF\_AAAM/jcr:content/guideContainer***12.11.8.3.14** `ubs_dor_body`[Up](#)**data (optional)**[String \(see page 0\)](#)*example: EEN001AFCXFA61391AAAM64044est testtest000000tetest 000000tetestest22222est**test[{"formcode":"AAAM","finalCdok":"64044","clpMandatory":""}]{ "isActive":false}**guide[0].guide1[0].guideRootPanel[0].PN\_Signatures[0]guideRootPanel[0].PN\_step\_1[0].PN\_Non\_US[0].txtMandator[0]**ue,guideRootPanel[0].PN\_step\_1[0].PN\_Non\_US[0].ST\_nonUS\_US\_001[0]=false,guideRootPanel[0].PN\_step\_1[0].PN\_Non\_US[0].ST\_nonUS\_US\_101[0]=true,guideRootPanel[0].PN\_Signatures[0].PN\_Signature[0].PN\_signature1[0].ST\_AH0[0]=fa***locale (optional)**[String \(see page 0\)](#)*example: en-us***resource (optional)**[String \(see page 0\)](#)

example: `/content/forms/af/afforms_ch_all/af_aa/AF_AAAM/jcr:content/guideContainer`

## 12.12 SoapUI and Postman afcomplib projects

The attached consist of the SoapUI and Postman project for afcomplib:



\*Do note that for several endpoints, there is a need to provide credentials for Basic Auth depending on the server or provide a valid CSIV2 token in the request header (for such, to test, you can copy the 'Cookie' from a request in the same environment through the normal open AF method). Some endpoints work differently depending on the type of AEM instance (author, publisher, processing).

| Servlet Name                | Endpoint                                       | Status  |
|-----------------------------|------------------------------------------------|---------|
| BrowserlessRenderingServlet | /bin/api/bdr/renderdor                         | SoapUI  |
| ComponentMetadataServlet    | /bin/com/ajila/forms/ubs/<br>componentmetadata | SoapUI  |
| ConsistencyCheckServlet     | /bin/com/ajila/forms/ubs/<br>consistencycheck  | SoapUI  |
| CreateDraftServlet          | /bin/api/bdr/createdraft                       | Postman |
| DorPreviewServlet           | /bin/com/ajila/forms/ubs/dor                   | SoapUI  |
| DownloadServlet             | /bin/com/ajila/forms/ubs/download              | SoapUI  |

|                                    |                                                       |         |
|------------------------------------|-------------------------------------------------------|---------|
| ExternalRedirectionResolverServlet | /bin/api/afforms/external                             | SoapUI  |
| FileServlet                        | /bin/com/ajila/forms/ubs/fileupload                   | SoapUI  |
| FormFragmentServlet                | /bin/ubsforms/formfragment                            | SoapUI  |
| HtmlTranslationCheckerServlet      | /bin/com/ajila/forms/ubs/translations/<br>htmlchecker | SoapUI  |
| InternalRedirectionResolverServlet | /bin/api/afforms/internal                             | SoapUI  |
| KeepAliveServlet                   | /bin/com/ajila/forms/ubs/keepalive                    | SoapUI  |
| RedirectionResolverServlet         | /bin/api/afforms                                      | SoapUI  |
| ReferenceResolverServlet           | /bin/com/ajila/forms/ubs/<br>referenceresolver        | SoapUI  |
| RetrieveDorServlet                 | /bin/api/afforms/retrievedor                          | Postman |
| RetrieveDorDataServlet             | /bin/api/bdr/retrievedordata                          | Postman |
| SaveAsDraftServlet                 | /bin/com/ajila/forms/ubs/saveasdraft                  | Postman |
| TranslationServlet                 | /bin/com/ajila/forms/ubs/translation                  | SoapUI  |
| XsdSchemaServlet                   | /bin/api/xsd/getAdaptiveFormSchema                    | SoapUI  |
| CountryListServlet                 | /bin/com/ajila/forms/ubs/rds/country                  | SoapUI  |
| ElectronicSignatureRdsServlet      | /bin/com/ajila/forms/ubs/rds/elsig                    | SoapUI  |
| NciHelperParamsServlet             | /bin/com/ajila/forms/ubs/rds/nci-<br>helperparams     | SoapUI  |
| NciIdentifierServlet               | /bin/com/ajila/forms/ubs/rds/nci-<br>identifier       | SoapUI  |



|                          |                                               |         |
|--------------------------|-----------------------------------------------|---------|
| SiapSubcategoryServlet   | /bin/com/ajila/forms/ubs/rds/siap-subcategory | SoapUI  |
| PrintPartialServlet      | /bin/com/ajila/forms/ubs/printpartial         | Postman |
| MetadataComponentServlet | /bin/com/ajila/forms/ubs/authoringmetadata    | SoapUI  |
| SessionCheck             | /bin/com/ajila/forms/ubs/ping                 | SoapUI  |

## 13 13) Known Issues

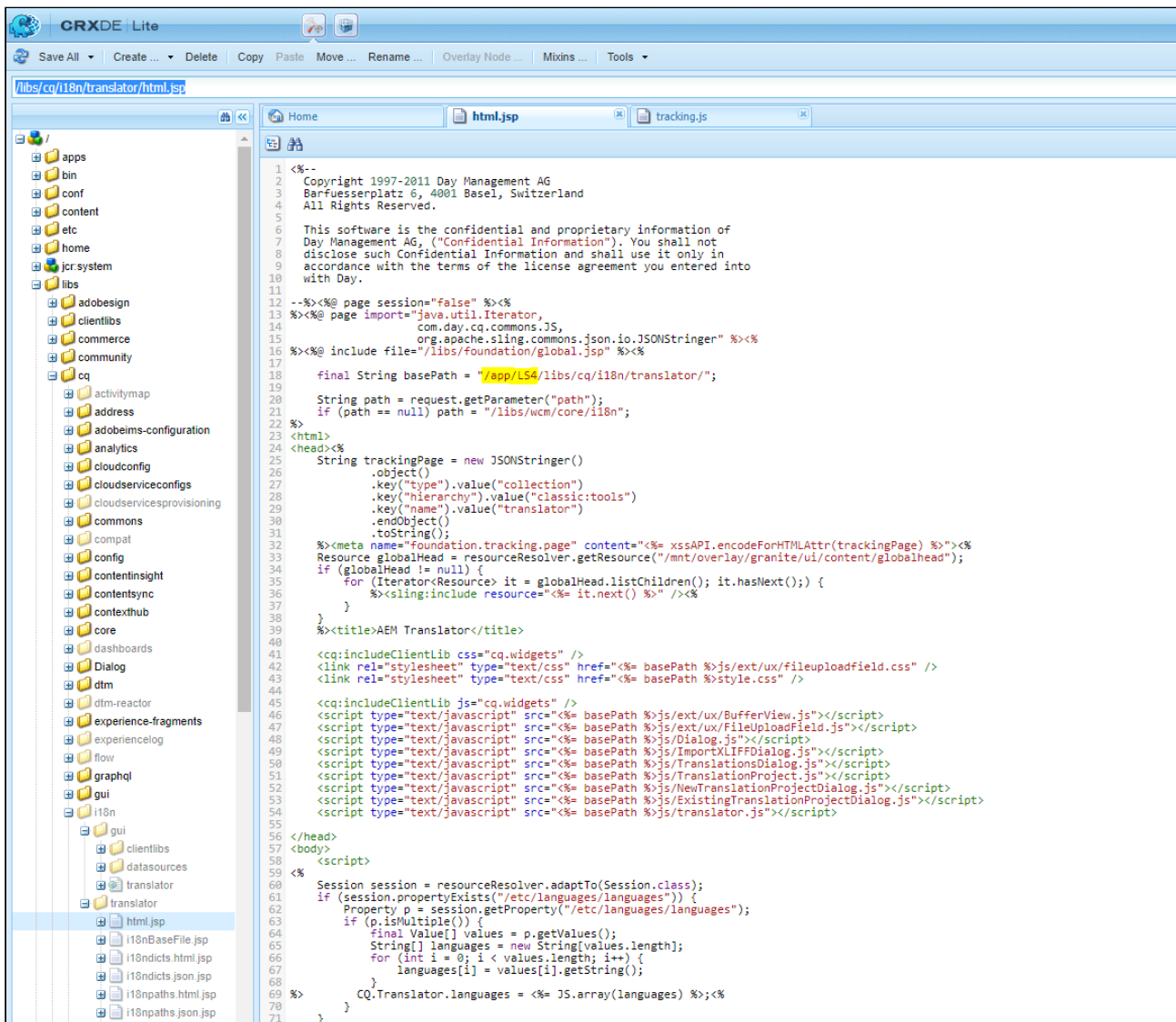
### 13.1 Translator not available (empty page)

**NOTE:** This issue will reoccur with every new AEM Forms installation / upgrade until the bug is fixed (the libs folder is overwritten by Adobe anytime).

With the added Application Context "/app/LS4", the following URL to the translator displayed a blank page:  
<https://a302-6991-0117.gdzd.ubs.net/app/LS4/libs/cq/i18n/translator.html>

This is a bug in the AEM Forms product. The internal JSP page doesn't consider the application context and is therefore unable to load the internal files. In order to fix this, the Application Context needs to be added to the following file in CRX:

/libs/cq/i18n/translator/html.jsp



## 13.2 Save and Cancel Button not visible

With service pack 6.5.5 there is an issue where you cannot save or cancel a configuration in the AEM authoring section. This issue will be fixed by Adobe with SP 6.5.6 which is released in September 2020. Meanwhile, follow this workaround:

1. Open the editor
2. On a component open the configuration mode
3. Change a property
4. Click on preview (top right)
5. An error message will appear like “changes need to be saved or cancelled”
6. The Buttons are coming back for the current sessions

## 13.3 Number of columns in content panel

In AEM 6.5 there is a problem with the number of columns and Colspan property on content panels. There is a fix for adaptive forms. But for form fragments, the configuration needs to be adjusted by the Ajila team. If there are fragments that need multiple columns, please contact Ajila.

A bug for this issue will also be reported to Adobe.

## 13.4 Dropdown list appears as Key instead of value for multi select

When dropdown list is used on DOR, it shows key instead of value which appears correctly on AF end. This is a known issue and reproducible on Adobe's template as well.

## 13.5 Translation

### 13.5.1 Translation not reflected for specific language

#### 13.5.1.1 Root Cause

In the normal process, a form designer adds the languages in the author instance via "Add Dictionary". This issue occurs when he misses out on a language and AEM doesn't create a dictionary entry for it. Lets assume an adaptive form needs the following three translation:

- DE-CH
- FR-CH
- IT-CH

The form designer forgets to add IT-CH in the process but requests it at ITT. ITT returns three language files which the form designer is importing. Since IT-CH doesn't have a pregenerated dictionary entry, AEM creates it on the fly. But this on-the-fly creation is not working properly. IT-CH will not work in that case.

The issue can be identified as follows:

1. Open CRX and navigate to the form (/content/forms/af/...
2. Display the language folder for the language that is not working. Example path:  
/content/forms/af/afforms\_global\_all/af\_ac/AF\_ACBH/jcr:content/guideContainer/assets/  
dictionary/de-ch

If the folder misses the attribute "sling:basename" and "jcr:primaryType" is "nt:folder" the translation does not work. A proper translation folder looks as follows:

| Properties |                 |        |                                                                                                 | Access Control | Replication | Console | Build Info |
|------------|-----------------|--------|-------------------------------------------------------------------------------------------------|----------------|-------------|---------|------------|
|            | Name ▲          | Type   | Value                                                                                           |                |             |         |            |
| 1          | jcr:created     | Date   | 2021-08-27T09:57:09.702+02:00                                                                   |                |             |         |            |
| 2          | jcr:createdBy   | String | translation-job-service                                                                         |                |             |         |            |
| 3          | jcr:language    | String | de-ch                                                                                           |                |             |         |            |
| 4          | jcr:mixinTypes  | Name[] | mix:language                                                                                    |                |             |         |            |
| 5          | jcr:primaryType | Name   | sling:Folder                                                                                    |                |             |         |            |
| 6          | sling:basename  | String | /content/forms/af/afforms_global_all/af_ac/AF_ACBH/jcr:content/guideContainer/assets/dictionary |                |             |         |            |

### 13.5.1.2 Solution

1. Delete the corrupted folder in CRX (for example the it-ch folder)
2. Click "Save All"
3. Switch to the Author Node
4. "Add Dictionary" again with the language (it-ch for example)
5. Open "Add Dictionary" again and check → If it didn't work, remove all other language and add the missing again. AEM will not really delete the other languages.
6. Reimport the XLIFF

## 14 14) Things you shouldn't do

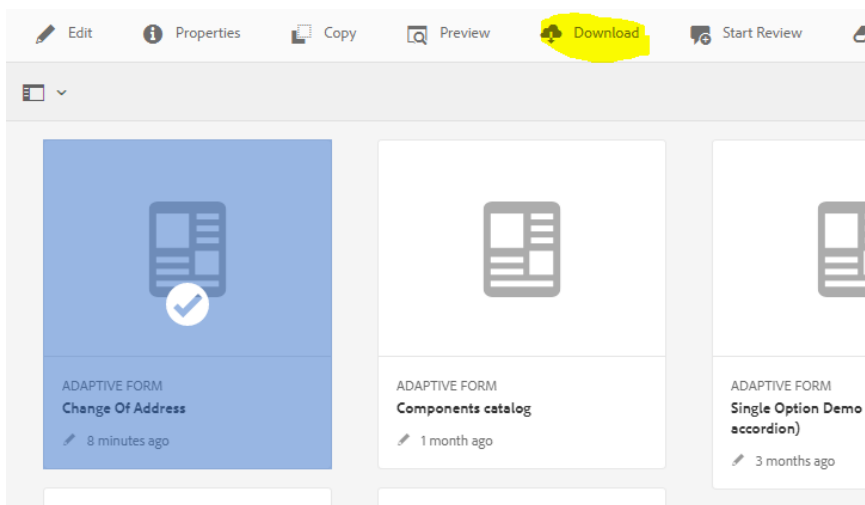
This chapter is about thing you shouldn't do in the AEM Forms

### 14.1 General topics

| No | What                                                                                                        | Why                                                                                                                                                                                          |
|----|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Do not delete the toolbar and its buttons                                                                   | The Toolbar and its related button are initialized by default while creating the form. Deleting the Toolbar or some of his button can cause unexpected behaviors and makes the form unusable |
| 2  | Do not delete the Error Text component                                                                      | As like the Toolbar the Error Text component is initialized by default while creating the form. Deleting the error text component can make the submit button unusable                        |
| 3  | Generally do not change the properties from the default initialized components                              | Changing the properties as example of the Next Button with "Hide Object" can cause to unexpected behavior.                                                                                   |
| 4  | Same name for multiple properties. The "name" attribute must be unique inside an adaptive form or form set. | AEM uses this name to generate the XML when submitting. Two with the same name can end up with none of them inside the XML.                                                                  |

### 14.2 Download / Upload adaptive forms

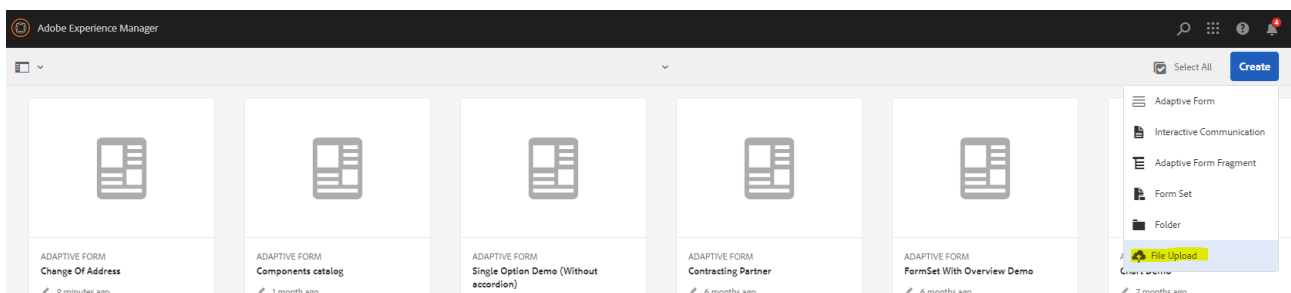
In AEM Forms, you can download and upload adaptive forms. In order to download a form, you can select it and then start the download:



The download of a ZIP starts that contains everything that is referenced by this form. Such as:

- The form itself
- All referenced fragments
- Referenced DoR templates
- ...

If you upload this ZIP again via "File Upload", all of these referenced fragments, ... will be uploaded again and overwrite the existing ones:



### Potential risk

The ZIP of an adaptive form with all its fragments, etc. is stored locally. When this ZIP is uploaded a month later for example, it will overwrite all the work in the fragments, etc. that was done in that time period.

Therefore uploading an adaptive form needs to be done with caution. Check in the upload dialog which components will be overwritten and only upload when you're 100% sure what you do.

## 15 15) Design deviations

1. Design deviations driven by CDD impact (automated processing in DPE) to specific CDOKs:

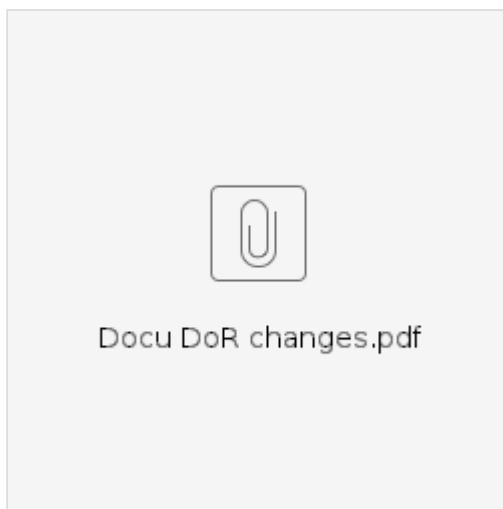
| CDOK  | Deviation                                                                                                                                              | Reason / comments                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 64925 | (potentially changed to AF code; no impact to DoR: "Contracting partner" field should not contain the address; potentially info box to be implemented) | 04.05.21 - In progress - discussions ongoing |
|       |                                                                                                                                                        |                                              |
|       |                                                                                                                                                        |                                              |



## 16 16) Design Guidelines for Adaptive Forms - Document of record (IT Forms team)

### 1. Adaptive Forms - Document of Record

The below design guidelines were prepared to define the rules for the adaptive forms implementation (versus classic forms implementation).



### 2. AF-Design Differences

Management Summary (author: Alex Wyss)

This Report shows pragmatically how certain elements will differ in the generated pdf of an adaptive form compared to the original pdf available in e-forms or generated by LoBs (Line of Business Applications). It includes constraints where AF cannot reproduce the existing pdf 1:1, elements which provide more flexibility in the pdf generation and on top shows the capabilities of validation that can be performed in the adaptive form.



## 16.1 APAC Standards (030, 046)

- Include the APAC Information text fragment on the bottom of the Step 1 only visible on AF. ([EFORMS-16589](#)<sup>17</sup>)
- Include the APAC Footnote fragment on the bottom of the Last step only visible on DoR. ([EFORMS-16534](#)<sup>18</sup>)
- For the APAC mandator forms, please use the APAC specific fragments only.
- Maximum number of instance for a repeating panel is 6 (Unless there is special request)
- Please use the UP section fragment specific to APAC only. ([EFORMS-14404](#)<sup>19</sup>)

## 16.2 CH Standards (001, 101)

- Maximum number of instance for a repeating panel is 4 (Unless there is special request)
- Please use the UP section fragment specific to CH only. ([EFORMS-6364](#)<sup>20</sup>)

<sup>17</sup> <https://sdic-agile.swissbank.com/jira/browse/EFORMS-16589>

<sup>18</sup> <https://sdic-agile.swissbank.com/jira/browse/EFORMS-16534>

<sup>19</sup> <https://sdic-agile.swissbank.com/jira/browse/EFORMS-14404>

<sup>20</sup> <https://sdic-agile.swissbank.com/jira/browse/EFORMS-6364>

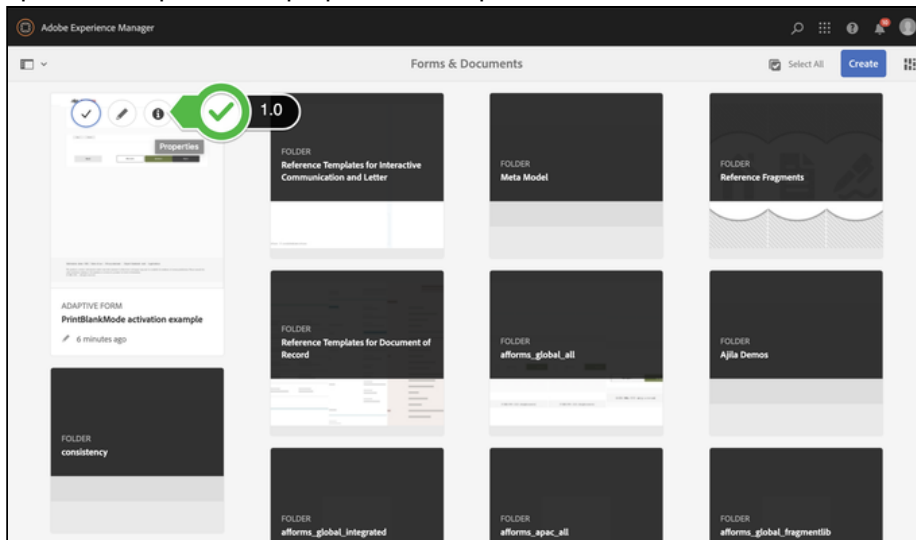
## 17 17) Incomplete Form Mode

Print Partial is used for preparing forms for downloading as a PDF.

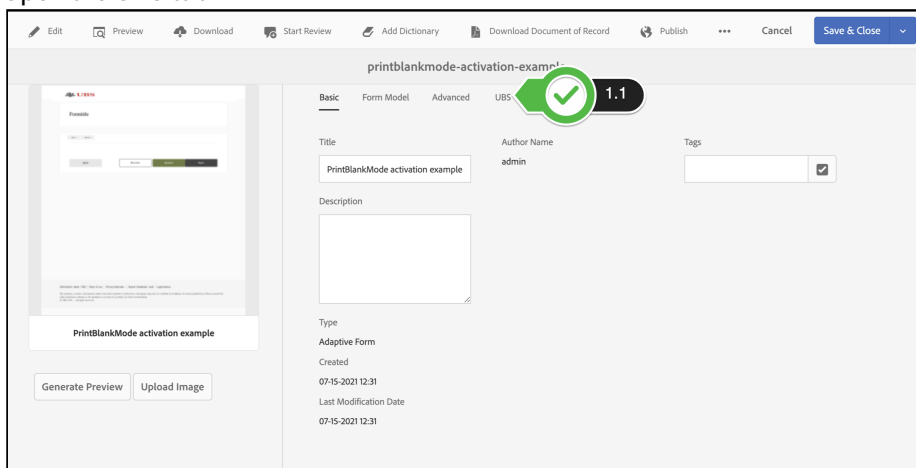
### 17.1 General Activation

For activating the "Print Partial" functionality in general the following steps have to be done:

- Open the adaptive form properties of a specific form



- Open the UBS tab



- Choose the "print partially" option

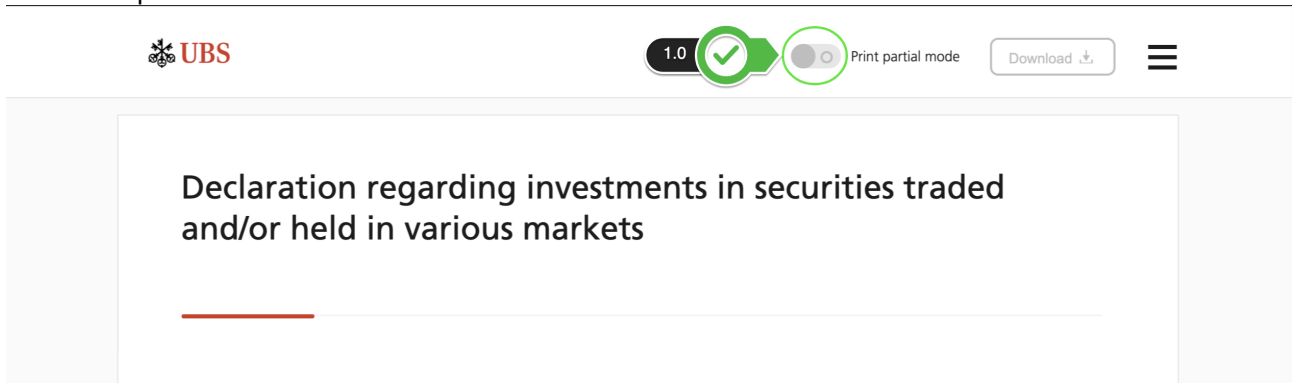
The screenshot shows the UBS Forms editor interface for a form titled "printblankmode-activation-example". The interface includes a top toolbar with buttons for Edit, Preview, Download, Start Review, Add Dictionary, Download Document of Record, Publish, Cancel, and Save & Close. The main workspace is divided into two panes. The left pane displays a preview of the form, which includes a "PrintBlankMode activation example" section. The right pane shows the form's configuration, with tabs for Basic, Form Model, Advanced, and UBS. The "Print partially" dropdown menu is open, showing options for "Active" and "Inactive". A green checkmark icon and a "1.2" label are overlaid on the "Active" option, indicating it is the selected choice.

- Save and close

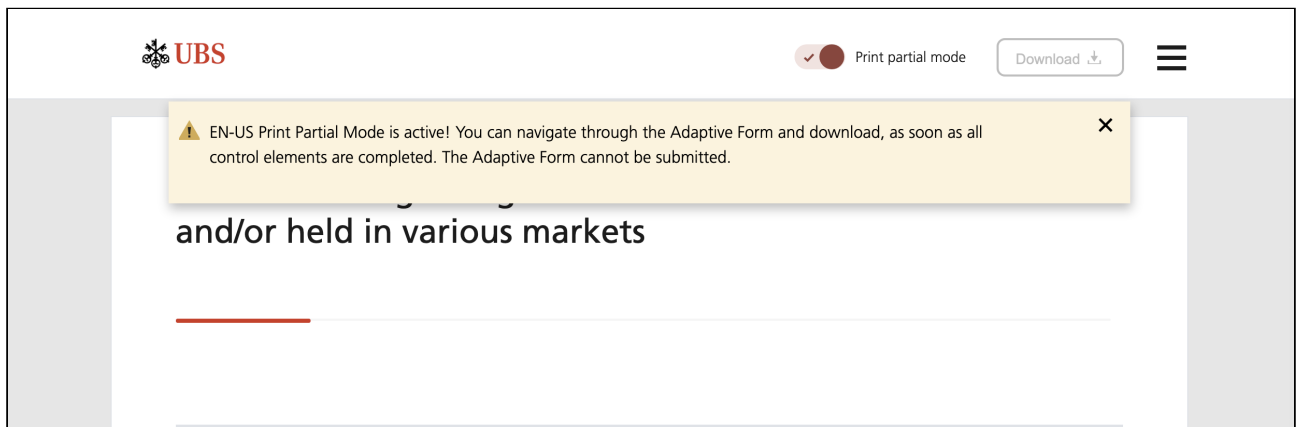
The screenshot shows the same UBS Forms editor interface as the previous one, but with the "Print partially" dropdown menu closed. The "Active" option is now the only visible option in the dropdown. A green checkmark icon and a "1.3" label are overlaid on the "Save & Close" button in the top toolbar, indicating it is the next step in the process.

## 17.2 Activate in Adaptive Form

If the Adaptive Form has the "print partially" configuration activated, the header of the adaptive form looks different in preview.



If the toggle "Print partial mode" is clicked by the user, the mode will be activated and a message is shown



The opposite happens if the toggle is clicked again. The mode will be deactivated and a message is shown.

UBS

Print partial mode

Download

EN-US Print Partial Mode is deactivated! The Adaptive Form can be submitted.

### Declaration regarding investments in securities traded and/or held in various markets

## 17.3 Scripts

### 17.3.1 Print Partial Active / Inactive

With the following script, a form designer can check whether print partial is active on the adaptive form or not. It can be used to execute specific logic in print partial mode or to skip it.

```
window.forms.ubs.printpartial.isPrintPartialActive()
```

### 17.3.2 Mandatory fields script

If we need to make the fields as mandatory dynamically, we should use below script:

```
window.forms.ubs.components.setMandatory(this, true);
```

### 17.3.3 Defining control elements

A control element is an input field which has to be filled before the PDF can be downloaded. To define a control element the following line of code can be added to the initial event in rule editor of a component:

```
// mark as control element
window.forms.ubs.printpartial.setControlElement(this);

// remove as control element
window.forms.ubs.printpartial.removeControlElement(this);
```

## 17.3.4 Events

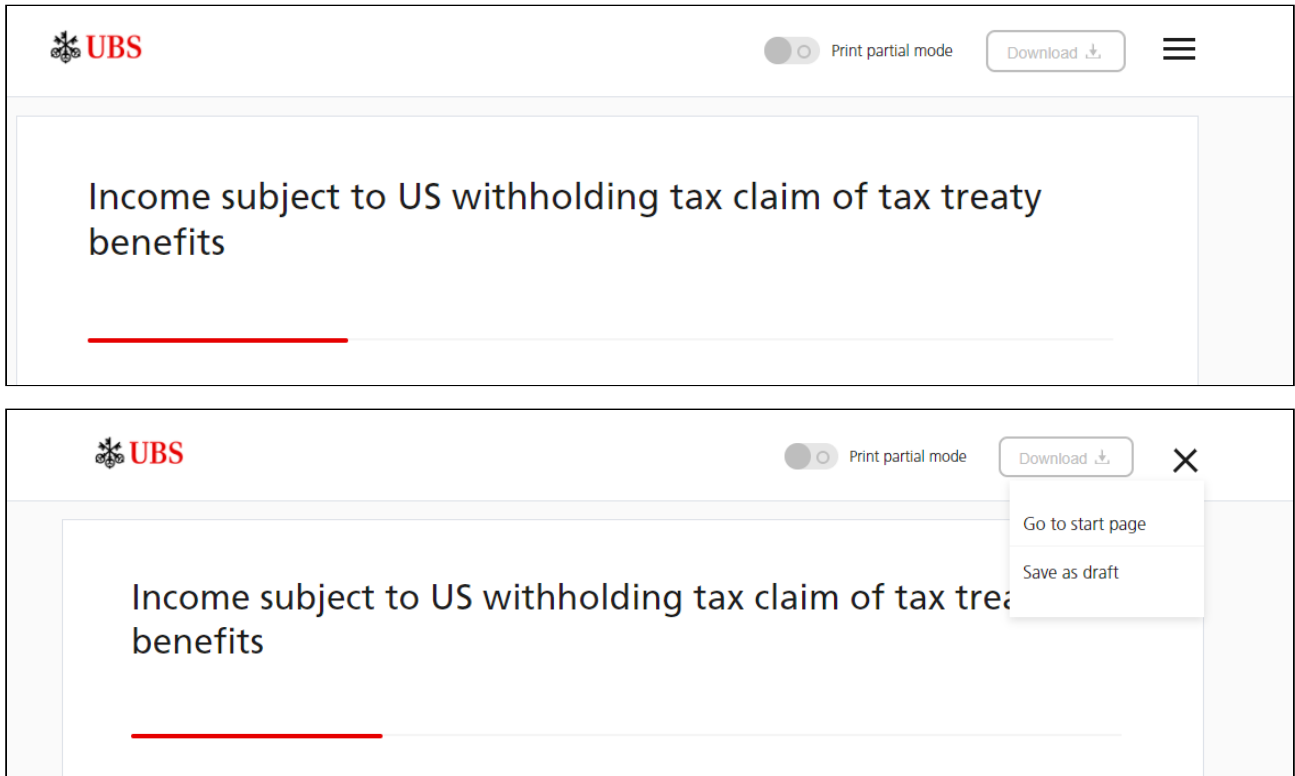
### 17.3.4.1 Activation / Deactivation

With the following code snippet, the form designer can add specific logic on activation or deactivation of the print partial mode. It can be placed on a initialize event of a component for example to register the callback.

```
document.addEventListener('printPartialChange', function(printPartial) {  
  try {  
    if (printPartial.active) {  
      window.forms.ubs.setFinalCdok('AAAA', '12345');  
    } else {  
      window.forms.ubs.setFinalCdok('AAAA', '54321');  
    }  
  } catch (err) {  
    console.error(err);  
  }  
});
```

## 18 18) Menu (Hamburger)

The adaptive forms have a hamburger menu which contains functionalities that the user can execute:



### 18.1 Configuration

Open the adaptive form properties and switch to the "UBS" section. In the field "Menu Options", the available features are listed and can be selected by the form designer. Every selected feature will be listed below the input field.

- If no feature is selected, the hamburger menu will not be shown in the adaptive form
- If one or more features are selected, the hamburger menu is shown in the adaptive form



Basic

Form Model

Advanced

UBS

Healthcheck Form

Select Option

▼

Print partially

Active

▼

Menu Options

Select Options

▼

×

Go to start page

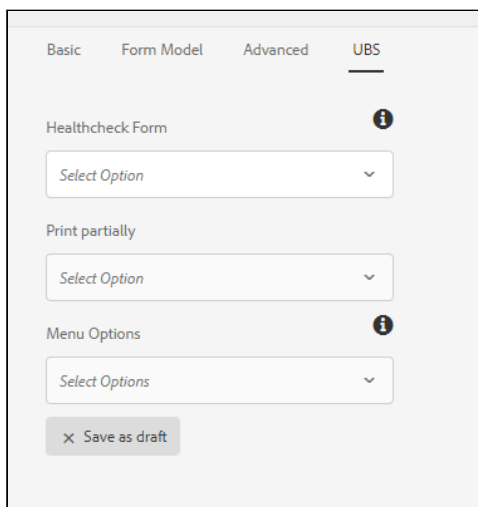
×

Save as draft

## 19 19) Save as Draft

### 19.1 Activation

In the properties of the Adaptive Form the menu option "Save as draft" must be activated. For further information on the Online Forms Menu, please visit the following documentation: [18\) Menu \(Hamburger\)](#) (see [page 224](#))



The screenshot shows the 'UBS' tab in the properties panel of an Adaptive Form. It contains three dropdown menus: 'Healthcheck Form' (with 'Select Option' selected), 'Print partially' (with 'Select Option' selected), and 'Menu Options' (with 'Select Options' selected). Each dropdown has an information icon (i) to its right. At the bottom of the panel, there is a button labeled 'x Save as draft'.

### 19.2 Online Form Behavior

#### 19.2.1 Menu option

When the Online Form has the "Save as draft" menu option active, the hamburger menu contains the corresponding entry. The user can execute this at every step of the Online Form Journey.

The screenshot shows the top of a UBS online form. The UBS logo is in the top left. In the top right, there is a toggle switch labeled 'Incomplete form mode' which is currently turned off. Next to it is a 'Download' button with a download icon. A 'Save as draft' button is also visible, which is the focus of the section. The main content area has the title 'Leasing of a safe-deposit box' and a progress bar below it.

### 19.2.2 Saving a Draft

Upon clicking the "Save as draft" entry, the user is prompted with the draft dialog:

The 'Save as draft' dialog box has a title bar with the text 'Save as draft' and a close button (X) in the top right corner. Below the title bar is a text input field labeled 'Draft Title'. At the bottom right of the dialog are two buttons: 'Cancel' and 'Save'.

The "Draft Title" will be stored in MyOnlineForms along with the form data. The title is currently restricted to 50 characters. By clicking on "Save":

1. The formData is stored in CLP and CLP generates a UUID. There is no PDF involved at this point.
2. The retrieved UUID from CLP is used to store the draft in MyOnlineForms.

From that point on, the user can see the draft in MyOnlineForms right away. The user gets a notification in the dialog box, whether the saving was successful or not.

### 19.2.3 Updating a Draft

When reopening a draft, the data is retrieved from CLP with the given UUID. The form is then prefilled with that data. When the user reaches the Online Form, the data is already there. At any point in the Online Form Journey, the user can "Save" the draft again. In case that there is already a UUID, no Draft Title Dialog is prompted. The data is updated right away in CLP.

MyOnlineForms is also updated with the new timestamp and eventually with an updated CDOK.

## 19.2.4 Submit

When an Online Form is submitted, the following actions are executed:

1. The generated PDF is stored under the draft UUID in CLP (with status "Final")
2. The MyOnlineForms entry is also updated (with Status "Final")

## 19.3 Form Designer Information

### 19.3.1 Get Draft Status of Form

With the following code snippet, a form designer can obtain whether the form is based on a draft or not. This means it's either opened from a draft, or the draft was created already in the current session.

```
window.forms.ubs.saveasdraft.isActiveDraft();
```

## 19.4 Steps to make the form ready for Save Draft

After activating the Save draft feature under the form properties, we need to check and update the existing scripts on the forms that are causing the data lose in the Draft form.

Scripts written in the Initialize event will trigger when we reopen the Draft form. So, we need to handle If any scripts written to hide panels/objects in the Initialize event.

**Please do not make any changes to the Form layout or design, it will cause the translation issues.**

### 19.4.1 Update the Print Blank Script

Please add this line of code in initialize event to the existing code to avoid executing the script in Draft form →

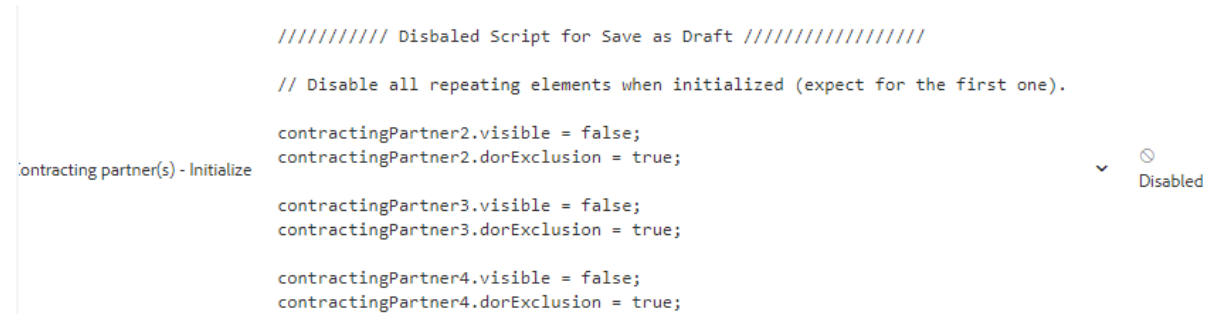
```
if(!window.forms.ubs.saveasdraft.isActiveDraft()) {
    TXT_FormCode.value = "XXXX";
    TXT_CDOK.value = "XXXXX";
}
```

### 19.4.2 Static repeatable panels

- If form contains the static repeating panels approach, we have to modify the scripts written in Initialize events to hide the additional instances. Otherwise, when draft is re-opened then Initialize event script will trigger and all additional instance will be set to hide in both AF and DoR which will

lead to data loss in draft form.

Remove if there is any script written in Initialize event on the parent/any panel to hide the additional instances



- To hide the additional instances, please configure the panel properties. Hide the panels from Adaptive form and Exclude from DoR.

- Insert a numeric box inside the static repeating parent panel as first component. This will be used to hold the instance count of static repeating panels.  
Hide the numeric box from both AF & DoR.  
Please name the numeric box as "NB\_Counter\_XXXX" (XXXX → Form code) and set the default value as 1.
- Increase the Numeric box value upon clicking the Add button, insert the below script in Click event of Add button.

```
NB_Counter_XXXX.value += 1;
```

- Decrease the Numeric box value upon clicking the Remove button, insert the below script in Click event of Remove button.

```
NB_Counter_XXXX.value -= 1;
```

- Based on the NB\_Counter\_XXXX value, show the required number of panels, add button and remove button. Write the script on the Add button Initialize event. (Please see example below)

☐ Add partner - Initialize

```

////////// Script for Save as Draft //////////

if (NB_Counter_ABHM.value == 1) {
    contractingPartner1.visible = true;
    contractingPartner1.dorExclusion = false;
} else if (NB_Counter_ABHM.value == 2) {
    contractingPartner1.visible = true;
    contractingPartner1.dorExclusion = false;
    contractingPartner2.visible = true;
    contractingPartner2.dorExclusion = false;
} else if (NB_Counter_ABHM.value == 3) {
    contractingPartner1.visible = true;
    contractingPartner1.dorExclusion = false;
    contractingPartner2.visible = true;
    contractingPartner2.dorExclusion = false;
    contractingPartner3.visible = true;
    contractingPartner3.dorExclusion = false;
    contractingPartner2.content.removebuttonPartner2.visible = false;
} else if (NB_Counter_ABHM.value == 4) {
    contractingPartner1.visible = true;
    contractingPartner1.dorExclusion = false;
    contractingPartner2.visible = true;
    contractingPartner2.dorExclusion = false;
    contractingPartner3.visible = true;
    contractingPartner3.dorExclusion = false;
    contractingPartner2.content.removebuttonPartner2.visible = false;
    contractingPartner4.visible = true;
    contractingPartner4.dorExclusion = false;
    contractingPartner3.content.removebuttonPartner3.visible = false;
}

```

- Handle the visibility of the Add, Remove button in Initialize event as well apart from Click event. (Please see example below)

☐ Add partner - Initialize

```

////////// Script for Save as Draft //////////

if (contractingPartners.contractingPartner4.visible === true) {
    this.visible = false;
} else {
    this.visible = true;
}

```

- If Signature section has dependency on this static repeating panel, please check the script and update accordingly.
- While testing the form, please check the AF and DoR for each instance.

### 19.4.3 Dynamic repeatable panels

As there is no issue of data loss with Dynamic repeating panels, we only need to handle the Add and remove buttons visibility.

Please add the script for the Add and remove button visibility in Initialize event as well apart from Click event.

#### Example code:

☐ Add Email address - Initialize

```

var Email_Len = PH_Delete_Email_Address_DYN.InstanceManager.instances.length;

for(var i = 0; i < Email_Len; i++){
    PH_Delete_Email_Address_DYN.InstanceManager.instances[i].BT_Remove_Delete_Email_Address.visible = (i == (Email_Len-1) && Email_Len > 1) ? true : false;
    PH_Delete_Email_Address_DYN.InstanceManager.instances[i].BT_Add_Delete_Email_Address.visible = (i == (Email_Len-1) && Email_Len < 6) ? true : false;
}

```

☐ Add Email address - Click

```

var Email_Len = PH_Delete_Email_Address_DYN.InstanceManager.instances.length;

for(var i = 0; i < Email_Len; i++){
    PH_Delete_Email_Address_DYN.InstanceManager.instances[i].BT_Remove_Delete_Email_Address.visible = (i == (Email_Len-1) && Email_Len > 1) ? true : false;
    PH_Delete_Email_Address_DYN.InstanceManager.instances[i].BT_Add_Delete_Email_Address.visible = (i == (Email_Len-1) && Email_Len < 6) ? true : false;
}

```

#### 19.4.4 APAC (030/046) forms with SC & TC translations

If the Form has SC & TC languages, while testing the form please also test the radio button group in the preview section (last step) whether selected radio button value is there or missing.

#### 19.4.5 Other Scenarios

Depending upon the form requirement we have to verify the script and if needed modify according to Save as Draft.

#### 19.4.6 Disable the Print incomplete feature (Based on the instruction/requirement)

If form is already enabled with Print incomplete mode, please disable and remove the scripts related to Print incomplete like setMandatory and setControlElement.

```

window.forms.ubs.components.setMandatory(this, true); change this script to ->
xxxxxxx_xxxxx.mandatory = true;

window.forms.ubs.printpartial.setControlElement(this); change this script to ->
xxxxxxx_xxxxx.mandatory = true;

```

#### 19.4.7 Fragments incorporated with Save as Draft script

Please update the below list upon implementing the Save as Draft on new fragments. If you have the below listed fragments in your form, no additional changes required.

| Fragment name | Folder |
|---------------|--------|
| EFORMS-21778  | APAC   |
| EFORMS-28967  | APAC   |
| EFORMS-21537  | APAC   |
| EFORMS-14914  | APAC   |
| EFORMS-22437  | APAC   |

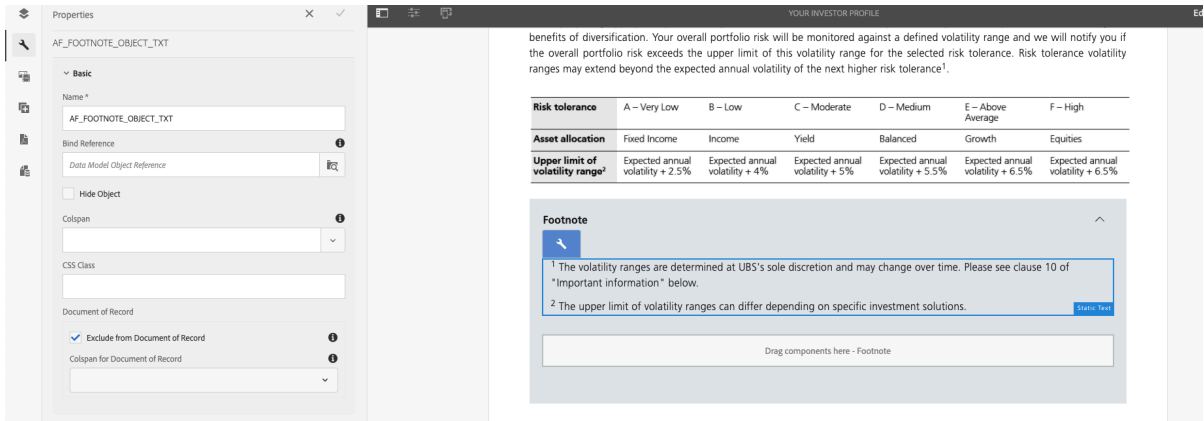
|              |        |
|--------------|--------|
| EFORMS-30401 | Global |
| EFORMS-21902 | APAC   |
| EFORMS-34259 | Global |
| EFORMS-24223 | Global |
| EFORMS-61783 | APAC   |



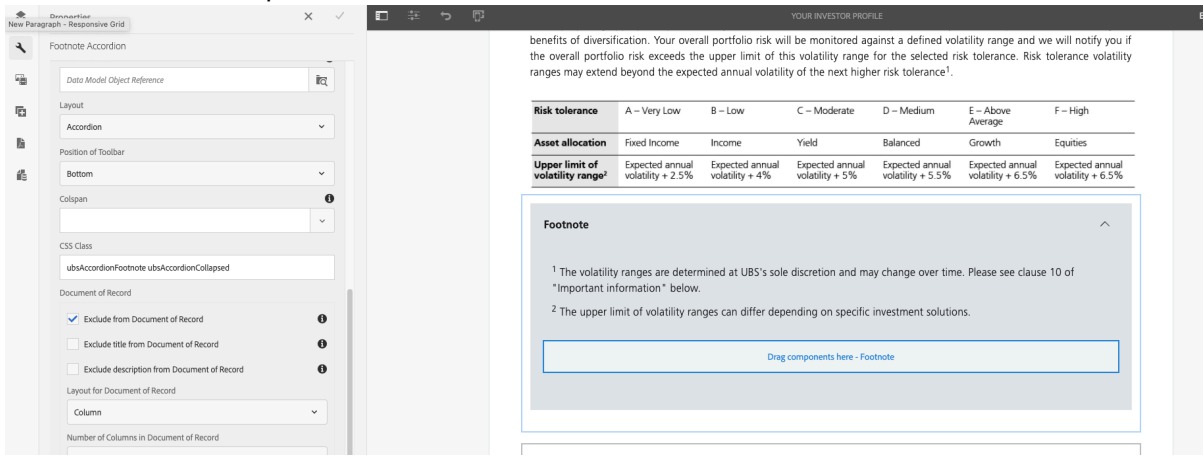
## 20 20) Footnotes

For displaying footnotes in DoR the DoR template "UBS\_Blank\_DoR\_Footnotes.xdp" has to be used.  
In the adaptive form the following steps has to be done:

1. All the static text objects, which contains footnotes, need to have the name "AF\_FOOTNOTE\_OBJECT\_TXT"



2. Exclude the footnotes panel from Document of Record



## 21 21) ELSIG eligibility check

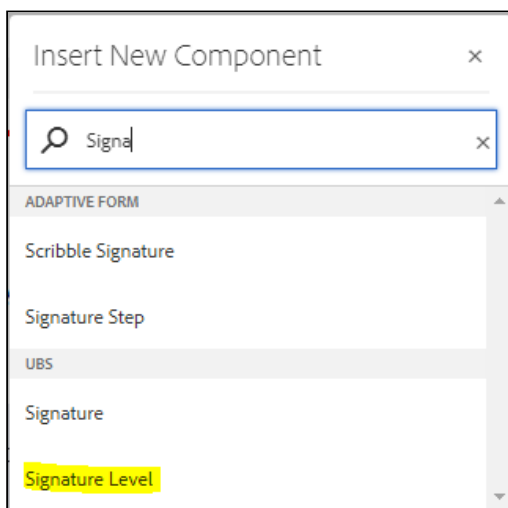
With the new component "Signature Level" (see below) you are now able to check if the form is ready for electronic signature.

There are 3 different states with 3 different resulting messages.

| State | Description                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| None  | Electronic signature cannot be used for this document                                                                                                                       |
| EACC  | Help reduce paper waste: The simple electronic signature can be used for this document (i.e., confirmation by clicking "Accept" in UBS Digital Banking)                     |
| EAQS  | Help reduce paper waste: The qualified electronic signature can be used for this document (i.e., confirmation in UBS Digital Banking and via the Access App or Access Card) |

### 21.1 Configuration

The E-Signature information can be added as component to the Online Form:



This "Signature Level" component adds an information text box to the form. In this section, the information whether the Online Form is E-Signature eligible be displayed.

**Important:** This component should be placed on the preview panel. There might be Online Forms, which have a dynamic CDOK. Only at the end of the form, the final CDOK is known.

## 22 22) Translation of the entity name

According to ITT "UBS Switzerland AG" should not be translated to other languages

[FW Wrong translations for UBS Switzerland AG.msg](#)<sup>21</sup>



| Deutsch            | Englisch           | Französisch        | Italienisch        | Spanisch           |
|--------------------|--------------------|--------------------|--------------------|--------------------|
| UBS AG             | UBS AG             | UBS SA             | UBS SA             | UBS SA             |
| UBS Group AG       | UBS Group AG       | UBS Group SA       | UBS Group SA       | UBS Group SA       |
| UBS Switzerland AG | UBS Switzerland AG | UBS Switzerland AG | UBS Switzerland AG | UBS Switzerland AG |

<sup>21</sup> <https://central-confluence.ubs.net/confluence/download/attachments/492786530/FW%20Wrong%20translations%20for%20UBS%20Switzerland%20AG.msg?api=v2&modificationDate=1645452807000&version=1>

## 23 23) AF Security

### 23.1 Background

In the first phase of the project, the Online Forms were not protected and only available to internal users. In the future, Online Forms will be exposed to:

- GFIM Users
- E-Banking Users
- ...

Therefore, the security feature for Online Forms was introduced. In addition, internal users will also be known via ISSO.

### 23.2 Technical Information

#### 23.2.1 Users

In an out of the box setting of AEM Forms, the Publisher is not protected. When no authentication is implemented, the user gets logged in as "Anonymous" user and has access to the Online Form. In order to restrict the access, this user is restricted to access the forms CRX paths (/content/forms/af and /content/dam/formsanddocuments).

This configuration is part of the "afcomplib\_all". When installed to an instance, this restriction takes immediate effect.

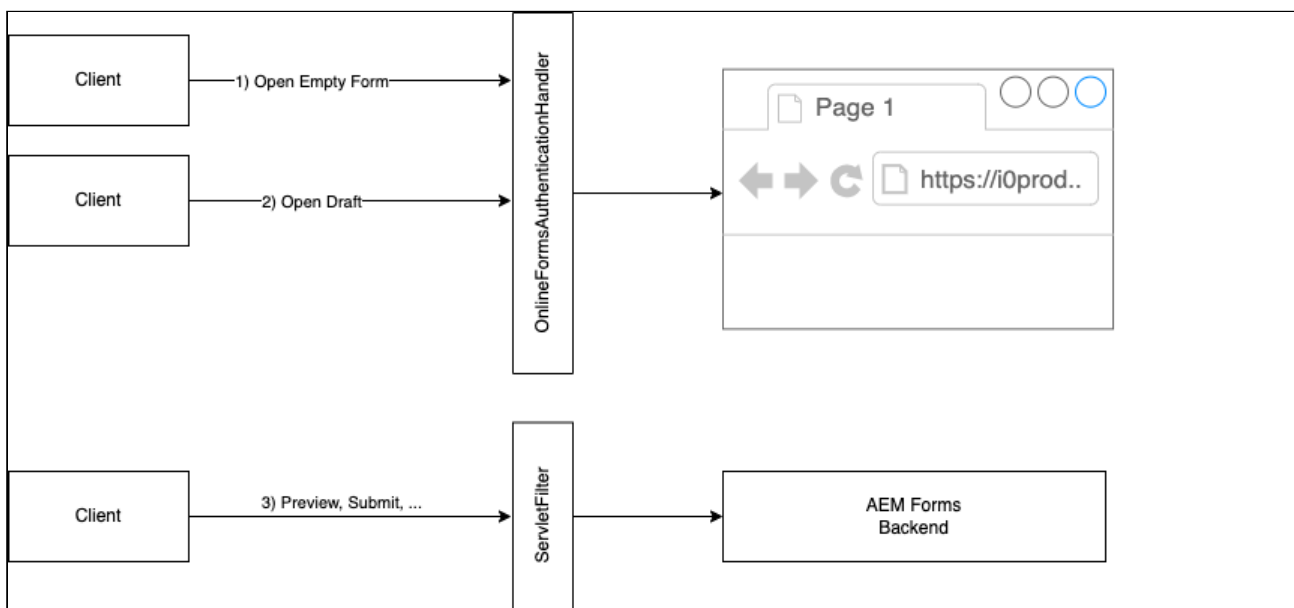
In addition to the restriction, two new users are introduced:

| User              | Description                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------|
| ubs-internal-user | Used when provided security context sent to MyOnlineForms indicates that the user is "Internal" |
| ubs-external-user | Used when provided security context sent to MyOnlineForms indicates that the user is "External" |

These users are also part of the "afcomplib\_all" and are installed with the "apps" part. They currently have the same AEM Forms rights (to see an Online Form).

### 23.2.2 Authentication

Every GET request on an Online Form passes the "OnlineFormsAuthenticationHandler". This class integrates to MU9 (MyOnlineForms). The following graphic illustrates this and more information on the cases can be found below:



| Case | Action          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Open Empty Form | <p>When an empty form is opened (from Kiosk or another place), the OnlineFormsAuthenticationHandler checks:</p> <ol style="list-style-type: none"> <li>1. If a CSIV_2 token is present (provided from ISSO or UBS Connect)</li> <li>2. Checks against MU9, if the provided token is valid (validateToken endpoint)</li> <li>3. Checks against MU9, if the returned user is allowed to access this form (formAccessCheck endpoint)</li> </ol> |

| Case | Action           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | Open Draft       | <p>When a draft is opened (from MyOnlineForms or another place), the OnlineFormsAuthenticationHandler checks:</p> <ol style="list-style-type: none"> <li>1. If a CSIV_2 token is present (provided from ISSO or UBS Connect)</li> <li>2. Checks against MU9, if the provided token is valid (validateToken endpoint)</li> <li>3. Checks against MU9, if the returned user is allowed to access this draft (draftAccessCheck endpoint)</li> </ol>             |
| 3    | Servlet Requests | <p>Every request from the client to the "afcomplib_all" bundle endpoints is secured in the "ServletFilter". The pattern for the URL is: */bin/com/ajila/*</p> <p>If the URL is a afcomplib_all Servlet Endpoint, the ServletFilter checks:</p> <ol style="list-style-type: none"> <li>1. If a CSIV_2 token is present (provided from ISSO or UBS Connect)</li> <li>2. Checks against MU9, if the provided token is valid (validateToken endpoint)</li> </ol> |

The idea is, to keep the Vendor Product AEM Forms and lean as possible when it comes to security. Therefore, we forward all access checks to MU9. When MU9 reports an error, no user login is done and the access is denied.

#### Successful Authentication Case

- The AEM Forms user is logged in according to the user-type (internal / external)
- The additional information from MU9 is stored on the HTTP session

## 24 24) Environmental Marker

The environmental marker implementation can be used to mark an AEM Forms instance as non-productive. When activated, the header bar appears in the defined color and all PDF documents have a "TEST" watermark.

### 24.1 Configuration

Component `com.ubs.OP2.forms.internal.service.EnvironmentMarkerServiceImpl`

Description for `com.ubs.OP2.forms.internal.service.EnvironmentMarkerServiceImpl`

Active For Test Env ☒ Turn on only for test environments (isActive)

Header Hex Colour  Provide the header hex colour for specific view (hexColour)

Text Hex Colour  Provide the text hex colour for specific view (textHexColour)

Environment  Provide the text hex colour for specific view (environmentVal)

**Configuration Information**

Persistent Identity (PID) `com.ubs.OP2.forms.internal.service.EnvironmentMarkerServiceImpl`

Configuration Binding `Unbound or new configuration`

Cancel Reset Delete Unbind Save

The relevant configuration can be found as OSGI config: `com.ubs.OP2.forms.internal.service.EnvironmentMarkerServiceImpl`

| Configuration       | Description                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Active For Test Env | Default is set to false, meaning no non-productive indicator is shown to the user. When activated, the watermark is applied and the headerbar is shown in the defined color. |
| Header Hex Colour   | Header Hex value which is applied to the headerbar when the marker is activated.                                                                                             |
| Text Hex Colour     | Hex value for the font-colour used in the header bar.                                                                                                                        |
| Environment         | UNUSED currently: Prepared if an indication of the test environment should be visualized (DEV, TE1, TE2).                                                                    |

### 24.2 Text Translation

The text shown to the user in the headerbar is configured in the `afcomplib_all`: **`apps\ajila-forms-customers\ajila-forms-ubs\i18n`**. The relevant key is: **`ajila-forms-ubs-environment-message`**.

The text is available in all languages. **Note:** The text could be changed in the AEM Forms translator during runtime if needed! But it would be overwritten with an installation.



## 25 25) SQL-2 Oak Index

### 25.1 Improve SQL-2 queries with Oak Indexes

Sometimes when you write sql2 queries which catches a lot of nodes you can run into multiple problems such as slow querying or even abortion of the thread. You even might get an error like these:

`3crConsistencyCheck error: The query read or traversed more than 1000000 nodes. To avoid affecting other tasks, processing was stopped.`

#### 25.1.1 Check your query

Since AEM 6.4 you can check your query performance under <http://localhost:4502/libs/granite/operations/content/diagnostools/queryPerformance.html>

Simply just paste your query in the rubric “Explain” and execute it. You will get a result (best case) or an error (worst case)

Usually you will get a result like this:

The screenshot shows the 'Query Explanation' tool interface. It displays the following information:

- Query Explanation**: The title of the tool.
- Indexes Used**: A list showing 'lucene(oak:index/lucene)' is used.
- Execution Time**: A summary of the query execution performance.
  - Total time: 4 ms
  - Query execution time: 1 ms
  - Get nodes time: 2 ms
  - Result node count time: 3 ms
  - Number of nodes in result: 28
- Execution Plan**: The SQL-2 query text: `[!structured] as [node] (*) lucene(lucene(oak:index/lucene) full $taglet" & ($taglet "")) where ([!descendantnode($node), ($content/forms/af)]) and ([contains($node($taglet), ""))]`
- Logs**: A section for viewing query execution logs.

In this case an index (lucene) is used. You might get a different result like “No indexes used”. If this is the case you can improve your performance by adding an Index.

#### 25.1.2 Create a new index

Under <http://oakutils.appspot.com/generate/index> (unfortunately restricted by ubs) you can generate a new index. Your result will look like this:

```

1 - compatVersion = 2
2 - async = "async"
3 - jcr:primaryType = oak:QueryIndexDefinition
4 - evaluatePathRestrictions = true
5 - type = "lucene"
6 + indexRules
7 + nt:unstructured
8 + properties
9 + metaTemplateRef
10 - name = "metaTemplateRef"
11 - propertyIndex = true
12 - notNullCheckEnabled = true
13

```

Now go to `/crx/de/index.jsp` and navigate to the node `oak:index`. Create a new subnode with type: `oak:QueryIndexDefinition`. Now add all the properties and subnodes based on your result.

For explanation: The values with the "-" sign are property values while the values with "+" sing are nodes

So in this case the node tree should look like

| Properties |                          | Access Control | Replication              | Console |
|------------|--------------------------|----------------|--------------------------|---------|
|            | Name                     | Type           | Value                    |         |
| 1          | async                    | String         | async                    |         |
| 2          | compatVersion            | Decimal        | 2                        |         |
| 3          | evaluatePathRestrictions | Boolean        | true                     |         |
| 4          | jcr.primaryType          | Name           | oak:QueryIndexDefinition |         |
| 5          | type                     | String         | lucene                   |         |

### 25.1.3 Increase node limit

Instead of creating indexes or optimize the query you can also change the limit.

Under <http://localhost:4502/system/console/jmx/org.apache.jackrabbit.oak%3Aname%3Dsettings%2Ctype%3DQueryEngineSettings> you are able to modify the limit but keep in mind that higher limits can cause other issues.

[illegible]

## 26 26) Accordion Layout - Best Practices (Legal/Terms and Condition)

When we have terms and conditions text or Legal text in Accordion layout. It is advised to populate both Title property as well as text description which appears on expanding the dropdown. In many cases, the issue was observed and reported about the layout missing header or numeration details logged in (EFORMS-44746).

There are different possible layout for Agreement panels used in different form, for example-

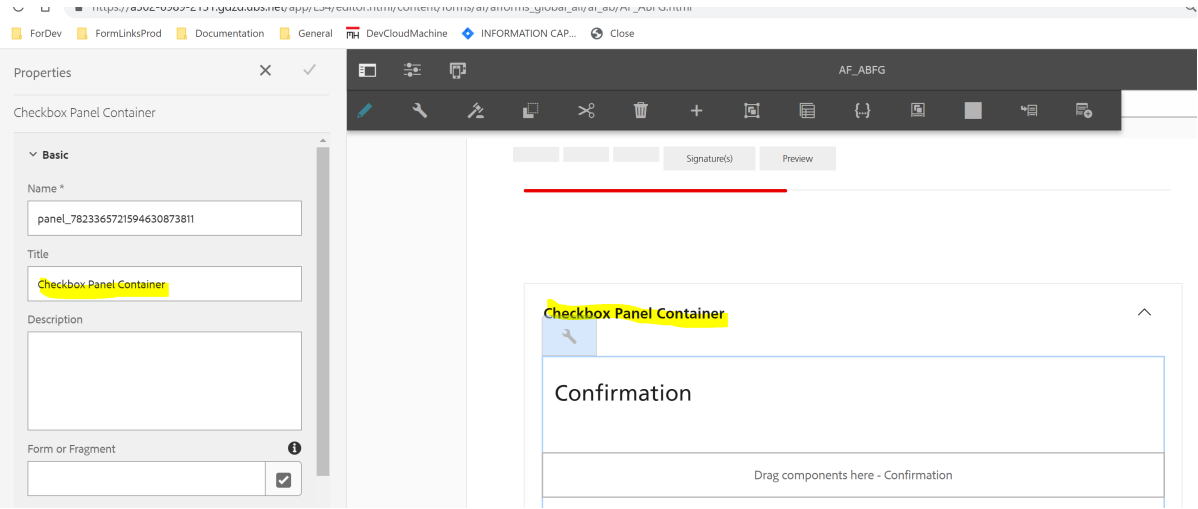
- 1) Accordion dropdown with Heading and on expansion description is visible..
- 2) Accordion layout with just numbering and user expands to see the text.
- 3) Accordion layout with no heading, the description text visible only on expansion. Ex. [https://a302-6991-0117.gdzd.ubs.net/app/LS4/content/forms/af/affirms\\_global\\_all/af\\_ab/AF\\_ABFG.html?wcmmode=disabled&mandator=001&afAcceptLang=it-ch](https://a302-6991-0117.gdzd.ubs.net/app/LS4/content/forms/af/affirms_global_all/af_ab/AF_ABFG.html?wcmmode=disabled&mandator=001&afAcceptLang=it-ch)

To add a title to any accordion item, please follow these steps-

If we need a result like the accordion dropdowns have label, we need to update the 'Title' property for panel name. This is the field from where the accordion dropdown picks value. In the example shared in the original request, the form had no value for title under panel properties and hence the dropdowns appeared empty.

### Solution screenshot-

The screenshot displays a form interface with a yellow warning banner at the top. The banner text reads: "The incomplete form mode has been activated. You can download the form once you have configured it. While this mode is active, you will not be able to submit the form digitally." Below the banner, the text "Trust, foundation, domiciliary company" is visible. Underneath this text is a red horizontal line. A dropdown menu is shown with the text "Checkbox Panel Container" and a downward arrow. At the bottom of the form, there are two buttons: "Back" and "Next".



## 27 27) Layout considerations / do's and don'ts

### 27.1 Introduction

#### Important

With the June 2024 release, **adaptive forms do not support initial styling** anymore. All **new forms** must add the e-banking style.

Because adaptive forms will be integrated in E-Banking, the strategic decision was made to match the [E-Banking style guide](#)<sup>22</sup>.

The ICC pod was responsible for applying the style guide to all components used, but we also noticed some issues with the choices that were made to design the forms by pod 8

#### Note

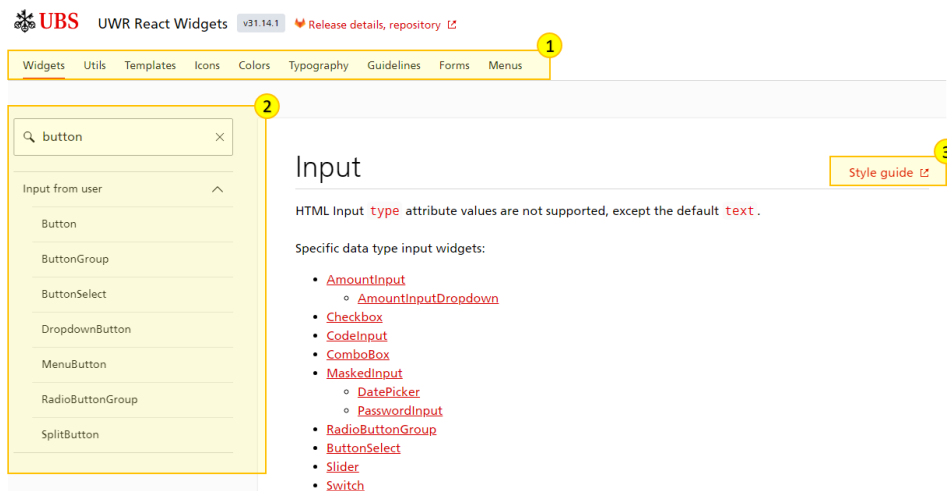
Online forms are not the same as PDFs. Please do not make the mistake to think you can migrate 1:1 from PDF to online and keep the same appearance in both.

I'm aware the strict style guide will require the designer to find new ways to deal with the requirements and the requests from the form owner. The BA and the designer should familiarize themselves with the E-Banking style guide. It gives the best arguments during a discussion with the form owner.

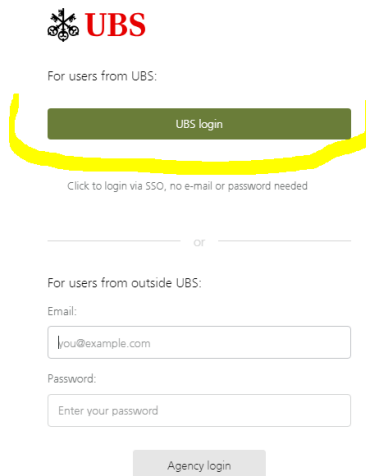
#### Introduction to the E-Banking style guide

---

<sup>22</sup> <https://cb-nexus.ubs.net/repository/hosted-raw-ubs-site/UWR/uwr-react-widgets/latest/docs/index.html#/widgets/Input>



1. In the top navigation you can see the top level topics of the style guide
2. On the left you have topics related to the selected option in the top navigation.
  - a. For widgets you can search for the widget you need, like: Button, Input, Radio button etc.
3. The link "Style guide" will open in a new tab with more information to the selected widget
  - a. After clicking, you will need to log in. Please use the "UBS Login" button.
  - b. **After you logged in, close the tab and click on the "Style guide" link again.**



### 27.1.1 Button position

According to the E-Banking style guide, buttons are positioned on the left side only.

**See:** [Button behavior](#)<sup>23</sup>

The praxis to center buttons should be avoided.

<sup>23</sup> <https://ubs.frontify.com/document/17#/input-from-user/button/behaviour>

Form selection "Power of attorney for the management of assets for financial intermediaries"

Selection...

Form

☒ 61218; with authorization to take out a loan; with fee direct debit authorization

☐ 61219; without authorization to take out a loan; without fee direct debit authorization

☐ 61518; with authorization to take out a loan; without fee direct debit authorization

☐ 61519; without authorization to take out a loan; with fee direct debit authorization

← Proceed to form

### 27.1.2 Misuse of checkboxes as radio buttons

A checkbox should never be used as a radio button.

See: [Radio button do's and don'ts](#)<sup>24</sup>

Example from form ACBA:

☐ 1. Principal only

☒ 2. Principal and interest

☐ 3. Principal plus negative cash balances or positive cash balances on current accounts:

☐ all current accounts irrespective of reference currency; or

☐ only current accounts of the original fixed advance currency; or

☐ the following current accounts irrespective of reference currency:

☐ 4. Principal and interest plus negative cash balances or positive cash balances on current accounts:

☐ all current accounts irrespective of reference currency; or

☐ only current accounts of the original fixed advance currency; or

☐ the following current accounts irrespective of reference currency:

### 27.1.3 Adding blank text elements as a means of increasing white space between elements

(feature not yet implemented) Please use the margin dropdown in the panel setting.

<sup>24</sup> <https://ubs.frontify.com/document/17#/input-from-user/checkboxes/do-s-and-don-ts>

## 27.1.4 Fake tabular layouts

Adaptive forms **doesn't support** tabular layouts.

Example from form ACCK:

| Do not use this form for                                                                                                                                                                                                                                                                                                                                                                                         | Instead, use the Form            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| • A US entity or US citizen or resident                                                                                                                                                                                                                                                                                                                                                                          | W-9                              |
| • A foreign individual                                                                                                                                                                                                                                                                                                                                                                                           | W-8BEN (Individual) or Form 8233 |
| • A foreign individual or entity claiming that income is effectively connected with the conduct of trade or business within the United States (unless claiming treaty benefits)                                                                                                                                                                                                                                  | W-8ECI                           |
| • A foreign partnership, a foreign simple trust, or a foreign grantor trust (unless claiming treaty benefits) (see instructions for exceptions)                                                                                                                                                                                                                                                                  | W-8IMY                           |
| • A foreign government, international organization, foreign central bank of issue, foreign tax-exempt organization, foreign private foundation, or government of a US possession claiming that income is effectively connected with US income or that is claiming the applicability of section(s) 115(2), 501(c), 892, 895, or 1443(b) (unless claiming treaty benefits) (see instructions for other exceptions) | W-8ECI or W-8EXP                 |
| • Any person acting as an intermediary (including a qualified intermediary acting as a qualified derivatives dealer)                                                                                                                                                                                                                                                                                             | W-8IMY                           |

## 27.1.5 Adding input fields in running text

Never add input fields as part of the text.

Example from form ACCK:

**9. Special rates and conditions** (if applicable see instructions):

The beneficial owner is claiming the provisions of article and paragraph

of the treaty identified on line 8a above to claim a \_\_\_\_\_ % rate of withholding on (specify type of income):

Explain the additional conditions in the article the beneficial owner meets to be eligible for the rate of withholding:

## 27.1.6 Input field label and placeholder

- Each input field **requires** a label
- Placeholders **are not** a replacement for labels
- Placeholder are always visible, not only after entering field



**See:** [Input field do's and don'ts](#)<sup>25</sup>

Example from form ACCK:

- Missing label

**9. Special rates and conditions** (if applicable see instructions):

The beneficial owner is claiming the provisions of article and paragraph

of the treaty identified on line 8a above to claim a \_\_\_\_\_ % rate of withholding on (specify type of income):

Explain the additional conditions in the article the beneficial owner meets to be eligible for the rate of withholding:

Example from form ABZE:

- Missing meaningful label
- Placeholder must be always visible

### Banking relationship



## 27.1.7 Keep related radio buttons / check-boxes together in a group

- Use always a **group** for multiple checkboxes or radio buttons
- Never place content between radio buttons / checkboxes in a group

Example from form: AABF (two checkbox components used instead of one group)

<sup>25</sup> <https://ubs.frontify.com/document/17#/input-from-user/input-fields/do-s-and-don-ts>

## Standing order Fisca

- ☐ UBS Fisca/Fiscalife with max. amount
- ☐ UBS Fisca/Fiscalife with individual amount

### 27.1.8 Each input element needs a label

#### 27.1.8.1 input field

See point 6

#### 27.1.8.2 Radio button / Checkbox group

- Labels are never **bold**
- Labels need to be closer to the radio button group (must be investigated by ICC pod)
- Labels can use 16px font size (standard E-Banking font size)

Example from form ABZE:

- Subtitle is misused as label
- Gap between label (subtitle) too big

## Request

Which unit should process your guarantee? Please select:

- ☐ Zurich ☐ Geneva ☐ Lugano

Next

Previous

### 27.1.9 Tooltip for radio button / check-box group

Adaptive forms doesn't allow to place tooltips for individual radio buttons / checkboxes in a group. The tooltip must be placed on the label for the entire group instead.

Example from form ABZE:

Is a third-party bank involved?

☐ No (direct issuance) ☐ Yes (indirect issuance) ☐ Yes, advice only



Next

Previous

### 27.1.10 Custom invalid messages

**Never** try to recreate standard components

### 27.1.11 Meaningful invalid messages

The invalid message should be helpful. Try to express what value is expected

### 27.1.12 Truncated labels

**Never** truncate labels. Use **tooltip (info icon)** to add additional information.

Example from form ACCK:

If you entered disregarded entity, partnership, simple trust, or grantor trust above, is the entity a hybrid making a treaty claim? If "Yes" complete Part II.

☐ Yes ☐ No

5. Permanent residence address (street, apt. or suite no., or rural route). Do not use a P.O. box or in-care-of address (other than a re...

Country \*

City or town, state or province. Include postal code where applicable

a

### 27.1.13 Accordion titles

Use **meaningful, short** titles. Avoid using only numbers.

Example from form ACCO:

Party A agrees to act as agent upon the terms and conditions set out below:

|   |   |
|---|---|
| 1 | ▼ |
| 2 | ▼ |
| 3 | ▼ |
| 4 | ▼ |
| 5 | ▼ |

## 28 28) AF Styling -E-Banking

- [Adding CSS classes to adjust margin \(see page 253\)](#)
- [Adding more flexible indentation options for Static Text component GitLab #1383 \(see page 254\)](#)
- [MessageBox component replaces Information Text and Error Text GitLab #1334 \(see page 255\)](#)
- [Migrating from initial styling GitLab #1940 \(see page 256\)](#)
- [Signature layout without using an empty Static Text element as placeholder GitLab #1372 \(see page 256\)](#)

### 28.1 Adding CSS classes to adjust margin

With the current setup of e-banking restyling the author is limited with the editor adjustments meaning that lot of "Styling" features not working.

As workaround we added new margin stylings in our CSS files. The designer can now adjust the margin-bottom as following by adding the corresponding class to the component.

Every component does have a field called CSS in the editor which allows the author to add as many classes he likes

| Class         | Value               |
|---------------|---------------------|
| ubs-margin-0  | margin-bottom: 0px  |
| ubs-margin-5  | margin-bottom: 5px  |
| ubs-margin-10 | margin-bottom: 10px |
| ubs-margin-15 | margin-bottom: 15px |
| ubs-margin-20 | margin-bottom: 20px |
| ubs-margin-25 | margin-bottom: 25px |
| ubs-margin-30 | margin-bottom: 30px |

### 28.1.1 Example usage

Adding margin-bottom of 30px to a static component:

Basic

Name \*

ST\_Guarantee\_Issue

Bind Reference ?

Data Model Object Reference 🔍

☐ Hide Object

CSS Class

ubs-margin-30

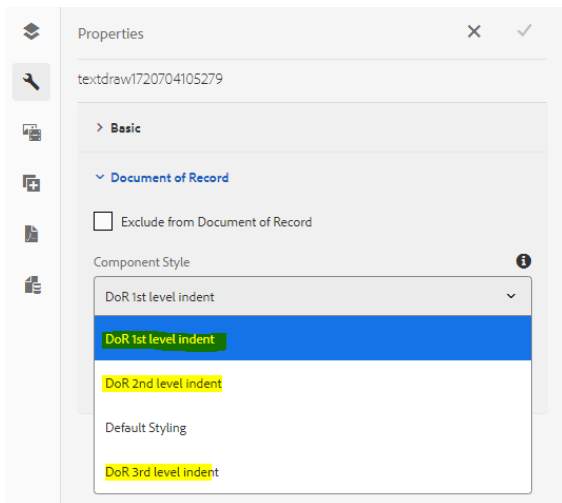
> Document of Record

## 28.2 Adding more flexible indentation options for Static Text component GitLab #1383

The designer can choose between 3 indents. The 1st level indent is 5mm, the 2nd level indent is 9mm and the 3rd level indent is 13mm.

This can be selected in the Settings of a Static Text components under Document of Record → Component Style

This feature doesn't apply to an adaptive form. Only the DoR will be affected





K

Banking relationship

FORM TITLE

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

## 28.3 MessageBox component replaces Information Text and Error Text GitLab #1334

In the June 2024 release we introduced a new component for ICS to use: **MessageBox**. Going forward, please do not use neither the "Information Text", the "Error Text" nor the "Errortext carousel preview" component in AF forms. **MessageBox** replaces both component and introduces Beside the Error and Info style, MessageBox introduces also styles for Confirmation and Warning.

ICS was informed during Pod X-Sync call on 08.07.2020.

**Update (XPod sync on 3.2.2025):** This also includes the Preview step on an Adaptive form.

## 28.4 Migrating from initial styling GitLab #1940

On the Preview step, the "Preview could not be created" component needs to be changed from "Error Text" to "MessageBox - carousel preview error".

### Aiila Forms Carousel

**Preview could not be created**  
Choose "Preview" below to try again, or submit the form without preview.

**Submission failed**  
The form could not be sent. Please try again later.

## 28.5 Signature layout without using an empty Static Text element as placeholder GitLab #1372

To change all existing forms at the same time, the existing signature fragment can be updated.

Steps needed to use this feature:

1. Delete empty Static Text
2. Change DoR settings for the element: Signature (s. screenshot) to: "DoR right aligned"

Signature of client

PN\_Signature\_Client

Mandator 033

EFORMS-49220 IT Client Signature (Signature, Place, Date, Name) (Single)

PN\_Signature\_Client

New fragment, without placeholder

Signature of client

Place

Date (dd.mm.yyyy)

Name of client \*



## 29 29) Logging

### 29.1 Background

With the Refactoring of the "FormSubmitServiceImpl", the logging for the Submit was reworked. Note: This change is currently implemented only in the submit. But the concepts and ideas can be taken into consideration in other areas of the afcomplib\_all. The idea is to have a comprehensible trace of logs throughout the submit. These logs should also be linked to a specific submit.

### 29.2 Submit

This section documents the logging during a Online Form / renderDor Submit.

#### 29.2.1 Concept

The main class for storing submit information is the class: FormSubmitContext. This class is instantiated at the beginning of a submit. It contains the following information:

##### FormSubmitContext Information

```
{
  "submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d",
  "browserlessDorRendering":false,
  "clpPersistenceAllowed":true,
  "userId":"43642842",
  "userType":"1",
  "draftUuid":"99e83180-79b8-418c-a130-664758753269",
  "xMandator":"",
  "applicationId":"AFC"
}
```

These information are available throughout the submit. The "submitTransactionId" is a UUID generated at the beginning of the submit. It is used to link all log entries to each other during the submission of the form. The log statements containing the FormSubmitContext are always having the JSON above attached.

Another important class is the DorPersistenceData class. This class is created per form / subform and handles an individual formcode and CDOK and stores the generated document. The information there looks as follows:

**FormMetadata Information**

```
{
  "formUuid":"74854d77-9c04-4ab3-95c7-36a93e832829",
  "formcode":"BAIU",
  "mandator":"030",
  "language":"EN",
  "masterCdok":"61710",
  "finalCdok":"61710",
  "documentType":"MAIN"
}
```

## 29.2.2 Technical information

The classes mentioned above overwrite the "toString()" method and return a JSON object out of it. This has the following advantages:

- In the toString() Method, we can steer what information should be in the log. Example: We don't want the CSIV2 token to be part of the log all the time. So we omit it there.
- The JSON can be extracted via Splunk and be formatted. The information gathering is easier.

### 29.2.2.1 Technical usage

**Sample Log**

```
log.debug("CLP Persistence completed. SubmitContext: {}, Persistence Data: {}",
submitContext, persistenceData);
```

By providing the "submitContext" and "persistenceData" to the Slf4j Logger, the toString() is automatically executed. The result in the logs looks as follows:

**Example log**

```
18.12.2023 17:37:15.123 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl CLP Persistence completed.
SubmitContext: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browser
lessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"
1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"
AFC"}, Persistence Data: {"formUuid":"74854d77-9c04-4ab3-95c7-36a93e832829","formcode
```

```
": "BAIU", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "MAIN"}
```

### 29.2.2.2 Log Levels

The log levels in the submit are used as follows (list is not complete and consists only of the loggings done in the refactorings - to be completed):

| Log Level | Usage                                                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INFO      | <ul style="list-style-type: none"> <li>The business traceability of the submit (PDF Documents generated, Watermarks applied, CLP Persistence done, ...)</li> </ul> |
| DEBUG     | <ul style="list-style-type: none"> <li>The technical traceability for further information and details about each step.</li> </ul>                                  |

### 29.2.3 Practical example

The following log extraction shows, how a submit of a business formset looks like in the log. This is also one of the more complex submits to illustrate this:

```
18.12.2023 17:37:02.776 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Form Submit started.
{"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRenderin
g":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"1","draftUuid":
"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"}
18.12.2023 17:37:02.779 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.dor.DorGenerationExecutor Root Panel Definition
indicates a business formset. SubmitContext: {"submitTransactionId":"c42edc71-
b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":false,"clpPersistenceAllowed":t
rue,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-
a130-664758753269","xMandator":"","applicationId":"AFC"}
18.12.2023 17:37:05.918 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl DoR Generation completed.
{"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRenderin
g":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"1","draftUuid":
"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"}
18.12.2023 17:37:05.918 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
```

com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Omit watermark because there is no document. Submit Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data: {"formUuid":"","formcode":"BAIS","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","documentType":"FORMSET\_DATA"}

18.12.2023 17:37:05.949 \*DEBUG\* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms\_apac\_integrated/af\_ba/AFS\_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Applied watermark to document. Submit Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data: {"formUuid":"","formcode":"BAIU","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","documentType":"MAIN"}

18.12.2023 17:37:05.980 \*DEBUG\* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms\_apac\_integrated/af\_ba/AFS\_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Applied watermark to document. Submit Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data: {"formUuid":"","formcode":"BAIV","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","documentType":"MAIN"}

18.12.2023 17:37:05.980 \*DEBUG\* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms\_apac\_integrated/af\_ba/AFS\_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Watermark applied to PDF documents. Environment DEV. {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"}

18.12.2023 17:37:05.980 \*DEBUG\* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms\_apac\_integrated/af\_ba/AFS\_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

com.ubs.OP2.forms.internal.service.dor.util.DorCustomPropertyService Skip applying custom properties because there is no document. Submit Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data: {"formUuid":"","formcode":"BAIS","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","documentType":"FORMSET\_DATA"}

18.12.2023 17:37:05.987 \*DEBUG\* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms\_apac\_integrated/af\_ba/AFS\_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

com.ubs.OP2.forms.internal.service.dor.util.DorCustomPropertyService Applied custom properties to PDF document. Submit Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":**false**,"clpPersistenceAllowed":**true**,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data:

```

{"formUuid":"","formcode":"BAIU","mandator":"030","language":"EN","masterCdok":"61710",
,"finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:05.992 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.dor.util.DorCustomPropertyService Applied custom
properties to PDF document. Submit Context: {"submitTransactionId":"c42edc71-
b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":false,"clpPersistenceAllowed":t
rue,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-
a130-664758753269","xMandator":"","applicationId":"AFC"} - Persistence Data:
{"formUuid":"","formcode":"BAIV","mandator":"030","language":"EN","masterCdok":"61710",
,"finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:05.993 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Custom Properties applied to
all PDF documents. {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","bro
wserlessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userTyp
e":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","application
Id":"AFC"}
18.12.2023 17:37:05.993 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl PDF Document generation and
attachment fetching completed. {"submitTransactionId":"c42edc71-b994-44ed-
b2f9-0b287b31021d","browserlessDorRendering":false,"clpPersistenceAllowed":true,"user
Id":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMa
ndator":"","applicationId":"AFC"}
18.12.2023 17:37:05.993 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl No CLP persistence for Top-
Level Formset. {"formUuid":"","formcode":"BAIS","mandator":"030","language":"EN","mas
terCdok":"61710","finalCdok":"61710","documentType":"FORMSET_DATA"}
18.12.2023 17:37:05.993 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl CLP Persistence starting.
SubmitContext: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browser
lessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"
1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"
AFC"}, Persistence Data: {"formUuid":"","formcode":"BAIU","mandator":"030","language":"
EN","masterCdok":"61710","finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:06.005 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.clp.FormDataPersistenceService Create new CLP
Record for Subform (Business Formset Case). SubmitContext {"submitTransactionId":"c42
edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":false,"clpPersistenceAll
owed":true,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-
a130-664758753269","xMandator":"","applicationId":"AFC"} - persistenceData

```

```

{"formUuid":"","formcode":"BAIU","mandator":"030","language":"EN","masterCdok":"61710",
,"finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:15.122 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.clp.FormDataPersistenceService CLP data processed
successfully for OnlineForm: Formcode: BAIU Mandator: 030 Language: EN MasterCdok:
61710 FinalCdok: 61710
18.12.2023 17:37:15.123 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl CLP Persistence completed.
SubmitContext: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browser
lessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"
1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"
AFC"}, Persistence Data: {"formUuid":"74854d77-9c04-4ab3-95c7-36a93e832829","formcode
":"BAIU","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","d
ocumentType":"MAIN"}
18.12.2023 17:37:15.123 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl CLP Persistence starting.
SubmitContext: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browser
lessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"
1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"
AFC"}, Persistence Data: {"formUuid":"","formcode":"BAIV","mandator":"030","language":"
EN","masterCdok":"61710","finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:15.137 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.clp.FormDataPersistenceService Create new CLP
Record for Subform (Business Formset Case). SubmitContext {"submitTransactionId":"c42
edc71-b994-44ed-b2f9-0b287b31021d","browserlessDorRendering":false,"clpPersistenceAll
owed":true,"userId":"43642842","userType":"1","draftUuid":"99e83180-79b8-418c-
a130-664758753269","xMandator":"","applicationId":"AFC"} - persistenceData
{"formUuid":"","formcode":"BAIV","mandator":"030","language":"EN","masterCdok":"61710",
,"finalCdok":"61710","documentType":"MAIN"}
18.12.2023 17:37:21.288 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.clp.FormDataPersistenceService CLP data processed
successfully for OnlineForm: Formcode: BAIV Mandator: 030 Language: EN MasterCdok:
61710 FinalCdok: 61710
18.12.2023 17:37:21.288 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl CLP Persistence completed.
SubmitContext: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browser
lessDorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"
1","draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"
AFC"}, Persistence Data: {"formUuid":"5a32273b-3676-4566-9045-19e0f2af525d","formcode

```



```

": "BAIV", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "MAIN"}
18.12.2023 17:37:21.289 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl PDF Documents are successfully persisted in CLP. {"submitTransactionId": "c42edc71-b994-44ed-b2f9-0b287b31021d", "browserlessDorRendering": false, "clpPersistenceAllowed": true, "userId": "43642842", "userType": "1", "draftUuid": "99e83180-79b8-418c-a130-664758753269", "xMandator": "", "applicationId": "AFC"}
18.12.2023 17:37:21.289 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.submit.MofRecordSubmitHandler Create or Update Top-Level Formset CLP Record for MOF. SubmitContext: {"submitTransactionId": "c42edc71-b994-44ed-b2f9-0b287b31021d", "browserlessDorRendering": false, "clpPersistenceAllowed": true, "userId": "43642842", "userType": "1", "draftUuid": "99e83180-79b8-418c-a130-664758753269", "xMandator": "", "applicationId": "AFC"} - MofRecordData: {"formUuid": "", "formcode": "BAIS", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "FORMSET_DATA"}
18.12.2023 17:37:24.096 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.submit.MofRecordSubmitHandler Prepare update of draft to final MOF record. SubmitContext: {"submitTransactionId": "c42edc71-b994-44ed-b2f9-0b287b31021d", "browserlessDorRendering": false, "clpPersistenceAllowed": true, "userId": "43642842", "userType": "1", "draftUuid": "99e83180-79b8-418c-a130-664758753269", "xMandator": "", "applicationId": "AFC"} - MofRecordData: {"formUuid": "", "formcode": "BAIS", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "FORMSET_DATA"}
18.12.2023 17:37:24.234 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.submit.MofRecordSubmitHandler Finalized existing draft in MOF. SubmitContext: {"submitTransactionId": "c42edc71-b994-44ed-b2f9-0b287b31021d", "browserlessDorRendering": false, "clpPersistenceAllowed": true, "userId": "43642842", "userType": "1", "draftUuid": "99e83180-79b8-418c-a130-664758753269", "xMandator": "", "applicationId": "AFC"} - MofRecordData: {"formUuid": "", "formcode": "BAIS", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "FORMSET_DATA"}
18.12.2023 17:37:24.241 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.dor.util.DorFileAssembler Add PDF to ZIP. SubmitContext: {"submitTransactionId": "c42edc71-b994-44ed-b2f9-0b287b31021d", "browserlessDorRendering": false, "clpPersistenceAllowed": true, "userId": "43642842", "userType": "1", "draftUuid": "99e83180-79b8-418c-a130-664758753269", "xMandator": "", "applicationId": "AFC"} - Persistence Data: {"formUuid": "74854d77-9c04-4ab3-95c7-36a93e832829", "formcode": "BAIU", "mandator": "030", "language": "EN", "masterCdok": "61710", "finalCdok": "61710", "documentType": "MAIN"}
18.12.2023 17:37:24.246 *DEBUG* [10.212.64.145 [1702917419556] POST /app/LS4/content/forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/guideContainer.af.submit.jsp HTTP/1.1]

```

```

com.ubs.OP2.forms.internal.service.dor.util.DorFileAssembler Add PDF to ZIP. Submit
Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDo
rRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"1","dr
aftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"}
- Persistence Data: {"formUuid":"5a32273b-3676-4566-9045-19e0f2af525d","formcode":"BA
IV","mandator":"030","language":"EN","masterCdok":"61710","finalCdok":"61710","docume
ntType":"MAIN"}
18.12.2023 17:37:24.252 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.dor.util.DorFileAssembler Save ZIP to CRX. Submit
Context: {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserlessDo
rRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"1","dr
aftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC"}
18.12.2023 17:37:24.278 *INFO* [10.212.64.145 [1702917419556] POST /app/LS4/content/
forms/af/affforms_apac_integrated/af_ba/AFS_BAIS/jcr:content/
guideContainer.af.submit.jsp HTTP/1.1]
com.ubs.OP2.forms.internal.service.FormSubmitServiceImpl Submit successfully
completed. {"submitTransactionId":"c42edc71-b994-44ed-b2f9-0b287b31021d","browserless
DorRendering":false,"clpPersistenceAllowed":true,"userId":"43642842","userType":"1","
draftUuid":"99e83180-79b8-418c-a130-664758753269","xMandator":"","applicationId":"AFC
"}

```

This is an extract of a submit. To trace it, there are two possibilities:

- An existing PDF Document has a ClpUuid attached to it. Searching the logs for this UUID will link back the submitTransactionId.  
Example: ClpUuid 74854d77-9c04-4ab3-95c7-36a93e832829 → In the logs, it can be traced back to the submitTransactionId → c42edc71-b994-44ed-b2f9-0b287b31021d
- Identity the start of the submit and find the submitTransactionId → c42edc71-b994-44ed-b2f9-0b287b31021d

With the submitTransactionId, the submit can be traced through the process.



# 30 30) RenderDor

- [FormMetadata Fragment](#) (see page 265)

## 30.1 FormMetadata Fragment

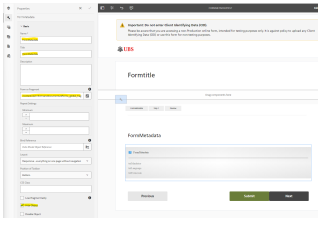
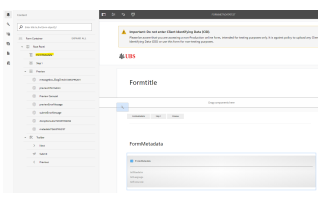
### 30.1.1 Background

The afcomplib\_all offers the following client lib to obtain the metadata of a form:

```
window.forms.ubs.getFormMetadata()
```

This method extracts the information from the URL in an Online Form. When this function is executed in the backend (renderDor), the URL is not present. To obtain those information, the "FormMetadata" Fragment needs to be added to the form as Step 0.

### 30.1.2 Step by Step

|   | Step                                                                                                                                                                  | Description                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 | Add a new "Step" to the form and name it "FormMetadata". The Fragment reference is:<br><br>/content/dam/formsanddocuments/<br>afforms_global_fragmentlib/formmetadata |  |
| 2 | Make sure, that it's the Step 0 (this is technically not needed but provides a consistency throughout all forms).                                                     |  |

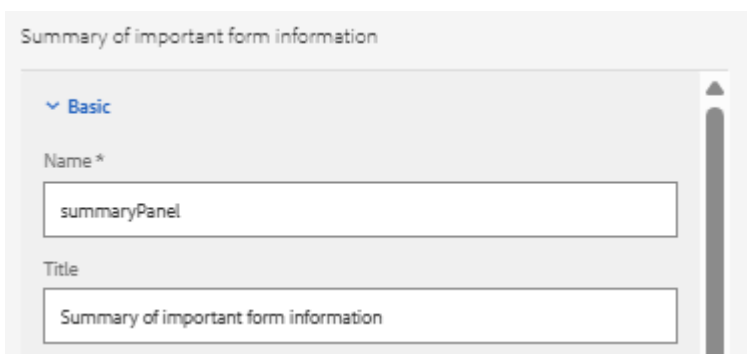
## 31 31) HTML Preview

### 31.1 Setting Up An Existing Form

All forms by default have the old preview style and will be updated to the HTML summary preview afterwards, and to do this we have three steps to achieve this.

Once we have taken these steps we can remove the old preview items such as the preview carousel, the messagebox relating to the preview too.

- 1) Change preview panel name to 'summaryPanel' with the Title 'Summary of important form information'



The screenshot shows a configuration panel titled "Summary of important form information". Inside the panel, there is a section labeled "Basic" with a dropdown arrow. Below this, there are two input fields: "Name \*" with the value "summaryPanel" and "Title" with the value "Summary of important form information".

- 2) Add the Summary component with the following Name 'summaryComponent' and Title 'Summary', and ensure the checkboxes at the end are checked.



The screenshot shows a configuration panel titled "Summary". Inside the panel, there is a section labeled "Basic" with a dropdown arrow. Below this, there are two input fields: "Name \*" with the value "summaryComponent" and "Title" with the value "Summary".

Summary

☐ Allow multiple lines

☐ Allow Rich Text

☐ Enable Autofill ⓘ

Maximum Number of Characters

Minimum Number of Characters

Exact Number of Characters ⓘ

☐ Value accessible in lazy loading context ⓘ

☐ Hide Object

☐ Disable Object

Field Layout

CSS Class

☒ Exclude From Summary Page

☒ Show Static Text in Summary Page

☒ Replace Empty Values in Summary Page

3) On the Next button add the following call

|                                             |                                                                                    |                                             |
|---------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
| <input type="checkbox"/> nextbtnnav - Click | <code>window.ajila.forms.ubs.control.summary.setSummaryData(guidRootPanel);</code> | <input checked="" type="checkbox"/> Enabled |
|---------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|

## 31.2 On a component level

Most components have two options, 'Exclude from summary page' and 'Show on summary page even if hidden'

By default any component hidden will **not** be shown on the summary page

Exclude will hide it from the summary in simple terms.

Show if hidden will still show on the summary even when hidden (Maybe used for Banking Relationship combined fields or anything in the future)

The screenshot shows a 'Summary' panel on the left with two checkboxes: 'Exclude from summary page' and 'Show on summary page even if hidden'. To the right, a form titled 'Static Title H2' contains two input fields: 'Last name' and 'First name(s)'. A blue 'Jump to field' button is visible next to the title.

On a panel level and Static text/title level there is an additional option for Show jump to field button. This will display the edit button on the Summary page itself to jump the user back to the part of the form.

For panels, please ensure we have a title value for the button be shown correctly.

The screenshot shows a 'Summary' panel on the left with three checkboxes: 'Exclude from summary page', 'Show jump to field button', and 'Hide the title on the summary page'. To the right, a form titled 'Static Title H2' contains two input fields: 'Last name' and 'First name(s)'. A blue 'Jump to field' button is visible next to the title. Below the input fields, a panel titled 'Drag components here - Panel Title' is shown with a 'Continue Page' button.